
GEOSPATIAL INTELLIGENCE

JUNE 2025

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This publication supersedes ATP 2-22.7, dated 26 March 2015.

HEADQUARTERS, DEPARTMENT OF THE ARMY

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GEOSPATIAL INTELLIGENCE

Contents

	Page
Preface	v
Introduction	vii
PART ONE	1
FUNDAMENTALS	1
CHAPTER 1	1
GEOSPATIAL INTELLIGENCE (GEOINT) OVERVIEW	1
What Is GEOINT?	1
The National System for Geospatial Intelligence	2
National Geospatial-Intelligence Agency.....	3
CHAPTER 2	5
GEOINT ROLES AND RESPONSIBILITIES	5
Army Organizations Supporting the National System for Geospatial Intelligence	5
Army Geospatial Enterprise.....	8
PART TWO	15
GEOINT TECHNIQUES	15
CHAPTER 3	15
PLANNING AND DIRECTING	15
Section I – Planning	15
GEOINT in the Military Decision-Making Process.....	15
GEOINT in the Intelligence Preparation of the Operational Environment Process	18
Intelligence Reach and Research.....	25
Developing the Information Collection Plan	26
Establishing a Storage and Distribution Capability.....	27
Section II – Directing	28
Collection Management.....	28
GEOINT Request for Information Process	29

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CHAPTER 4.....	31
COLLECTION AND PROCESSING	31
Section I – Collection.....	31
GEOINT Collection.....	31
Managing GEOINT Collection Requirements	37
GEOINT Collection Activities	38
GEOINT Collection Methods.....	38
Multisensor Collection and Multidiscipline Intelligence Collection	39
Section II – Processing.....	40
Data Management.....	40
Data Manipulation	41
CHAPTER 5.....	45
PRODUCTION AND DISSEMINATION AND INTEGRATION.....	45
Section I – Production	45
GEOINT Analysis	45
Product Type.....	45
Consumer Product Feedback	46
Section II – Dissemination and Integration	47
GEOINT Dissemination and Integration.....	47
Dissemination Methods and Techniques	47
PART THREE	51
GEOINT OPERATIONAL CONSIDERATIONS.....	51
CHAPTER 6.....	51
GEOINT ACROSS THE ARMY STRATEGIC CONTEXTS.....	51
Competition Below Armed Conflict	51
Crisis	52
Armed Conflict.....	53
CHAPTER 7.....	55
JOINT AND MULTINATIONAL OPERATIONS AND UNIQUE ENVIRONMENTS	55
Joint and Multinational Operations.....	55
Unique Environments.....	57
APPENDIX A	61
IMAGERY TYPES AND CHARACTERISTICS	61
APPENDIX B	71
POLICY CONSIDERATIONS	71
APPENDIX C	73
GEOINT COLLECTION ASSETS.....	73
APPENDIX D	77
GEOINT SUPPORT TO TARGETING.....	77

APPENDIX E	81
GEOINT INTEGRATED ARCHITECTURE	81
Glossary	85
References	91
Index	95

Figures

Figure 2-1. Army GEOINT and geospatial integration.....	9
Figure 3-1. GEOINT request for information process (example)	29
Figure 4-1. Grid reference box (example)	42
Figure A-1. Portions of the electromagnetic spectrum used for GEOINT collection	61
Figure C-1. Manned aircraft systems	75
Figure C-2. MQ-1C Gray Eagle unmanned aircraft system	76

Tables

Table 1-1. National System for Geospatial Intelligence members	3
Table 5-1. Tailored GEOINT products.....	46
Table 5-2. Types of imagery intelligence reports	48
Table 6-1. GEOINT activities during competition below armed conflict	51
Table 6-2. GEOINT activities during crisis	52
Table 6-3. GEOINT activities during armed conflict	53
Table A-1. Characteristic matrix of imagery types.....	69
Table D-1. Items in an electronic target folder (example)	78
Table D-2. Physical damage assessment confidence levels	80

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Preface

ATP 2-22.7 provides doctrinal guidance concerning geospatial intelligence (GEOINT); it complements guidance provided in ATP 3-34.80. ATP 2-22.7 focuses on the fundamentals of GEOINT as well as specific tasks and techniques for performing GEOINT activities.

The principal audience for ATP 2-22.7 is all members of the profession of arms. Commanders and staffs of Army headquarters serving as joint task force or multinational headquarters should also refer to applicable joint or multinational doctrine concerning the range of military operations and joint or multinational forces. Trainers and educators throughout the Army will also use this publication.

To understand the doctrine in ATP 2-22.7, readers must understand the following:

- Intelligence doctrine described in ADP 2-0 and FM 2-0.
- Collection management described in ATP 2-01.
- Operational doctrine described in ADP 3-0 and FM 3-0.
- Geospatial engineering doctrine described in ATP 3-34.80.

Commanders, staffs, and subordinates, with assistance from their servicing judge advocates, ensure their decisions and actions comply with applicable U.S. laws and policy, including but not limited to Executive Order 12333 as amended; relevant DOD instructions; DODD 2310.01E; DODD 3115.09; DODD 5240.01; DODM 5240.01; AR 381-10; FM 2-22.3; FM 6-27; the Uniform Code of Military Justice; military orders, including fragmentary orders; and international treaties. Commanders at all levels ensure their Soldiers operate in accordance with the law of armed conflict and the rules of engagement. (See FM 6-27.)

ATP 2-22.7 uses joint terms where applicable. Selected joint and Army terms and definitions appear in both the glossary and the text. For other definitions shown in the text, the term is italicized, and the number of the proponent publication follows the definition.

ATP 2-22.7 applies to the Active Army, Army National Guard/Army National Guard of the United States and U.S. Army Reserve unless otherwise stated.

The proponent of ATP 2-22.7 is the U.S. Army Intelligence Center of Excellence. The preparing agency is the Directorate of Training and Doctrine, U.S. Army Intelligence Center of Excellence. Send comments and recommendations on DA Form 2028 (*Recommended Changes to Publications and Blank Forms*) to Commander, U.S. Army Intelligence Center of Excellence, ATTN: ATZS-DOT-D (ATP 2-22.7), 550 Cibique Street, Fort Huachuca, AZ 85613-7017; by email to usarmy.huachuca.icoe.mbx.doctrine@army.mil; or submit an electronic DA Form 2028.

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Introduction

GEOINT, in its most basic form as an intelligence discipline, includes—

- Imagery.
- Imagery intelligence.
- Geospatial information.

This composition forms the foundation of Army GEOINT products and allows the Army to build on that foundation by fusing and layering various intelligence and command and control information with a geospatial component. ATP 2-22.7 comprises seven chapters and five appendixes:

- **Chapter 1** summarizes GEOINT, including imagery, imagery intelligence, and geospatial information. It also provides an overview of the National System for Geospatial Intelligence (NSG) and the National Geospatial-Intelligence Agency.
- **Chapter 2** discusses GEOINT roles and responsibilities—from those organizations and programs supporting the NSG to the Army Geospatial Enterprise, through which the Army integrates with other NSG members to improve the quality of intelligence support to operations.
- **Chapter 3** discusses GEOINT in the plan and direct step of intelligence process.
- **Chapter 4** discusses GEOINT in the collect and process step of the intelligence process.
- **Chapter 5** discusses GEOINT production and its dissemination and integration.
- **Chapter 6** describes GEOINT activities across the Army strategic contexts.
- **Chapter 7** provides considerations for joint multinational operations and unique environments.
- **Appendix A** details imagery types and characteristics.
- **Appendix B** provides policy considerations and domestic imagery requirements.
- **Appendix C** details GEOINT collection assets and systems.
- **Appendix D** discusses GEOINT support to targeting, including the decide, detect, deliver, and assess functions of the Army targeting process.
- **Appendix E** discusses the GEOINT integrated architecture, including Army GEOINT architecture, processing systems, and software and application tools.

This publication—

- Uses the term *threat*, which includes all enemies and adversaries that are part of the operational environment.
- Uses the word *theater* to indicate theater of operations.
- Uses the generic term *intelligence analysis system* due to the variety of fielded and partially fielded systems and applications.
- Introduces acronyms at their first use in the front matter of this publication (preface and introduction), and again in the body of the publication (chapters and appendixes).
- Introduces the *G-X*, *S-X*, and *J-X* (such as *G-2*, *S-3*, *J-2*) acronyms at their first use without defining them as it hinders readability. Definitions for these acronyms can be found in the glossary of this publication.
- Uses *G-2/S-2*, *G-3/S-3*, and *G-6/S-6* to denote division or corps/battalion or brigade intelligence, operations, and signal staff officers, respectively.
- Uses *U.S.* as a modifier (for example, *U.S. forces*) and *United States* as a noun (for example, the *United States*, a country in North America).

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PART ONE

FUNDAMENTALS

Chapter 1

Geospatial Intelligence (GEOINT) Overview

WHAT IS GEOINT?

1-1. *Geospatial intelligence* is the exploitation and analysis of imagery and geospatial information to describe, assess, and visually depict physical features and geographically referenced activities on or about the Earth. Geospatial intelligence consists of imagery, imagery intelligence, and geospatial information (JP 2-0).

1-2. GEOINT is one of the single-source intelligence disciplines nested within the intelligence process (plan and direct, collect and process, produce, and disseminate and integrate), which generates information, products, and knowledge about threats, terrain and weather, and civil considerations. This supports the conduct (plan, prepare, execute, and assess) of operations across the Army strategic contexts (competition below armed conflict, crisis, and armed conflict). GEOINT provides geospatial information and intelligence that supports the commander and staff and the development of all-source intelligence. (FM 2-0 provides doctrine on the intelligence disciplines and the intelligence process; FM 3-0 provides doctrine on the Army strategic contexts.)

1-3. GEOINT provides supported commanders with information that provides situational awareness and facilitates the military decision-making process (MDMP). GEOINT operations and information are also instrumental in corroborating other intelligence discipline reporting and cueing other discipline operations.

IMAGERY

1-4. *Imagery* is a likeness or presentation of any natural or man-made feature or related object or activity, and the positional data acquired at the same time the likeness or representation was acquired, including: products produced by space-based national intelligence reconnaissance systems; and likeness and presentations produced by satellites, airborne platforms, unmanned aerial vehicles, or other similar means (except that such term does not include handheld or clandestine photography taken by or on behalf of human intelligence collection organizations) (JP 2-0). Imagery is the visual basis of GEOINT, enabling the intelligence staff to see the operational environment (OE) without physically being there. (See appendix A for imagery types.)

IMAGERY INTELLIGENCE

1-5. *Imagery intelligence* is the technical, geographic, and intelligence information derived through the interpretation or analysis of imagery and collateral materials (JP 2-0). Imagery intelligence (IMINT) provides concrete, detailed, and precise image-based information on the location and activity of a threat and the environment's physical characteristics. It detects key natural and terrain features, installations, and infrastructure—information that is used to develop intelligence studies, reports, and target folders.

1-6. IMINT assists commanders in applying and protecting their combat power and allows commanders to see the area of operations (AO) in near real time as the operation progresses. IMINT supports decision making and facilitates situational understanding by reducing uncertainty about the threat and OE. It also facilitates the intelligence preparation of the operational environment (IPOE) process, is included in all-source analysis, and is used to develop analytical and IPOE products.

GEOSPATIAL INFORMATION

1-7. Joint doctrine defines *geospatial information and services* as the collection, information extraction, storage, dissemination, and exploitation of geodetic, geomagnetic, imagery (both commercial and national source), gravimetric, aeronautical, topographic, hydrographic, littoral, cultural, and toponymic data accurately referenced to a precise location on the surface of the Earth (JP 2-0).

1-8. *Geospatial information* is information that identifies the geographic location and characteristics of natural or constructed features and boundaries on or about the Earth, including: data and information derived from, among other things, remote sensing, mapping, and surveying technologies; and mapping, charting, geomatics data, and related products and services (JP 2-0).

THE NATIONAL SYSTEM FOR GEOSPATIAL INTELLIGENCE

1-9. The *National System for Geospatial Intelligence* is the combination of technology, policies, capabilities, doctrine, activities, people, data, and organizations necessary to produce geospatial intelligence in an integrated, multi-intelligence environment (JP 2-0). The mission of the National System for Geospatial Intelligence (NSG) is to create an integrated, collaborative community of common practice, promoting GEOINT across a diverse group of producers and consumers.

1-10. The functional manager for GEOINT, serving concurrently as the Defense Intelligence Enterprise Manager for GEOINT, manages the NSG. The NSG manages operations through guidance, policy, programs, and organizations. (See appendix B for policy considerations.) It fosters collaboration among U.S. Government departments and agencies, the Services, echelons, and multinational partners to provide decision makers, commanders, intelligence users and producers, and civil authorities a better understanding of GEOINT.

1-11. During information collection planning, intelligence staffs leverage support from each of these elements. For example, when intelligence staffs develop intelligence communications networks, the NSG can assist them in identifying those organizations or agencies that should be included in the command's intelligence architecture. The National Geospatial-Intelligence Agency (NGA), under the guidance of the NGA director and GEOINT Enterprise management through the National GEOINT Committee, can advise on integrating NSG members and partners (see table 1-1) on information collection strategies. This collaboration leverages GEOINT for the Army, resulting in more comprehensive and tailored intelligence products across the warfighting functions.

1-12. The NSG, in providing GEOINT to the Nation's decision makers and Soldiers, depends on its close collaboration with multinational partners. The NGA director, serving as the U.S. GEOINT representative to the Allied System for Geospatial Intelligence (ASG) Senior GEOINT Council, may advise NGA on the input from multinational partners, through the ASG, to provide integrated, unique, and assured GEOINT to the Nation's warfighters and policy makers to inform decisions and enable action. Additionally, as a member of the U.S. intelligence community (IC), NGA executes the tasking of imagery collection, coordination of imagery processing, and exploitation activities, ensuring the timely dissemination of imagery to authorized recipients.

1-13. Commanders generally employ organic GEOINT resources against time-sensitive and critical information needs. They rely on nonorganic resources to provide medium to long-term collection and analysis. Intelligence staffs assist commanders in understanding how to use nonorganic GEOINT resources that contain or collect data related to a theater.

1-14. The NSG comprises members from U.S. IC and DOD organizations under GEOINT functional management authorities, as outlined in Executive Order 12333 as amended, DODD 5105.60, and Intelligence Community Directive 113. (See table 1-1.) NSG members prioritize, plan, program, budget, acquire, collect, analyze, produce, share, store, and process GEOINT across national, military, and civil domains.

Table 1-1. National System for Geospatial Intelligence members

<i>U.S. intelligence community</i>	<i>Joint staff</i>
Intelligence agencies	Military Departments
- Central Intelligence Agency - Defense Intelligence Agency - National Geospatial-Intelligence Agency - National Reconnaissance Office - National Security Agency	- Department of the Army - Department of the Navy (including the U.S. Marine Corps) - Department of the Air Force (including the U.S. Space Force)
Intelligence components	Combatant commands
- Army Intelligence - Marine Corps Intelligence - Naval Intelligence - Air Force Intelligence - Space Force Intelligence - Coast Guard Intelligence - Federal Bureau of Investigation (Intelligence Branch) - Department of Energy (Office of Intelligence and Counterintelligence) - Department of Homeland Security (Office of Intelligence and Analysis) - Department of State (Bureau of Intelligence and Research) - Department of the Treasury (Office of Intelligence and Analysis) - Drug Enforcement Administration (Intelligence Program)	- U.S. Africa Command - U.S. Central Command - U.S. Cyber Command - U.S. European Command - U.S. Indo-Pacific Command - U.S. Northern Command - U.S. Southern Command - U.S. Space Command - U.S. Special Operations Command - U.S. Strategic Command - U.S. Transportation Command
U.S.	United States

NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY

1-15. NGA, established in Section 441, Title 10, United States Code, is a DOD combat support agency that collects, analyzes, and disseminates GEOINT data and products. NGA manages satellite collection requirements and develops distribution protocols for the NSG according to the National Intelligence Priorities Framework (also known as NIPF), which is managed by the Director of National Intelligence (DNI).

1-16. The NGA director serves as the functional manager for GEOINT in accordance with applicable laws and guidance from the DNI as well as DOD directives and agreements. The NGA director—

- Serves as principal adviser to the DNI for performance of the GEOINT function.
- Exercises functional management and oversight of the NSG, including technical oversight of NSG tactical elements, to ensure interoperability between existing and future NSG systems, connectivity between national and tactical systems, and modernization of tactical systems.
- Manages GEOINT activities.
- Sets standards for the GEOINT architecture and products. **Note.** A reachback component at NGA Headquarters focuses on NGA production support. Commanders and staffs can access NGA products directly from the NGA website on NIPRNET.
- Provides technical guidance for systems using GEOINT.

NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY SUPPORT TEAM

1-17. NGA directly supports joint force intelligence operations centers through its National Geospatial-Intelligence Agency support teams (NSTs) and maintains these teams for DOD agencies, several non-DOD agencies, and each of the Services.

1-18. An NST is the primary mechanism of interaction between NGA, federal departments, the Services, combatant commands (CCMDs), and other organizations. NSTs provide timely, accurate, and tailored GEOINT to CCMDs during peacetime, throughout exercises, and during war. They act as a conduit for ensuring the fusion of available information between NGA and CCMDs.

1-19. NSTs coordinate NGA's operational, policy, and training support to its customers. A typical NST includes a senior representative (a senior military intelligence [MI] officer or a defense intelligence senior leader), staff officers, and imagery and geospatial analysts.

NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY–ARMY

1-20. NGA-Army provides embedded GEOINT support to Army combat formations, enhancing operational readiness, warning intelligence, and targeting; strengthening partnerships; and accelerating GEOINT technology and integration. NGA-Army coordinates numerous activities between NGA and the Service, including mission-partner exercises, combat training center rotations, policy development, advanced field craft, and support to Army deployments through GEOINT support teams during contingency and combat operations. NGA-Army advocates for Army GEOINT positions, ensuring compliance with GEOINT functional manager responsibilities while integrating NSG priorities. NGA-Army has embedded analysts across combat formations at corps and division headquarters in and outside the continental United States.

Chapter 2

GEOINT Roles and Responsibilities

ARMY ORGANIZATIONS SUPPORTING THE NATIONAL SYSTEM FOR GEOSPATIAL INTELLIGENCE

2-1. The following are integrated with and/or support the NSG:

- United States Army Forces Command (FORSCOM).
- United States Army Intelligence and Security Command (INSCOM).
- United States Army Special Operations Command (also called USASOC).
- Army GEOINT Office.
- Army National to Theater Program.
- Army Geospatial Center (AGC).
- Army Tactical Exploitation of National Capabilities Program (TENCAP) Office.
- National Ground Intelligence Center (NGIC).
- Army Geospatial Intelligence Battalion (AGB).
- U.S. Army MI School.
- U.S. Army Engineer School.

U.S. ARMY FORCES COMMAND

2-2. FORSCOM trains and prepares the total force to build and sustain readiness to meet CCMD requirements. FORSCOM provides expeditionary, regionally engaged, campaign-capable land forces to combatant commanders. When tasked by the Army Deputy Chief of Staff for Operations, Plans, and Training (also known as DA G-3/5/7), FORSCOM provides GEOINT capabilities from subordinate units for intelligence support to combatant commanders.

U.S. ARMY INTELLIGENCE AND SECURITY COMMAND

2-3. INSCOM is the Army's operational intelligence organization. INSCOM—

- Executes command and control (C2) of operational intelligence and security forces.
- Conducts and synchronizes worldwide multidisciplined and all-source intelligence and security operations.
- Delivers linguist support and intelligence-related advanced skills training, acquisition support, logistics, communications, and other specialized capabilities to support Army, joint, and coalition commands and the U.S. IC.

2-4. Army intelligence operates interdependently—through the integration across intelligence disciplines and between echelons of command—and occurs across a range of activities. INSCOM assists MI units and staffs at all echelons in leveraging Army intelligence assets, capabilities, and services by—

- Managing the Army's contribution to the U.S. IC.
- Bridging intelligence-related gaps or boundaries and leveraging placement, access, and interior lines of communications (LOCs).
- Scaling advanced or complex intelligence operations or activities and managing specialized, low-density intelligence warfighting function capabilities.
- Maintaining Army-wide intelligence readiness for an expeditionary force (for example, developing foundational intelligence; managing intelligence databases; developing architectures; linking training and certification to Army, joint, and/or U.S. IC standards; supporting intelligence readiness for Reserve Component MI forces).
- Driving intelligence and security processes and operations to support the—
 - Protection of Army modernization (for example, requirements determination, acquisition, future force design).
 - Army's ability to protect and secure its forces, information, technologies, and other resources.

U.S. ARMY SPECIAL OPERATIONS COMMAND

2-5. U.S. Army Special Operations Command special forces groups and its 75th Ranger Regiment have both GEOINT imagery analysts and geospatial engineers. Civil affairs brigades have geospatial engineers, whereas the 160th Special Operations Aviation Regiment only has GEOINT imagery analysts. An NGA GEOINT imagery analyst also supports each of these organizations; NGA experts assist in interpreting and exploiting GEOINT products and services.

2-6. U.S. Army Special Operations Command units may collect full-motion video (FMV) using available or theater GEOINT collection assets tasked to support the unit. The unit's task-organized GEOINT cell uses this FMV along with existing intelligence to conduct geospatial analysis and develop GEOINT products.

Note. *Available collection assets* describe organic, assigned, and attached assets as well as those assets under operational control or in direct support of that unit. The unit lists organic, assigned, attached, operational control, and supporting units and the overall task organization in Annex A (Task Organization) to the order.

ARMY GEOINT OFFICE

2-7. The Army GEOINT Office is the onsite representative at NGA Headquarters and the formal Army point of contact for NGA. The office integrates with the Army Deputy Chief of Staff for Intelligence (DA G-2), FORSCOM, INSCOM, the U.S. Army Intelligence Center of Excellence, and all relevant Army GEOINT stakeholders to produce fully coordinated Army-NSG positions, ensure active participation in NSG governance forums, and perform integrated mission management. It also supports the planning, programming, budgeting, acquisition, technology investment, policy, doctrine, training, tradecraft, personnel, tasking, collection, production, and processing, exploitation, and dissemination (PED) functions related to military GEOINT functions and activities.

ARMY NATIONAL TO THEATER PROGRAM

2-8. The Army National to Theater Program serves as the NSG's primary producer of GEOINT supporting theater information requirements for CCMDs. The program contains the Army's portion of GEOINT support to the U.S. IC as designed by the NGA director. The Army National to Theater Program, in conjunction with NSG member organizations, executes unified GEOINT operations across the U.S. IC and CCMDs.

2-9. Functional management authority for the Army National to Theater Program resides with the DA G-2 or its appointed agent. The operations manager, located at Headquarters, INSCOM G-3, implements instructions from the program functional manager and assists with program execution reviews, as required.

2-10. Five integrated GEOINT divisions (also known as IGDs) execute the Army National to Theater Program. These divisions reside within the six INSCOM military intelligence brigades-theater (MIB-Ts)—66th, 207th, 470th, 500th, 501st, and 513th—and geographically align with Army Service component commands (ASCCs). They respond to intelligence requirements by converting national and theater GEOINT collection into intelligence products using various GEOINT phenomenologies.

National Ground Intelligence Center Support

2-11. NGIC Applied Remote Sensing (formerly known as the Ground Spectral Intelligence Center) is a specialized subordinate organization of the Army National to Theater Program. It is the Army's center for image science, including spectral and overhead persistent infrared (OPIR) expertise; it conducts OPIR research and development. The AGB OPIR PED capability also resides within NGIC.

U.S. Army Space and Missile Defense Command

2-12. The U.S. Army Space and Missile Defense Command (also known as USASMDC), located at Petersen Space Force Base, is also an Army National to Theater Program specialized subordinate organization. It provides a range of national satellite systems and commercial GEOINT services and products to designated civil agencies, other DOD forces, U.S. Space Command, U.S. Northern Command, U.S. Strategic Command, and their subordinate joint forces worldwide.

2-13. The U.S. Army Space and Missile Defense Command GEOINT Division is a GEOINT unit specializing in advanced production using nonliteral analysis of imagery data obtained from visible and invisible portions of the electromagnetic spectrum (EMS). (See appendix A.) The GEOINT Division—

- Creates detailed geospatial information and services (GI&S) and mapping data.
- Conducts analysis of airborne and satellite imagery, including synthetic aperture radar (SAR), panchromatic, multispectral, hyperspectral, thermal, FMV, OPIR, and light detection and ranging (LIDAR) imagery.
- Fields requests for support and access to its products via the Geospatial Intelligence Enterprise Tasking, Processing, Exploitation, and Dissemination Service (GETS). (See paragraph 5-23.)

ARMY GEOSPATIAL CENTER

2-14. AGC coordinates, integrates, and synchronizes geospatial data and standards across the Army. It develops and fields geospatial enterprise-enabled systems and capabilities to the Army and DOD and provides direct geospatial support and products to Soldiers. The AGC director is the Army Geospatial Information Officer (an Army staff position), who develops and implements the Army Geospatial Enterprise.

ARMY TACTICAL EXPLOITATION OF NATIONAL CAPABILITIES PROGRAM OFFICE

2-15. The Army TENCAP Office is an Army acquisition organization subordinate to the Program Executive Office for Intelligence, Electronic Warfare, and Sensors (also known as PEO IEW&S). It executes the TENCAP for the Army. The office integrates TENCAP systems and ensures interaction with NGA, the National Reconnaissance Office (NRO), and other national agencies, as well as DOD, non-DOD, and U.S. commercial science, technology, and laboratory entities and institutions.

NATIONAL GROUND INTELLIGENCE CENTER

2-16. NGIC is an INSCOM functional command with operational control exercised by the DA G-2. NGIC provides all-source intelligence and GEOINT on foreign ground force capabilities and related military technologies, and integrates with mission partners to ensure DOD, Army, joint, and national-level decision makers maintain decision advantage to protect the United States and interests abroad. It provides scientific and technical intelligence (also known as S&TI) and general MI to support senior defense and Army leaders, warfighting commanders, and force and materiel developers. NGIC also has an assigned NST.

2-17. NGIC focuses on maintaining detailed knowledge of current foreign ground force capabilities from operational through small-unit levels. NGIC supports the visualization that is critical to MI by providing imagery. NGIC develops and produces a range of sophisticated GEOINT products using experienced imagery analysts and scientists specializing in physics, chemistry, and mechanical engineering.

2-18. NGIC analyzes information and data to assess foreign force equipment, weapons, capability development, countermeasures, and operational tactics. The analysis considers near-term, medium-term, and long-term foreign force technology development and adaptive tactics, techniques, and procedures (TTP). Intelligence activities include irregular and conventional warfare analysis and examining foreign ground forces based on the warfighting functions, doctrine, TTP, training, sustainment, and threat characteristics.

ARMY GEOINT BATTALION

2-19. AGB is directly subordinate to NGIC. It is integrated within NGA and a tenant unit at NGA Headquarters. AGB Headquarters and a third of the battalion are at Fort Belvoir, VA; the rest of the battalion is at facilities across the United States. AGB—

- Provides GEOINT sustainment training and products to support military operations and national-level requirements.
- Manages the Army Departmental Requirements Office (ADRO), which advocates for Army-specific GEOINT collection.
- Serves as the functional lead for Army GEOINT Foundry training.
- Advocates for Army requirements for broad-spectrum imagery products and support while contributing to NGIC's scientific and technical intelligence mission and general MI.
- Provides GEOINT tactical overwatch of deployed forces.
- Provides crisis support for DOD agencies and the Army.

ADRO

ADRO ensures the Army's operational readiness by managing and coordinating requirements related to imagery and geospatial information. ADRO helps Army echelons—

- Create and adjudicate national and commercial imagery, data, and product requests for collection (RFCs).
- Disseminate imagery, data, and products.
- Implement policies and processes concerning national and commercial imagery, data, and products.

ADRO coordinates with the AGC Imagery Office's collection manager to submit imagery requirements through the NGA Geospatial Intelligence Information Management Services (GIMS). The AGC Imagery Office serves as the Army's commercial imagery acquisition agent. Army personnel located at disparate organizations without a local or regional G-2/S-2 collection management section must contact ADRO through the ADRO portal on Intelink. (See AR 525-95.)

U.S. ARMY MILITARY INTELLIGENCE SCHOOL

2-20. The U.S. Army MI School at the U.S. Army Intelligence Center of Excellence trains, develops, and educates the Nation's Army intelligence Soldiers and Civilian leaders. It designs, develops, and integrates intelligence capabilities, concepts, and doctrine to support operations in joint, interagency, intergovernmental, and multinational environments.

U.S. ARMY ENGINEER SCHOOL

2-21. The U.S. Army Engineer School at the Army Maneuver Support Center of Excellence synchronizes and integrates the doctrine, organization, training, materiel, leadership and education, personnel, and facilities (also called DOTMLPF) domains to ensure the Engineer Regiment can provide engineer support. The school trains Soldiers and DA Civilians at all levels to generate, manage, analyze, and disseminate GI&S that allows commanders and staffs to visualize and understand the physical aspects of an OE and their effects on operations within the OE.

ARMY GEOSPATIAL ENTERPRISE

2-22. The Army Geospatial Enterprise, detailed in ATP 3-34.80—

- Leverages—
 - An integrated system of technologies and processes that provides a standard and shareable geospatial foundation (also known as SSGF) upon which data from all warfighting functions can be used to display the common operational picture (COP).
 - A comprehensive framework for exploiting and sharing GI&S (to include associated spatial and temporal data) to support military operations across the Army strategic contexts.
- Governance Board addresses geospatial issues at the Army level. The DA G-2 and the Army Chief of Engineers cochair the board.

2-23. The DA G-2 is the Army's central point of contact for GEOINT and coordinates all GEOINT issues with NGA, the other Services, the joint staff, and the U.S. IC. The DA G-2 also—

- Serves as the Service GEOINT element in accordance with DODD 5105.60 and AR 525-95.
- Establishes and maintains the Army GEOINT Office to serve as the Service GEOINT element representative at NGA Headquarters. The Service GEOINT element provides representation, coordination, liaison, and interface on Army GEOINT issues and actions.
- Collaborates with the Army Chief of Engineers on coordinating, assessing, and synchronizing Army policies and standardized requirements for geospatial information.
- Represents the Army (in coordination with the Army Chief of Engineers) at NSG senior management councils and GEOINT committees.
- Prepares and coordinates Army positions on joint actions and documents affecting GEOINT.
- Serves as the Army voting representative to the Geospatial-Intelligence Standards Working Group.

- Designates an Army GI&S officer to coordinate with each CCMD and ASCC to submit geographic GI&S statements of requirements (excluding requirements managed by CCMDs to support operational needs) to NGA for training exercises, materiel system development, and deployment readiness.

ARMY GEOINT INTEGRATION

2-24. The Army integrates with other NSG members to improve the quality of intelligence support to operations. Army GEOINT data and reporting are tailored for ground force commanders and complement NGA reporting and joint operations. Several Army units and organizations task, collect, process, exploit, and disseminate geospatial data and reporting to support Army GEOINT. The Army has two primary GEOINT providers:

- MI units and organizations** provide access to imagery and GEOINT collection assets.
- Geospatial engineer units and organizations** provide geospatial data and information assets.

2-25. Several Army and U.S. IC organizations provide functions and resources, ensuring the Services can conduct their warfighting mission in training and deployed environments. Therefore, some GEOINT activities occur within the intelligence warfighting function while others occur outside of intelligence channels. Figure 2-1 illustrates Army GEOINT operational relationships.

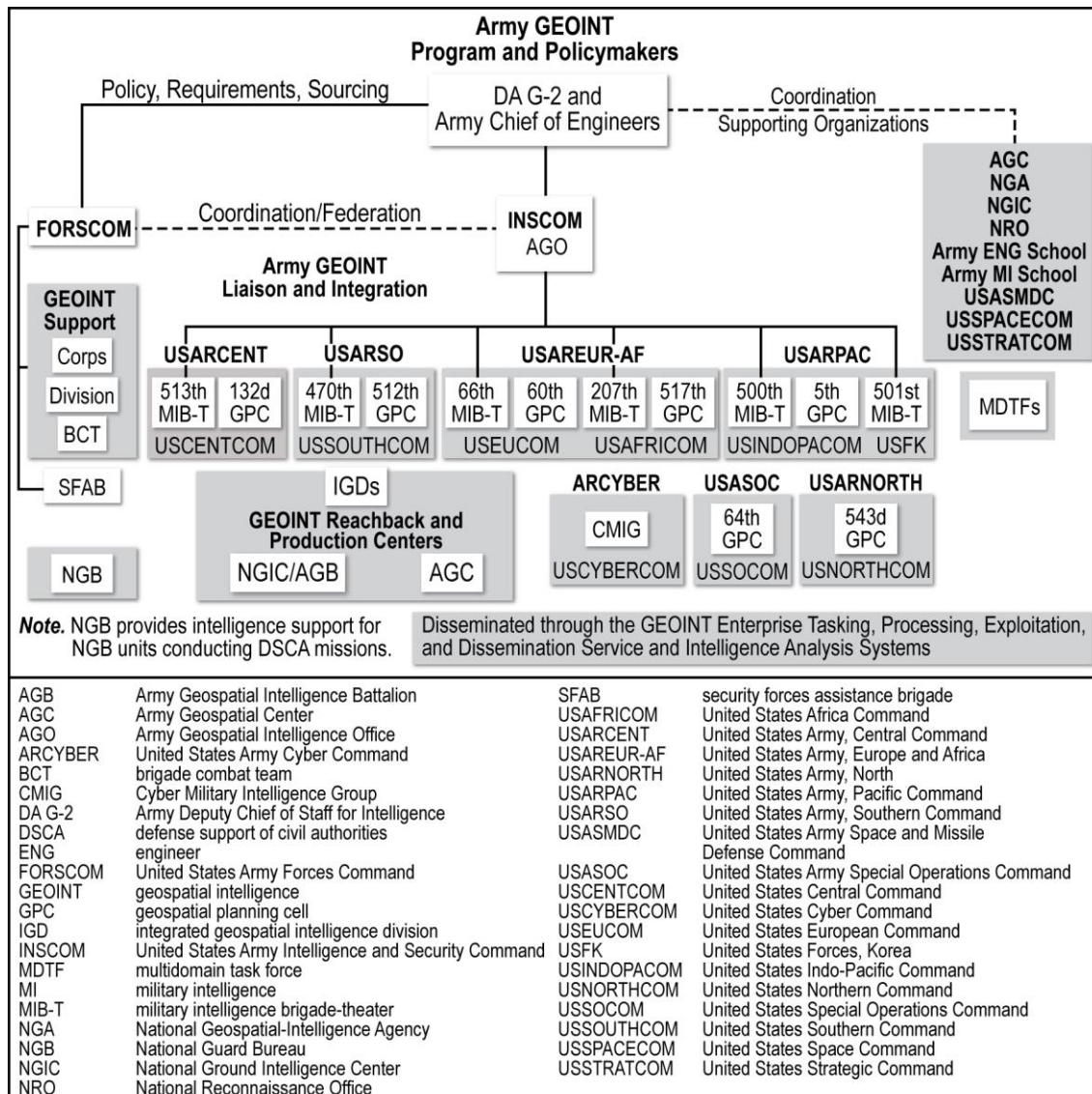


Figure 2-1. Army GEOINT and geospatial integration

2-26. Army GEOINT cells and geospatial engineer teams provide imagery and IMINT and GI&S capabilities to ASCCs, corps, and divisions. GEOINT cells are organic to corps, divisions, and brigade combat teams (BCTs) and provide commanders and staffs with GEOINT and geospatially derived information to satisfy the commander's priority intelligence requirements (PIRs). In addition to assigned personnel, GEOINT cells may include other attached personnel, as required.

THEATER ARMY GEOINT ELEMENTS

2-27. Army GEOINT organizations include theater elements within regionally focused ASCCs as well as units within tactical organizations at echelons corps and below. At the theater level, GEOINT components reside within ASCC staffs and INSCOM MIB-Ts. These elements receive GI&S support from geospatial engineers in geospatial planning cells (GPCs) assigned to each ASCC. (See ATP 3-34.80.)

Army Service Component Command

2-28. At the ASCC level, GEOINT supports—

- Corps by producing GEOINT data to satisfy GEOINT requirements.
- ASCC planning and decision-making processes and operational requirements.
- Warfighting functions through the development of GEOINT products.

2-29. The ASCC G-2 and engineer staff officer provide support for missions outside the reasonable capacity of a MIB-T and GPC. Support activities include but are not limited to—

- CCMD or geographic CCMD requirement support.
- ASCC, forward, or contingency command post operations.
- Geospatial joint or multinational partner operations.
- Subordinate unit or joint task force (JTF) operations or exercises.

Military Intelligence Brigade-Theater

2-30. Imagery exploitation is conducted at INSCOM MIB-Ts, which are aligned with geographic CCMDs to give each an operational-level ground intelligence capability. These brigades are incorporated into the intelligence structure of each ASCC and CCMD. They support CCMDs by providing access to intelligence, intelligence infrastructure, and training. Each MIB-T manages ground intelligence collection, production, and knowledge management for ASCC intelligence and operations staffs. MIB-Ts receive support from GPCs that conduct geospatial analysis and generate geospatial data to fill data gaps in theaters. MIB-Ts—

- Fuse the production of all-source intelligence, fully integrating regionally aligned force elements into daily production requirements focused on CCMD priorities.
- Align analytic capabilities and enable reachback to national resources for specific informational needs.
- Ensure regionally aligned force commanders' intelligence requirements are nested in ASCC PIRs.

Military Intelligence Brigade-Theater and the Focal Point Concept

2-31. The MIB-T serves as the focal point for its respective CCMD; it is the contact organization that regionally assigned, aligned, and designated forces use to access CCMD opportunities. Designated forces may include global response force units, other Service forces, and multinational partners.

2-32. The focal point concept provides a unifying framework for intelligence organizations across each CCMD area of responsibility. The concept assists units in focusing their organic intelligence capabilities and enhancing their situational awareness and mission readiness. The focal point concept uses MIB-T resident and reachback capabilities to—

- Access CCMD GEOINT expertise, support, and products.
- Integrate into the CCMD's intelligence architecture.
- Provide training opportunities for regionally aligned forces' intelligence organizations.
- Leverage expertise resident in CCMD organizations and INSCOM functional commands.
- Provide theater reception, staging, onward movement, and integration (also called RSOI). (See FM 2-0.)
- Provide target development and maintenance to support the CCMD J-2 and inform the ASCC G-2.

Increasing Strength and Agility

2-33. This concept facilitates the national military shift toward strengthening INSCOM's support to regional treaty partners and allies. It also reinforces global response force units' ability to respond to global crisis. The concept assists these forces in maintaining the adaptability and agility needed to support each CCMD's specific mission requirements.

Assisting Regionally Aligned Forces

2-34. Regionally aligned FORSCOM units will be able to focus mission and training requirements on specific CCMD areas of responsibility. Most regionally aligned forces are based in their homelands and require an intelligence point of contact at their gaining CCMD to gain situational awareness and tailor training to meet the current requirements of their projected theater. Under the focal point concept, MIB-Ts assist in meeting these requirements and increasing regionally aligned force readiness.

2-35. MIB-Ts reduce the likelihood of forces deploying unprepared by providing units, both deployed and in garrison, with training and access to current CCMD intelligence production. MIB-Ts coordinate with respective ASCCs to identify regionally aligned and global response force unit requirements and facilitate their training and intelligence capabilities.

2-36. As the focal point, the MIB-T integrates regionally aligned forces into the ASCC collection and requirements management process. This integration enables the ASCC intelligence structure to consolidate requests for information (RFIs) and leverage CCMD and U.S. IC resources for answers. The MIB-T also incorporates regionally aligned forces' intelligence elements into the appropriate battle rhythm events.

CORPS AND BELOW GEOINT ELEMENTS

2-37. Corps and below GEOINT elements consist of GEOINT cells and expeditionary-military intelligence brigades (E-MIBs). GEOINT cells are in the G-2 analysis and control element (ACE) and include both GEOINT imagery analysts and geospatial engineers. This combination facilitates collaboration, enabling the cell to develop GEOINT products to support planning and operations while reducing redundancies. GEOINT cells are assigned to corps and divisions and may augment BCTs. E-MIBs provide multidiscipline intelligence support to corps and division elements.

GEOINT Cell

2-38. GEOINT cells focus on converging, correlating, and analyzing imagery, IMINT, and geospatial information to create products or display timely, accurate, and relevant intelligence to provide battlefield commanders a combat advantage. GEOINT cells combine current OE information, combat information, and intelligence to generate products and assessments that support all warfighting function areas (C2, movement and maneuver, intelligence, fires, sustainment, and protection) within corps, divisions, combat aviation brigades, and BCTs, and with mission partners. This GEOINT collaboration solidifies and enhances analysis and production relationships.

Functions

2-39. GEOINT cell functions include but are not limited to the following:

- Exploit motion imagery.
- Conduct moving target indicator (MTI) analysis and produce supporting intelligence.
- Provide and update the foundation layer for IPOE.
- Tag and prepare GEOINT data for discovery.
- Use predictive analysis applicable to the mission.
- Manage and provide the geospatial foundation of the COP.
- Visualize the AO.
- Identify gaps in existing GEOINT databases and develop nominations for the information collection plan that answer intelligence requirements and other information requirements.
- Research external databases to determine if existing products answer intelligence requirements and other information requirements.
- Develop RFIs that answer intelligence requirements and other information requirements.

- Collect and process data and imagery to extract GEOINT and geospatial information to answer intelligence requirements and other information requirements.
- Ensure every C2 system has the same map background.
- Analyze GEOINT and geospatial data and create products to directly support military operations.
- Disseminate GEOINT data, products, and reports to national agencies and commanders.
- Coordinate GEOINT requirements within the area of responsibility and the appropriate requirements management personnel.
- Provide and update overlays.
- Conduct analysis of terrain characteristics to determine phenomenon that will affect friendly and enemy military operations.

Capabilities

2-40. GEOINT cells enable understanding spatial and temporal relationships across the OE. They can query, filter, and retrieve a variety of combat information, such as spot reports, significant activities, reconnaissance reports, and patrol debriefs. GEOINT imagery analysts use IMINT and imagery as well as data from geospatial engineers to develop and disseminate GEOINT products to supported consumers. GEOINT imagery analysts can identify and relay environmental hazards, civil considerations, and threat activity to units maneuvering through the battlefield.

2-41. GEOINT cells possess the equipment and expertise to process, exploit, disseminate, and display data from multiple sources in local or cloud computing environments. These resources allow the cell to layer and merge all relevant data capable of being georeferenced to a specific location, target, or event.

2-42. Collocating GEOINT imagery analysts and geospatial engineers facilitates GEOINT production to satisfy the commander and staff's GEOINT requirements. GEOINT cells enable—

- Centralized GEOINT that focuses on fusing intelligence data into GEOINT data, layering production for C2 and production coordination.
- Synchronized production, reducing potential product or mission duplication.
- Improved, collaborative intelligence and geospatial analysis, enabling advanced, combined, multidiscipline, three- and four-dimensional, and production capabilities.

2-43. Corps and divisions are consumers of GEOINT products produced by NGA and other NSG and ASG members. They develop GEOINT products that assist theater army, corps, division, and BCT intelligence cells in performing tasks associated with IPOE, warning intelligence, situation development, information collection, and targeting:

- **Corps capabilities.** The corps does not have an organic imagery collection capability. The corps receives imagery collection support from NSG members, MIB-Ts, and E-MIBs. Organized under the G-2 ACE, the corps GEOINT cell develops GEOINT products and collection tasking to support corps operations. The corps GEOINT cell—
 - Conducts geospatial and imagery analysis, developing specialized geospatial products and imagery-based products to support corps planning and operations.
 - Develops intelligence based on motion imagery video from E-MIB and echelons-above-division GEOINT collection assets.
 - Leverages national- and theater-level direct downlinks to satisfy GEOINT production requirements.
 - Supports the warfighting functions by exploiting imagery, analyzing MTIs, and providing GI&S support.
 - Supports the division by producing GEOINT data to satisfy GEOINT requirements.
 - Receives field-generated or collected geospatial data updates from subordinate division and separates brigade units for inclusion in corps data holdings.
 - Maintains awareness of the OE and provides updates to GPCs for inclusion in the theater geospatial database, thereby ensuring consistency of geospatial data from ASCC to division levels.

- **Division capabilities.** The division receives imagery collection support from MIB-Ts, E-MIBs, and combat aviation brigades. Organized under the division G-2 ACE, the GEOINT cell develops GEOINT products to support division operations. The division GEOINT cell, organized under the G-2 staff—
 - Conducts geospatial and imagery analysis.
 - Develops intelligence based on motion imagery from E-MIB and echelons above division GEOINT collection assets.
 - Develops specialized geospatial and imagery-based products to support planning and operations.
 - Conducts imagery exploitation, MTI, and GI&S missions to support the warfighting functions.
 - Produces GEOINT data and products to satisfy GEOINT requirements and develops operation orders (OPORDs) to support the BCT.
 - Maintains the COP and distributes data to the corps for consolidation and forwarding to GPCs for inclusion in its national geospatial data holdings, ensuring consistency of geospatial data from corps to BCTs.

2-44. BCT GEOINT capabilities reside in the—

- BCT headquarters intelligence cell's S-2X element. A GEOINT imagery analyst and geospatial engineers provide timely, accurate, and relevant GEOINT and geospatial information to support operations.
- BCT MI company's intelligence processing team (Tactical Intelligence Ground Station [TGS]). GEOINT imagery analysts provide a responsive intelligence processing capability using TGS and stored secondary imagery.

GEOINT Cells and the Intelligence Staff

2-45. At the tactical level, corps and division GEOINT cells directly support their respective G-2/S-2. They respond to RFIs, integrate GEOINT collection assets into the unit's information collection plan, and provide GEOINT products to support IPOE and all-source analysis products.

2-46. **Coordinating with the intelligence staff.** GEOINT cells coordinate with intelligence staff to identify and properly task members comprising the cell. Forming an effective GEOINT cell and synchronizing its efforts require teamwork between the intelligence and engineer staffs. The G-2/S-2 is responsible for the GEOINT cell and collaborates with the engineer staff officer to capture geospatial requirements. The G-2/S-2 and the engineer staff officer must coordinate with their respective technicians and the G-3/S-3 to ensure clearly established priorities for the geospatial effort throughout the operations process.

2-47. **Staff responsibilities.** The G-2/S-2 directs all intelligence activities in the headquarters, including GEOINT. The G-2/S-2 is also the principal staff officer responsible for collecting and analyzing knowledge on threats, geospatial analysis, weather, and civil considerations for the commander. The G-2/S-2—

- Leads the staff in the IPOE process and assists the G-3/S-3 in developing and executing the information collection plan.
- Provides intelligence support to lethal and nonlethal targeting operations, ensuring targeting priorities are developed based on intelligence threat assessments and included in the information collection plan. (See FM 3-60 for doctrine on targeting.)
- Is the principal adviser to the commander regarding the intelligence warfighting function.

2-48. G-2/S-2 key responsibilities related to GEOINT include the following:

- Provides guidance on creating GEOINT products to support IPOE, information collection, and targeting.
- Prioritizes requests for GEOINT support from the staff and subordinate commands.
- Ensures GEOINT support is performed according to the established battle rhythm.
- Develops GEOINT collection tasking to support operational requirements.
- Coordinates GEOINT support to electromagnetic warfare.

2-49. The engineer staff officer is responsible for integrating geospatial engineering and—

- Coordinates with the G-2/S-2, G-3/S-3, other staff elements, and the geospatial engineering technician for geospatial products that describe the physical environment.
- Coordinates with the G-2/S-2 for the production and distribution of maps and geospatial products based on established priorities.
- Recommends adjustments to geospatial engineering technician priorities in concert with the G-2.
- Establishes procedures for effectively curating and distributing updated standard and shareable geospatial foundation data. (See ATP 3-34.80.)

2-50. If a corps or division functions as part of a JTF, the G-2 is designated as the J-2—who is responsible for providing the JTF and subordinate components and subordinate task forces with the complete land, air, maritime, and space threat situation by integrating threat assessments developed by the combatant commander's intelligence organization. The G-2's primary function when acting as a JTF J-2 is supporting the multinational JTF and staff by ensuring the availability of reliable intelligence and timely warning intelligence on the threat situation in the joint operations area. Additionally, the JTF J-2 supports the multinational JTF protection mission. (See JP 2-0 for doctrine on joint GEOINT operations.)

Expeditionary-Military Intelligence Brigade

2-51. An E-MIB is the primary collection asset assigned or attached to the corps, designed to conduct intelligence operations to support operations. It comprises two intelligence and electronic warfare (IEW) battalions and a headquarters company. E-MIBs perform PED of collected information, conduct analysis to produce intelligence, and integrate operations. IEW battalions collaborate with GEOINT cells in the G-2 ACE to accomplish the GEOINT mission—providing geospatial analysis and management and disseminating GEOINT. (See ATP 2-19.3 for doctrine on E-MIBs.)

PART TWO

GEOINT TECHNIQUES

Chapter 3

Planning and Directing

SECTION I – PLANNING

3-1. GEOINT supports operational planning during the plan and direct step of the intelligence process by providing the commander and staff with a visual, near real-time representation of the OE. GEOINT provides insight into how the environment will affect both threat and friendly courses of action (COAs), assisting the commander in selecting a COA for approval. This occurs during each step of the MDMP and the IPOE process—which supports the mission analysis step of the MDMP and lays the foundation for the commander and staff to develop, analyze, and approve COAs.

GEOINT IN THE MILITARY DECISION-MAKING PROCESS

3-2. The MDMP is the Army's primary decision-making model; it gives the commander a better understanding of the situation and mission. (FM 5-0 provides doctrine on the MDMP.) GEOINT supports this by giving the commander and staff a visual representation of the OE at each step of the MDMP:

- **Step 1:** Receipt of mission.
- **Step 2:** Mission analysis.
- **Step 3:** COA development.
- **Step 4:** COA analysis.
- **Step 5:** COA comparison.
- **Step 6:** COA approval.
- **Step 7:** Orders production, dissemination, and transition.

STEP 1: RECEIPT OF MISSION

3-3. Upon receiving the mission, the GEOINT cell prepares its personnel, data, and systems requirements assessment to support the commander and staff. The cell integrates and coordinates with the staff to develop requirements for operational variables.

Personnel

3-4. GEOINT leaders identify specific GEOINT support required for the mission. They determine how many GEOINT imagery analysts are needed and whether they require any specialized training or experience. Identifying C2 and support relationships for GEOINT imagery analysts supporting the mission is key. GEOINT leaders must also consider logistical, supply, equipment, and administrative issues required for GEOINT imagery analysts to perform the mission.

Note. Army GEOINT training occurs at all levels to provide Soldiers with initial familiarity of the discipline and additional, more advanced education as their careers progress. Access Army training requirements information on the Army Training Requirements and Resources System (also known as ATRRS) via NIPRNET.

Data

3-5. GEOINT imagery analysts organize existing data and use several methods to acquire information needed for planning. The GEOINT cell—

- Identifies information gaps and verifies existing data and products.
- Conducts research and collects products from libraries, holdings, and external support agencies.
- Submits requests for foundation vector and data extraction.
- Provides and manages foundational geospatial data to support the COP within C2 systems.
- Develops image and terrain data collection requirements and submits them to the collection manager.
- Continuously updates and tracks RFI status.

Supporting the COP

A *common operational picture* is a single, identical display of relevant information shared by more than one command that facilitates collaborative planning and assists all echelons to achieve situational awareness (JP 3-0). Geospatial information and imagery provide the foundation for the COP, which facilitates situational understanding for commanders and the warfighting functions. The COP is critical in integrating Army operations because it provides a common view of operations, established command standards, and aspects of the OE.

Controlled image base (CIB) is a visual planning tool that supports the COP, mission planning, and C2 systems. CIB is a standard NGA dataset of panchromatic orthorectified (see paragraph 4-74) digital imagery from national satellite systems or other adequate commercial imagery. CIB also supports weapon systems, regional overviews, and mapping where maps are nonexistent or outdated.

CIB imagery is useful for image map backgrounds and displaying features not represented on digital map backgrounds. CIB can be used for urban areas, but it may not provide the detailed resolution needed for urban analysis. In this situation, a higher resolution from commercial or national sources is better suited. National or commercial imagery is also more appropriate when up-to-date imagery is vital to the intended use.

CIB conforms to both the raster product format and National Imagery Transmission Format standard, and it uses World Geodetic System 1984 (WGS 84) datum (see paragraph 4-71). It is distributed via classified NGA networks and the Army supply system from the Defense Logistics Agency, which can be accessed via NIPRNET.

Systems

3-6. The GEOINT cell ensures its systems are functional and integrated. The GEOINT noncommissioned officer in charge and officer in charge identify necessary equipment and networks and identify available GEOINT collection assets in theater. This applies to GEOINT collection assets along with their capabilities. (See appendix C for more on GEOINT collection assets.) They also coordinate with the intelligence officers and the G-6/S-6 to ensure connectivity to all necessary networks and raise technical concerns with the systems maintainer or integrator and the G-6/S-6.

STEP 2: MISSION ANALYSIS

3-7. Mission analysis allows commanders to visualize and describe how the operation may unfold in their initial commander's intent and planning guidance. GEOINT supports mission analysis by leveraging new and existing information to meet the commander's intelligence requirements. The GEOINT cell creates products to assist commanders in visualizing the terrain and explaining its military significance; the cell presents the products in a briefing during the mission analysis, enabling commanders and staffs to effectively evaluate operations and threat COAs. GEOINT actively supports the two primary outputs of the intelligence staff during mission analysis: preparing the initial information collection plan and performing IPOE.

STEP 3: COURSE OF ACTION DEVELOPMENT

3-8. The staff develops options for subsequent analysis and comparison to satisfy the commander's intent and planning guidance. The staff references updated IPOE products and running estimates to develop friendly COAs that meet evaluation criteria developed during mission analysis. GEOINT supports the commander and staff during COA development by continuously updating IPOE products, responding promptly to new RFIs, and disseminating new GEOINT products to the intelligence staff. This provides the intelligence staff with a continuous flow of GEOINT to reference as the intelligence staff develops, analyzes, and compares COAs. For example, GEOINT can assist in determining the strengths and disposition of enemy maneuver elements, which are an aspect of relative combat power and a key consideration during friendly COA development.

3-9. During COA development, the GEOINT cell—

- Refines vector and terrain databases.
- Develops sequel and branch plan products.
- Refines GEOINT products based on the requesting unit's feedback.
- Collects, tasks, and analyzes imagery for named areas of interest (NAIs) and other areas of interest.
- Coordinates with outside agencies, other units, and fusion elements to refine products.
- Conducts line of sight (LOS) analysis.
- Updates and tracks the status of outstanding RFIs.
- Maintains foundation datasets.

STEP 4: COURSE OF ACTION ANALYSIS

3-10. COA analysis begins once the commander selects one or more acceptable COAs from the COA development step. As part of COA analysis, the staff prepares to war-game the COAs.

Role-Playing the Enemy

3-11. During the war game, the G-2/S-2 role-plays the enemy commander, other threat organizations, and critical civil considerations in the AO. The G-2/S-2 captures the results of each enemy, threat group, and civil consideration action and counteraction as well as the corresponding friendly and enemy strengths and vulnerabilities. By trying to win the war game for the enemy, the G-2/S-2 ensures the staff addresses friendly responses for each enemy COA. This enables the G-2/S-2 to refine friendly intelligence requirements, the event template and corresponding decision points, target areas of interest, and high-value targets (HVTs).

Reacting to Enemy Courses of Action

3-12. As new potential enemy reactions to friendly COAs are considered during the war game, the GEOINT cell assists in filling correlated intelligence gaps with current imagery and geospatial data. The cell supports the war-gaming effort by maintaining current products that depict enemy locations and movements, refining operations graphics, coordinating with the fusion cell to refine and revise products to the intelligence staff's needs, and responding to new RFIs as they arise. During war gaming, the GEOINT cell—

- Creates data from enemy locations and movements.
- Updates friendly and enemy force unit positions.
- Coordinates with outside agencies, other units, and the fusion cell to refine products.
- Refines and revises products to requesting unit's requirements.
- Submits for imagery collection, if needed.
- Updates and tracks RFI status.

STEP 5: COURSE OF ACTION COMPARISON

3-13. COA comparison refers to the objective process of evaluating COAs independently and against set evaluation criteria approved by the commander and staff. Once the staff identifies its preferred COA, it delivers a decision briefing to the commander concerning possible COAs and war-game results and provides current IPOE products.

3-14. The GEOINT cell supports COA comparison by refining and updating GEOINT portions of IPOE products, refining identified target locations, and coordinating with other staff sections as intelligence requirements change. For example, as the intelligence staff determines the most likely and most dangerous enemy COAs, existing requirements may change or new RFIs may be submitted to aid decision making.

STEP 6: COURSE OF ACTION APPROVAL

3-15. The commander selects the best COA after the decision briefing, then issues the final planning guidance. This guidance includes a refined commander's intent (if necessary) and new PIRs to support execution. The GEOINT cell supports COA approval by—

- Coordinating with the staff to ensure products meet the requesting unit and commander's intent.
- Adjusting products to the commander's final guidance.
- Continuing to develop NAIs.
- Adjusting products to current imagery to provide current analysis.
- Producing and disseminating hardcopy products.
- Updating and tracking RFI status.

STEP 7: ORDERS PRODUCTION

3-16. The staff prepares the order or plan by converting the selected COA into a clear, concise concept of operations and the required supporting information. The COA statement becomes the concept of operations for the plan. The COA sketch becomes the basis for the operation overlay. If time permits, the staff may conduct a more detailed war game of the selected COA to synchronize the operation and complete the plan.

3-17. The GEOINT cell integrates and coordinates with the staff to write and refine Appendix 5 (GEOINT) to Annex B (Intelligence). The cell supports the final step of the MDMP by—

- Producing and disseminating information and hardcopy and digital products to the community and requesting unit.
- Communicating with the community and requesting unit for any discrepancies with the data, information, and products.
- Performing intelligence briefs.
- Updating and tracking the RFI status.

GEOINT IN THE INTELLIGENCE PREPARATION OF THE OPERATIONAL ENVIRONMENT PROCESS

3-18. IPOE, detailed in ATP 2-01.3, supports the MDMP by providing an understanding of the OE during mission analysis. Conducting pre-mission analysis of the OE is the foundation for performing IPOE and mission analysis. (See FM 2-0.) The following are the four steps of the IPOE process:

- **Step 1:** Define the OE.
- **Step 2:** Describe environmental effects on operations.
- **Step 3:** Evaluate the threat.
- **Step 4:** Determine threat COAs.

3-19. The GEOINT cell supports this effort by exploiting and analyzing imagery from many sensors to detect and characterize observable elements associated with the intelligence staff's RFIs. Integrating GEOINT into the IPOE process requires coordination and synchronization between the intelligence staff, geospatial engineers, and higher, lower, and adjacent units. (See ATP 3-34.80 for doctrine on geospatial engineering operations during IPOE.)

3-20. The following results from initial IPOE:

- Imagery derived products.
- Terrain products and weather products (including the modified combined obstacle overlay).
- Likely threat COAs.
- HVT lists.
- An explanation of how key civil considerations affect the operation.

STEP 1: DEFINE THE OPERATIONAL ENVIRONMENT

3-21. During IPOE step 1, the GEOINT cell assists in visualizing geographic boundaries and organizational handover lines. Current and historic GEOINT data allow analysts to depict the commander's AO, as prescribed by higher headquarters, as well as the area of interest and area of influence. They exploit GEOINT to identify significant characteristics of the enemy, terrain, weather, and civil considerations within the AO. This analysis supports the primary output of IPOE step 1, the significant characteristics brief.

3-22. For example, historic map data depicts three significant waterways that may prevent movement within the AO and eight total bridges identified as crossings. Analysis of up-to-date imagery determines the usability of the bridges and assesses whether there are additional wet gap crossings in the area. This analysis informs the significant characteristics brief by establishing the importance of each body of water in the AO. Similarly, initial estimates of enemy strength in the AO consist of a brigade element supplemented with additional fires and reconnaissance capabilities. GEOINT analysis will confirm or deny these numbers and provide the initial estimates of enemy numbers and equipment type throughout the AO.

3-23. GEOINT analysis of significant characteristics of the AO identifies initial intelligence gaps regarding the enemy, terrain, weather, and civil considerations. During IPOE step 2, these gaps drive the development of intelligence requirements and subsequent NAIs to support the development of the initial information collection plan.

STEP 2: DESCRIBE ENVIRONMENTAL EFFECTS ON OPERATIONS

3-24. During IPOE step 2, GEOINT imagery analysts visualize enemy terrain, weather, and civil considerations and describe their effects on threat and friendly capabilities. They do so by exploiting current and historical GEOINT data to depict characteristics such as enemy equipment, prepared battle positions, key terrain, chokepoints, road networks, infrastructure, maritime (bathymetry), and mountainous terrain. The resulting products support the terrain effects matrix, threat overlay, identifying weather impacts, and describing characteristics of the areas, structures, capabilities, organizations, people, and events (also called ASCOPE) within the AO.

Obtain Detailed Terrain Information and Intelligence

3-25. Analysts obtain detailed terrain information and GEOINT and categorize it by environment type and observable metrics that may impact the offense or defense. These observable characteristics can distinguish urban from rural or jungle from desert environments, aiding analysts in describing the environment during mission analysis.

3-26. The GEOINT cell considers the five military aspects of terrain—observation and fields of fire, avenues of approach, key terrain, obstacles, and cover and concealment (OAKOC)—when evaluating battlefield landscape features. Assessing OAKOC within a prospective OE provides commanders an understanding of the limitations and opportunities of the terrain in which they must operate.

Observation and Fields of Fire

3-27. *Observation* is the condition of weather and terrain that permits a force to see the friendly, enemy, and neutral personnel and systems, and key aspects of the environment (FM 1-02.1). Examples include but are not limited to fortified locations on higher elevation with a cleared field of fire or lookout towers. Some variables to consider when obtaining data related to observation are topography, vegetation, urban development, and man-made obscurants such as smoke or defensive perimeters.

3-28. A *field of fire* is the area that a weapon or group of weapons may cover effectively from a given position (FM 3-90). Observation directly relates to a unit's field of fire. One example is open land with a clear view within available weaponry's range. Observation determines the field of fire for direct fire weapons and lines of fire determine the field of fire for indirect capabilities. The GEOINT cell identifies observation and fields of fire by—

- Supporting dynamic LOS analysis by obtaining detailed elevation data through research and intelligence reach databases.
- Obtaining available terrain categorization data.

- Researching and obtaining baseline infrastructure geospatial information.
- Obtaining current enemy situation and threat templates for the OE by coordinating with internal unit intelligence and staff elements.

3-29. GEOINT uses digital terrain elevation data (DTED) to support observation and fields of fire. DTED has varying levels of detail:

- **DTED level 0:** Roughly 1,000-meter post spacing, bare earth.
- **DTED level 1:** Roughly 100-meter post spacing, bare earth.
- **DTED level 2:** 30-meter post spacing, bare earth.
- **DTED level 3:** 10-meter post spacing, bare earth.
- **DTED level 4:** 3-meter post spacing, bare earth.
- **DTED level 5:** 1-meter post spacing, bare earth.
- **Shuttle radar topography mission 1:** 90-meter post spacing, reflective surface, or treetop data.
- **Shuttle radar topography mission 2:** 30-meter post spacing, reflective surface, or treetop data.
- High-resolution elevation data.

3-30. Shuttle radar topography mission data refers to elevation data obtained on a near-global scale to generate the most complete high-resolution digital topography of Earth. Shuttle radar topography mission is an international project spearheaded by NGA and the National Aeronautics and Space Administration.

3-31. DTED is intended for lower resolution viewing of large areas. While high-resolution data is available for specific small areas, DTED is not appropriate for tactical planning requiring higher resolution viewing. DTED can be downloaded from NGA websites. Most government and commercial software applications that read raster data also read DTED.

Avenues of Approach

3-32. An *avenue of approach* is a path used by an attacking force leading to its objective or to key terrain. Avenues of approach exist in all domains (ADP 3-90). The breadth and difficulty of an attacking unit's size limits its avenue of approach (AA). Variables that can affect an AA include surface conditions, topography, and drainage characteristics. AAs include but are not limited to LOCs, such as roads, rail lines, rivers, and areas where movement of troops, weapons, and equipment could occur. AAs also include mobility corridors (areas that are relatively free of obstacles where a force will be canalized due to terrain restrictions allowing military forces to capitalize on the principles of mass and speed). To identify AAs, the GEOINT cell researches and obtains GEOINT data on current LOCs through the OE, catalogs mobility corridors within the OE, and categorizes AAs supporting either threat or friendly capabilities.

Key Terrain

3-33. *Key terrain* is an identifiable characteristic whose seizure or retention affords a marked advantage to either combatant (ADP 3-90). Two factors can render an area key terrain: how a commander wants to use the area and whether the enemy can use it to defeat friendly forces. Key terrain typically offers control of a local objective or an important transportation route. The GEOINT cell determines key terrain by identifying—

- High ground with good observation and fields of fire.
- Transportation chokepoints such as a water crossing, mountain gap, or road junction.
- Dense woods or rivers that anchor a battle line's flank.

3-34. *Decisive terrain* is key terrain whose seizure and retention is mandatory for successful mission accomplishment (ADP 3-90). Also referred to as critical terrain, decisive terrain is relatively rare and not present in every battle. The GEOINT cell determines decisive terrain by identifying—

- Urban areas.
- LOCs and supply routes.
- Topography.
- Bridges and chokepoints.
- Key military installations.

Obstacles

3-35. An *obstacle* is any barrier designed or employed to disrupt, fix, turn, or block the movement and maneuver, and to impose additional losses in personnel, time, and equipment (JP 3-15). The presence and relative difficulty of obstacles determine whether terrain is unrestricted, restricted, or severely restricted. Analysts categorize the hindrance level of obstacles as *go*, *slow go*, or *no go*. The GEOINT cell identifies obstacles and determines their effects on the OE by analyzing existing obstacle layers. They identify additional obstacles in the OE by analyzing new GEOINT data.

3-36. Obstacles include but are not limited to the following:

- Vegetation (swamps, dense forests, open fields, and farmland).
- Topography.
- Fences.
- Bridges.
- Reinforcing and fortification features (parapets, stone walls, and vehicle barriers).
- Ditches.
- Battle events.
- Urban areas.
- Water (streams, rivers, lakes, and irrigation canals).
- Drainage characteristics (natural and man-made).
- Wet and dry surface materials (ravines and bluffs).

Note. Friendly or threat forces place reinforcing obstacles within the OE to slow, stop, or control opposition movement. Examples include but are not limited to barriers and revetments, blown bridges, and cratered runways.

Cover and Concealment

3-37. *Cover* is protection from the effects of fires (FM 3-96). Examples include ditches, riverbanks, buildings and structures, walls, and entrenchments. *Concealment* is protection from observation or surveillance (FM 3-96). Examples include forests, ravines, dense vegetation, and reverse slopes. To determine cover and concealment, the GEOINT cell identifies terrain that offers protection from the effects of fire and from observation or surveillance.

Obtain Detailed Weather and Weather Effects Information and Intelligence

3-38. When researching, obtaining, and developing weather and weather effects information and intelligence, the GEOINT cell should consider how terrain features affect weather elements such as visibility, temperature, humidity, precipitation, winds, and clouds. Conversely, weather conditions such as temperature, winds, and precipitation affect terrain in a definitive manner. These conditions can enhance or limit operations by impacting trafficability, water crossing (fording), and visibility.

Note. The most common example of terrain affecting weather is the higher rainfall rate on the windward side of high terrain (such as mountains) versus the leeward (opposite) side.

Forecast the Effect of Climatology on Operations

3-39. The GEOINT cell and the United States Air Force (USAF) staff weather officer (SWO) assess a general location's climatology and forecast its impact on future operations. The GEOINT cell should leverage the USAF SWO and intelligence reach organizations to acquire weather support products, information, and knowledge. If the USAF SWO is not available, analysts should use publicly available information (PAI) and resources. The GEOINT cell obtains detailed weather effects information and intelligence by analyzing meteorological, oceanographic, and space environmental information as well as seasonal or cyclic weather events or trends, such as monsoon seasons and tropical storms.

Identify Extreme Weather Effects

3-40. Extreme weather events, such as tropical storms or monsoons, can cause delays and disruptions in plans for activities, including survivability, mobility, LOCs, points of embarkation and debarkation, and sustainment support. Identifying potential effects early is critical; it gives the command and staff time to develop mitigation strategies and evaluate COAs. Some of this data is available for developed regions of the globe, but analysts must develop other data from secondary sources. Examples of extreme weather events include but are not limited to heavy precipitation and flooding; heavy snowfalls, blizzards, and ice storms; dust storms and heat waves; and droughts and tropical storms.

Obtain Detailed Civil Considerations Information and Intelligence

3-41. The GEOINT cell obtains information and intelligence on civil considerations by researching accessible databases and leveraging external agencies, echelons, and partners to define the civil considerations within the OE. The cell catalogs different aspects of civil areas, structures, capabilities, organizations, people, and events for further refinement and development during IPOE. The GEOINT cell obtains information and intelligence about the OE by executing these tasks:

- Develop geospatial information and intelligence of civil areas.
- Obtain GEOINT/imagery for structures.
- Obtain updated GEOINT/imagery for the capabilities of public and commercial services.
- Obtain updated GEOINT/imagery of organizations.
- Research GEOINT/imagery of the local population.
- Obtain GEOINT/imagery of historical, cultural, and religious events.

Civil Areas

3-42. Key civil areas are locations or aspects of the terrain within the OE that are not often highly secured or protected. These areas can be heavily influenced by military operations or have significant effects on them. Key civil areas include but are not limited to the following:

- Areas defined by political boundaries, such as districts within a city or municipalities within a region.
- Locations of government centers.
- Social, political, religious, or criminal enclaves.
- Agricultural and mining locations.

Structures

3-43. Some structures play a significant role in answering sustainment RFIs. Others provide insight about life support or threat intent, leading to potential targeting. Identifying critical infrastructure before combat operations could shape COA development within the area. Examples of these structures include but are not limited to bridges, communications towers, harbors, power plants, dams, and piers/docks.

3-44. Structures can also refer to cultural sites protected by international law or civil and government agreements. These structures are often prohibited from being targeted by fires and effects. Examples of protected structures include but are not limited to churches, mosques, temples, hospitals, clinics, and cemeteries.

3-45. Structures that serve practical applications useful for protection or support purposes include larger facilities, such as warehouses, jails, television broadcast facilities, radio stations, airports, and train stations.

3-46. It is important to obtain information and intelligence on locations of essential services, infrastructure, and key logistics and sustainability sites, such as water distribution and pumping stations, sewage treatment plants and sewer systems, and refineries and storage facilities for oil, propane, or other fuels.

Capabilities of Public and Commercial Services

3-47. Capabilities refer to those services required to save, sustain, or enhance life, as well as the ability of local host-nation authorities, aggressors, state and nonstate actors, or other participants in the area to provide a populace with key functions and services, such as emergency services, food distribution, technology, public administration, and public safety. Capabilities include public works and utilities, public health, commerce, and other areas in which the populace may need help after combat operations.

3-48. Capabilities also refer to resources and services that can be contracted to support operations or reconstruction efforts, such as construction sites containing materials and equipment, agricultural equipment locations, and financial institutions.

Organizations

3-49. The GEOINT cell should research the locations and actions of noncombatant elements and nongovernmental and other civilian organizations in the OE that may impact operations. They should also research the role of clans, tribes, religious organizations, criminal elements, militias, and other groups active in the OE. Analysts can identify organizations using multiple indicators, such as flags and banners, equipment, vehicles, and base or camp configuration. Organizations that may impact operations include but are not limited to paramilitary organizations, host-nation forces, civic organizations, government organizations, religious groups, insurgent groups, and intergovernmental organizations.

Local Population

3-50. The local population includes all civilians within the OE whose actions can collectively shape or affect military operations. For the GEOINT cell to develop an understanding of this local population, analysts gather detailed information and intelligence regarding the population's ethnic breakdown, languages, religion, and historical, cultural, and religious events.

Historical, Cultural, and Religious Events

3-51. Events are routine, cyclical, planned, or spontaneous activities that significantly affect organizations, people, and operations. The GEOINT cell can identify and confirm these events by reviewing available imagery. GEOINT imagery analysts should consider the following when conducting research:

- National and religious holidays.
- Elections.
- Celebrations.
- Civil disturbances.
- Agricultural, crop, livestock, and market cycles.
- Other events, such as disasters caused by natural, man-made, or technological sources. These create civil hardship and the events that enable future analysis of their implications.

Complete Studies

3-52. GEOINT imagery analysts do not have a specific operation, mission, or area of responsibility when conducting pre-mission analysis of the OE, but they can use completed studies in potential contingency and combat operations. Completed studies or products obtained during pre-mission analysis of the OE include country briefs, written assessments, or graphics. These products inform the commander and staff on current and historical situations that may affect future operations when a mission is received. The GEOINT cell completes studies to—

- Develop observable trends in threat activities that visually depict consistent threat trends and new or emerging threats and trends.
- Develop GEOINT assessment, reporting, and products to support a threat assessment rollup to confirm or deny estimated threat unit strengths and intelligence reporting of threat systems.
- Catalog and develop GEOINT reports of cyclic or recurring historical, cultural, and religious events occurring within the OE.
- Describe the impacts of civil considerations on operations to—
 - Visually depict the ethnic demographics of a region, province, or district.
 - Provide imagery derived products of paramilitary organizations operating in the OE.
 - Identify and analyze critical infrastructure.
- Describe the weather and weather effects within the OE by—
 - Analyzing visibility and light data.
 - Determining wind effects.
 - Developing precipitation and flood plain charts.
 - Providing historic templates for temperature.

STEP 3: EVALUATE THE THREAT

3-53. Step 3 of the IPOE process determines threat force capabilities and the doctrinal principles and TTP threat forces prefer to employ. GEOINT supports this effort by analyzing current imagery and geospatial data to characterize threat composition, disposition, and operational status.

Identify Threat Characteristics

3-54. While IPOE steps 1 and 2 focus on all aspects of an operation, including the enemy, terrain, weather, and civil considerations, IPOE step 3 focuses specifically on identifying threat characteristics. To do so, the intelligence staff analyzes all threats in the OE and develops a threat model for each one. GEOINT supports this process by monitoring the ongoing situation within the AO and using collected data to identify the threat's composition, disposition, and current strength.

Determine Threat Disposition

3-55. GEOINT uses cross-cueing of multiple sensors to determine threat disposition within the OE. For example, the cell leverages MTI to analyze vehicle movement and to discover abnormalities that may be used to identify threat activity. Based on this information, FMV GEOINT collection assets can collect near real-time coverage of the movement.

Identify Threat Forces

3-56. The GEOINT cell identifies threat forces at the tactical level. The cell can use imagery to monitor threat installations and the status of its associated equipment. This enables the command and staff to evaluate the threat's overall strength and disposition and select HVTs.

Identify Areas of Vulnerability

3-57. The GEOINT cell uses current imagery and geospatial data to identify likely and potential enemy attack zones, support zones, AAs, and mobility corridors. GEOINT imagery analysts conduct LOS analysis using current imagery, LIDAR, and terrain data. This analysis assists in determining areas of vulnerability to friendly forces and gives the commander and staff additional information to determine feasibility of potential threat COAs.

Counter Threat Activity

3-58. The GEOINT cell leverages multiple sensors to counter explosive hazards threats such as mine manufacturing and emplacement. GEOINT imagery analysts use multispectral imagery to analyze NAIs likely associated with homemade explosive manufacturing signatures. Multispectral imagery determines hot spots associated with disturbed earth, indicating caching activity, mine emplacement, and command wire emplacement. It can also detect certain chemical compounds against the soil background that match detailed signatures developed during IPOE. GEOINT imagery analysts use thermal imaging to detect cooling and heating signatures. Thermal imaging can detect signatures of disturbed earth or seek buried objects that cool or heat at a different rate from the surrounding soil. The cell links suspected cache areas to support areas using imagery, change detection, and MTI. GEOINT takes multiple products relating to points of interest and links them to refine areas requiring further attention from collection or operations.

Develop the Foundation to Define Threat Characteristics

3-59. The GEOINT cell, in coordination with other intelligence disciplines and analysts, gathers available data, information, and intelligence about threats affecting operations by—

- Conducting research and intelligence reach.
- Leveraging databases (internal and external).
- Data mining U.S. IC web services and portals.
- Reviewing academic materials from publicly available resources.
- Studying finished intelligence products.

3-60. This task develops detailed information specific to various aspects of threat characteristics. Developing an understanding of where opposition forces are historically postured, what environmental factors exist within the AO, and which civil considerations may impact military operations is critical to generating a baseline of shared knowledge. The information, intelligence, products, and materials obtained are refined for future use in mission analysis, IPOE, and other planning tasks. This refinement occurs through functional analysis and other analytic techniques. To develop the foundation for defining threat characteristics, the GEOINT cell obtains analysis of recent observable threat actions and trends and researches the composition and disposition of threats based on available imagery.

Analyze Recent Observable Threat Actions and Trends

3-61. The GEOINT cell obtains recent and historical threat reporting by researching online intelligence web services, submitting RFIs through the G-2 ACE, and leveraging outside organizations such as NGA and NGIC. Observable actions and trends include but are not limited to roadblocks, blown bridges, trenching, defensive perimeters, and minefield emplacement.

Research the Composition and Disposition of Threat Forces

3-62. The GEOINT cell conducts research through available threat characteristic databases and references existing threat models and templates to analyze threat composition and disposition. Using GEOINT (including but not limited to panchromatic, SAR, spectral, polarimetric imagery, LIDAR, FMV, MTI, or derivatives of these datasets), the GEOINT cell develops a threat characteristic baseline to prepare for follow-on mission analysis and COA development. Examples of include but are not limited to—

- Military threat characteristics at templated garrison locations.
- Threat characteristics (referred to as order of battle at the tactical level) in tactical assembly areas.
- Transport activity in train stations or airports.
- Observable activity at staging locations.

STEP 4: DETERMINE THREAT COURSES OF ACTION

3-63. During IPOE step 4, the intelligence staff identifies and develops possible threat COAs that can affect the friendly mission. GEOINT supports this effort by defining elements of the OE that affect threat COAs and aid in determining their likelihood and danger.

3-64. The analysis of environmental effects on operations and their relation to the threat assists the staff in developing threat COAs. The staff combines this analysis with an understanding of the threat's operational art and tactics to predict the threat's most likely immediate and subsequent objectives. GEOINT can detect trends over time, increasing the intelligence staff's understanding of enemy TTP and helping to determine the most likely threat COA.

3-65. Because objectives are usually terrain- or force-oriented, GEOINT provides the staff valuable support in visualizing how a threat may attempt to achieve its goal. Using satellite imagery and FMV, the GEOINT cell monitors threat disposition relative to key terrain and friendly forces, allowing the staff to assess all possible COAs as the situation unfolds and to help identify branches and sequels to the threat COA.

INTELLIGENCE REACH AND RESEARCH

3-66. Through intelligence reach, GEOINT imagery analysts explore satellite imagery product databases and libraries maintained by different agencies and higher, adjacent, and subordinate echelon units. This capability can assist them in satisfying requirements without further collection. In conducting research, GEOINT imagery analysts acquire data from multiple sources, such as classified and unclassified national information libraries and commercial imagery footprint servers. They also search archives for imagery and evaluate the accuracy, currency, and suitability of existing information. If analysts determine new coverage is needed to satisfy the requirements, they forward the request to the responsible collection management team.

National Data Library (NDL)

Numerous archives and libraries, both public and internal to NGA, store national and commercial satellite system imagery products. NGA uses the NDL (formerly known as the NSG Consolidated Library), which provides—

- Greater flexibility to adapt to changing mission needs and new data types (such as unstructured) with no degradation to mission timelines.
- Assistance in locating and retrieving library data (imagery, image-based products, geospatial information, and video data) to satisfy NSG and U.S. IC data-access needs.

Access the NDL through the ADRO portal on Intelink.

DEVELOPING THE INFORMATION COLLECTION PLAN

3-67. The G-2/S-2 and G-3/S-3, in coordination with the rest of the staff, develop an integrated information collection plan that uses many capabilities, including imagery and geospatial collection, to satisfy intelligence requirements, support the targeting process (see appendix D), and inform maneuver. Once developed, the plan becomes Annex L (Information Collection) of the supported unit's OPOD. GEOINT provides input to and validates the GEOINT portions of the initial information collection plan. (See ATP 2-01 and FM 3-55 for doctrine on the information collection plan.)

PROVIDING GEOINT INPUT

3-68. The GEOINT cell provides input to develop the initial information collection plan by advising the intelligence staff on available GEOINT collection assets at the current echelon and echelons above and below. The cell advises the staff on capabilities and limitations of available GEOINT collection assets to support the commander's intelligence requirements. This process should address high-payoff targets (HPTs) as determined by the staff. The cell manages and amends inputs to the information collection plan as requirements change.

VALIDATING GEOINT PORTIONS

3-69. The GEOINT cell validates GEOINT portions of the initial information collection plan. Validation is usually in a matrix or spreadsheet format, referred to as the information collection synchronization matrix. The collection manager creates the matrix considering all available GEOINT collection assets. The cell uses the matrix to synchronize information collection activities with the current threat assessment and friendly force COA, and to show tasked and requested GEOINT collection assets. (See ATP 2-01.)

3-70. The GEOINT cell validates the information collection synchronization matrix by reviewing GEOINT collection requirements and advising the intelligence staff. The cell ensures collection requirements coincide with specific information requirements that support intelligence requirements. The GEOINT cell confirms the compatibility of proposed GEOINT collection assets and timelines with GEOINT requirements. The cell also advises the intelligence staff of potential conflicts that may arise from GEOINT collection asset limitations, terrain, and weather.

3-71. The collection management team uses the information collection synchronization matrix to develop the information collection overlay, which the operations staff uses to graphically depict the information collection effort. Annex L (Information Collection Plan) of the OPOD normally includes the information collection overlay as an appendix. (See ATP 2-01.)

3-72. The GEOINT cell verifies that the GEOINT portions of the information collection plan address all required collection. The cell validates the information collection overlay, ensuring the location and status of all GEOINT collection assets, and verifies control measures are in place to support friendly COAs and the scheme of maneuver. The collection manager, with input from the GEOINT cell, revises the information collection plan as requirements are answered, new requirements are added, and mission requirements change. The cell provides feedback to the collection management team to improve collection strategy as needed and validates revisions to the information collection plan with the same fidelity as the initial one.

ESTABLISHING A STORAGE AND DISTRIBUTION CAPABILITY

3-73. GEOINT operations require extensive bandwidth and include FMV, full frame imagery, chip imagery, imagery derived products, spectral imagery, geospatial information and data, and geospatial products. GEOINT operational communications must be optimized within the existing communications architecture. The specific organization of the OE, command relationships, staffs, and units drive the interconnecting communications architecture. The C2 system must complementarily support this organization and provide for the uninterrupted flow of information to and from commanders at all levels.

BANDWIDTH SUPPORT

3-74. The GEOINT cell leader collaborates with the staff to obtain the required bandwidth support for GEOINT operations. The GEOINT cell leader—

- Leverages the unit's communications systems to provide redundant, diverse, and resilient means to transport data.
- Conducts an initial evaluation of the communications architecture.
- Uses a bandwidth tool to determine if the current bandwidth supports GEOINT requirements within the current structure.
- Evaluates the current communications architecture's ability to provide the necessary interoperability for mission success, focusing on commonality, compatibility, and standardization of communications.
- Evaluates the current communications architecture to ensure it provides the flexibility required to meet dynamic and changing situations and diverse operations with minimal disruption.
- Ensures the communications architecture can instantaneously respond to demands for information during operations.

SYSTEMS IMPROVEMENTS

3-75. To be responsive, communications systems must be reliable, redundant, and timely. After evaluating the communications architecture, the GEOINT cell leader identifies improvements to better facilitate GEOINT. The GEOINT cell leader requests implementation of these changes via unit staff elements and coordinates points of contact for any specialized units to ensure the communications architecture—

- Continues to meet C2 system objectives and principles.
- Produces unity of effort.
- Exploits the capabilities of the total forces.
- Positions critical information properly.
- Creates information fusion opportunities.
- Provides the interoperability necessary to ensure success.

ARCHITECTURE EVALUATION

3-76. The GEOINT cell must continually evaluate the communications architecture to ensure it is as mobile as supported forces, elements, or organizations without degrading information quality or flow. The GEOINT cell leader ensures the communications architecture is secure, disciplined, and survivable. Procedural and technical measures and capabilities detect and isolate system intrusions to protect information and major components of the C2 system. Incident reaction measures include the following:

- Using offensive actions to eliminate threats.
- Closing system component vulnerabilities.
- Using alternate frequencies.
- Changing communications security keying materials.
- Building and implementing a viable primary, alternate, contingency, and emergency (also called PACE) plan.

SECTION II – DIRECTING

3-77. GEOINT supports the direction of intelligence efforts by requesting collection support to fill information gaps and by managing GEOINT RFIs to answer the commander's intelligence requirements. These processes require the GEOINT cell leader to make recommendations concerning tasking of assets and to leverage support from higher, lower, and adjacent units.

COLLECTION MANAGEMENT

3-78. During the direct step of the intelligence process, GEOINT leaders focus on collection management and managing the RFI process. Collection management—

- Analyzes requirements.
- Evaluates available GEOINT collection assets (internal and external).
- Recommends GEOINT collection asset tasking to the operations staff.
- Submits RFIs for adjacent and higher collection support.
- Assesses the information plan's effectiveness.

3-79. The continuous functions of collection management identify the best way to satisfy the requirements of the supported commander and staff. These functions are not necessarily sequential. Collection management requires an inherent understanding of the priority of incoming RFCs and PED and the technical capabilities of GEOINT collection assets. Additionally, collection management includes the staff vetting requirements against current intelligence holding to ensure resources are not wasted by collecting information that is already available. (See ATP 2-01.)

3-80. GEOINT requirements, methods of collection and associated systems, and products vary widely based on the supported echelon and type of operation. The GEOINT cell directly communicates with mission requesting units to define user requirements. It then interfaces with the NSG to obtain the best quality GEOINT and provide it to the consumer. This fulfills the broad range of requirements associated with various mission functions. In defining GEOINT requirements, the cell collects and prioritizes GEOINT mission requirements.

3-81. Managing collection requirements for GEOINT involves satisfying the commander's needs through evolving GEOINT capabilities. Requirements managers compare requirements to available data at all levels to determine collection requirements. Requirements then direct GEOINT collection assets to collect GEOINT data based on specific information derived from RFIs submitted to the GEOINT cell. This information may include but is not limited to time, location, collection frequency, image quality, priority, GEOINT collection asset type, and sensor type. (See chapter 4.)

3-82. If data available to the GEOINT cell does not satisfy a requirement, the senior GEOINT imagery analyst coordinates with the collection management team for the collection of classified and publicly available geospatial information and imagery through reconnaissance, topographic survey, site survey, data mining, and satellite and airborne imagery. The senior GEOINT imagery analyst performs numerous tasks:

- Determines the collection requirement.
- Supports the collection manager with imagery collection requirements.
- Submits imagery nominations for collection.
- Submits requests for digital geospatial information directly after determining mission requirements.
- Reviews mission requirements and applicable IPOE products, including the decision support template with the synchronization matrix, the modified combined obstacle overlay, and the event template and matrix.
- Analyzes requirements to determine how GEOINT can best satisfy each one by—
 - Contributing to the development of the initial information collection plan by determining available GEOINT collection assets and addressing HPTs as determined by the staff.
 - Identifying unique support requirements, including maintenance, training, and connectivity.
 - Identifying reporting criteria and GEOINT collection asset capabilities and limitations.
 - Obtaining weather effects on all GEOINT collection assets being considered.
 - Ensuring production is completed by the latest time information is of value for each task and identifies, prioritizes, and validates requirements.

- Contributes to the preparation of the information collection overlay by—
 - Depicting the location and status of all GEOINT collection assets.
 - Depicting collection control measures that support friendly COAs and the scheme of maneuver.
 - Depicting primary and alternate locations for each GEOINT collection asset, including pertinent mission information such as scan sectors, start and stop times, and routes.
 - Managing and amending the information collection plan as requirements change.
- Supports the S-2, who collaborates with the S-3.

GEOINT REQUEST FOR INFORMATION PROCESS

3-83. A *request for information* is any specific, time-sensitive, ad hoc requirement for intelligence information or products to support an ongoing crisis or operation not necessarily related to standing requirements or scheduled intelligence production (JP 2-0). An RFI is the normal procedure for obtaining intelligence information that is not readily available to satisfy an existing information requirement.

3-84. An RFI may be assigned to the GEOINT cell for additional collection, processing, exploitation, analysis, and production. The cell's senior GEOINT imagery analyst is responsible for receiving requests from adjacent and subordinate units. The cell creates specialized and focused visualization products that allow consumers to meet operational requirements. Figure 3-1 illustrates a GEOINT RFI process example.

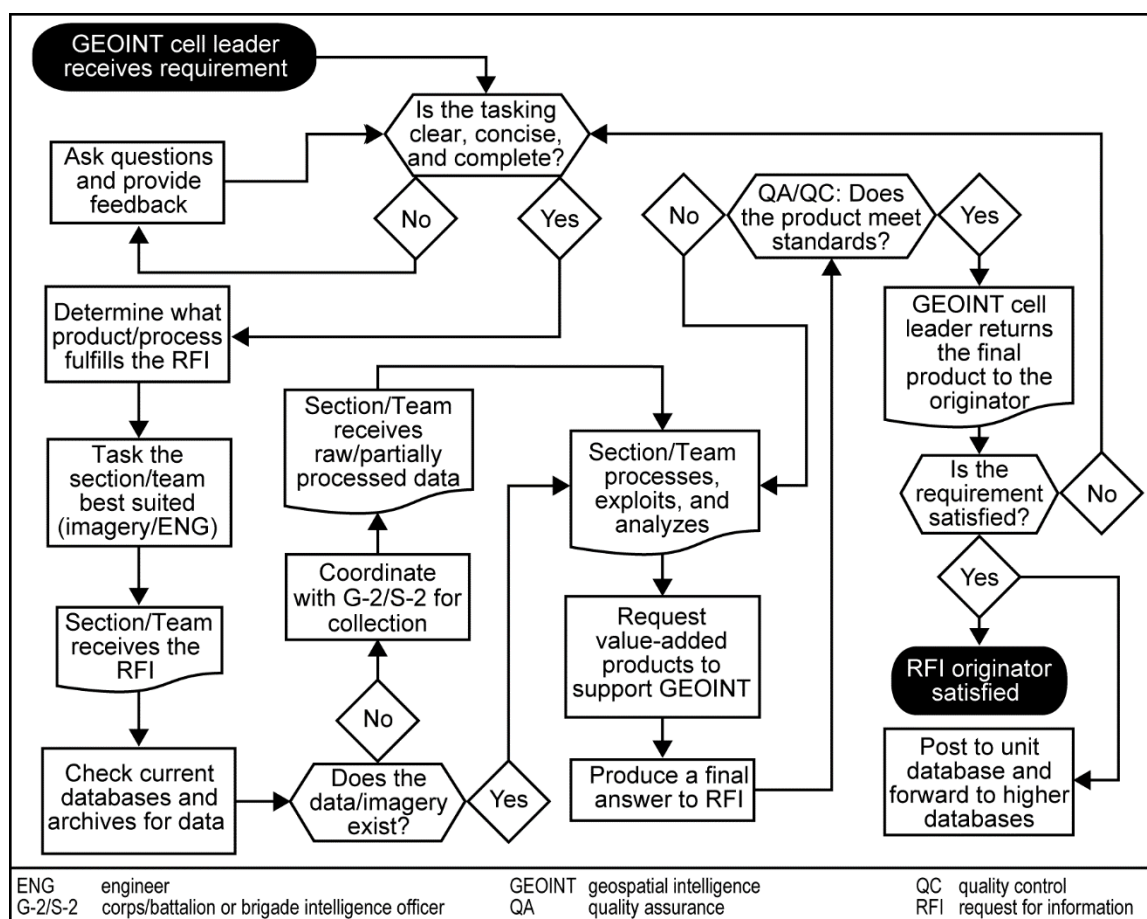


Figure 3-1. GEOINT request for information process (example)

FULFILLING THE REQUEST

3-85. Upon receiving an RFI, the GEOINT cell reviews the request to ensure comprehension. GEOINT cell members check current databases and archives to determine whether the request can be answered with existing information. If the request can be fulfilled with existing data, the cell creates a product in the requested format. If the request cannot be fulfilled with existing data, the cell notifies the originator, who may decide to modify the request, allowing the use of existing data to create the product. In some instances, the RFI leads to a request for additional collection. In such situations, the collection manager may initiate the tasking of available GEOINT collection assets or requests collection from higher echelons.

DOCUMENTING AND MANAGING THE REQUEST

3-86. Units enter RFIs into an RFI management system where every other system user can see it. This allows a unit several echelons above the requester to see the request and potentially answer it. The GEOINT cell and responsible G-2/S-2 should include guidelines and procedures on submitting RFIs in their standard operating procedures.

Chapter 4

Collection and Processing

SECTION I – COLLECTION

4-1. *Collection* is, in intelligence usage, the acquisition of information and the provision of this information to processing elements (JP 2-0). The commander focuses collection by establishing priorities, asking questions of intelligence relevance, and approving PIRs recommended by the staff during the MDMP. The commander's information needs and intelligence gaps drive GEOINT collection operations. GEOINT requirements necessitate tasking and collecting imagery, geospatial data, and completed products. GEOINT collection operations use satellites, manned and unmanned aircraft, and reach to collect data from existing requirements and derive new requirements.

GEOINT COLLECTION

4-2. GEOINT collection entails developing RFCs; validating, prioritizing, and tasking collection requirements; assessing the collection; and providing feedback. The collection management team performs this process to support the commander's intelligence and operational priorities. The team prioritizes collection requirements and plans, schedules, and tasks GEOINT collection assets. GEOINT collection results in either the direct tasking of GEOINT collection assets, over which the supported commander has authority, or the generation of an RFC that can be supported at a higher, lower, or adjacent echelon.

4-3. In joint doctrine, collection management has two subordinate functions anchored under collection management authority:

- *Collection requirements management* is the authoritative development and control of collection, processing, exploitation, and/or reporting requirements that normally result in the direct tasking of requirements to units over which the commander has authority, or the generation of tasking requests to collection management authorities at a higher lower, or lateral echelon to accomplish the collection mission (JP 2-0).
- *Collection operations management* is the authoritative direction, scheduling, and control of specific collection operations and associated processing, exploitation, and information reporting resources (JP 2-0).

4-4. In Army doctrine, collection management has three functions that are based on a tactical level point of view (see ATP 2-01):

- **Requirements management** identifies what requirements to collect against and when to collect. This function is performed through collaboration by the collection management team, commander, requesters, and the rest of the staff.
- **Mission management** specifies what assets should collect against intelligence requirements and defines when, where, and how to employ those assets and PED capabilities. This function is performed through collaboration by the collection management team, operations staff, and collection assets or the assets' C2 element.
- **Execution management** ensures the continuous integration and synchronization of ongoing information collection. This function is performed through collaboration by commanders and staffs that control current operations with support from the collection management team.

DEVELOP THE REQUEST FOR COLLECTION

4-5. RFCs differ from RFIs. RFIs require researching and analyzing existing data to answer intelligence problem sets; RFCs drive collection assets to generate new collection. RFCs must be submitted through the collection management team to be validated, prioritized, and processed for action. The submission process begins when the requester generates an RFC and ends when the collection requirement is satisfied.

RFC Example

Before an RFC can be leveraged against a GEOINT collection asset, the collection management team must determine if it is a valid requirement. The collection management team reviews the request for feasibility and ensures the requested capability can fulfill the collection requirement. Once validated, the team assigns the RFC a collection requirement number to track and manage it. The RFC is now an approved requirement.

If available GEOINT collection assets can satisfy the requirement, the BCT collection management team processes the request to task the available GEOINT collection asset (for example, an unmanned aircraft system [UAS]). If BCT or division GEOINT collection assets cannot support an RFC, the corps collection management team will leverage the theater army collection management team for action.

The theater army collection management team may choose to leverage other GEOINT collection assets in the theater. Requests requiring support from national GEOINT collection assets are routed through the theater army collection management team to the theater CCMD collection management team for final validation and processing.

Prioritization is the final step before an active requirement can be answered. A unit's active requirement competes with higher, adjacent, and lower unit active requirements for priority—for example, several units conducting operations in a joint operations area and few available joint, theater, or national GEOINT collection assets can be leveraged to support approved requirements. The order of priority determines when a GEOINT collection asset is tasked to collect for an approved requirement.

Theater army collection management teams prioritize many collection requirements based on factors such as current intelligence priorities, collection suspense, ongoing operations across the OE, or when the last date-time-group of collection against the object of interest occurred. Each factor has a different weight, which affects where the active requirement appears on the collection requirement prioritization list.

The RFC is not satisfied once it has been validated and actioned. The GEOINT cell must review the resulting collection to determine if the collected data, information, imagery, or associated product meets the criteria articulated in the request. If the collection does not satisfy the requirement, the GEOINT cell provides feedback to the collection management team requesting follow-on collection. The cell can recycle the original request for follow-on action. If the new collection satisfies the requirement, the GEOINT cell confirms the RFC's successful actioning and closes the request with the collection management team.

Unit standard operating procedures and higher headquarters policy determine the method for submitting RFCs. The GEOINT cell should consult its organization's collection management team for specific instructions before generating an RFC.

4-6. The GEOINT cell leader submits an RFC to the responsible collection manager to obtain information that does not reside in unit information holdings. When developing an RFC, the GEOINT cell should—

- Research GEOINT information holdings.
- Determine the required GEOINT capability.
- Define collection parameters.
- Justify the collection requirement.
- Submit the RFC.

Research GEOINT Information Holdings

4-7. Researching theater and national databases can determine if existing information holdings satisfy current collection requirements, saving time and resources. Existing collection must meet all necessary requirements for the intelligence problem. If collection does not exist, the GEOINT cell searches collection requirement management portals to see if it can benefit from another organization's active collection requirement. (The collection management team can assist with accessing websites and portals displaying active collection requirements.) GEOINT imagery analysts can research current imagery holdings and collection postures using tools and applications, including but not limited to—

- AGC. (Access via NIPRNET.)
- GEOINT Search and Retrieval. (See appendix E.)
- NGA's Map of the World. (Access via NGA on NIPRNET.)
- Office of Naval Intelligence. (Access via NIPRNET.)

Determine the Required GEOINT Capability

4-8. Different factors within the initial collection requirement determine which capability can answer each problem set. For example, the tactical identification of ground order of battle requires a different capability than determining nighttime activity on a mountainside. Panchromatic, radar, and infrared collection are a few examples of capability types.

Note. A unit cannot request specific GEOINT collection assets. Units may only request a capability. The responsible unit collection management team will assign GEOINT collection assets based on the sensor's ability to satisfy a requirement.

Define Collection Parameters

4-9. RFC parameters determine how a sensor will be used. The RFC articulates where the GEOINT collection asset will go, when it will be there, and what will happen once it arrives over the objective.

Target Location

4-10. The location of the objective or target is the most important information when developing an RFC. The requester cannot research or direct the sensor to the point of collection without grid coordinates. The target or point of collection for theater and below sensors is identified using the military grid reference system (MGRS). (See paragraph 4-69.) The target or point of collection for national level sensors is identified using the national technical means latitude and longitude.

4-11. When submitting requests for new GEOINT collection, a requester can select from five target types to determine the target location and size. The requester chooses the target type based on size, desired image type, and priority and timeline for collection. The five types of targets are—

- Point.
- LOC.
- Directed search area (also known as DSA).
- Broad area search (also known as BAS).
- Global geospatial information services (also known as GGIS), formerly known as mapping, charting, and geodesy.

Time Requirements

4-12. The activity or intelligence the requester is attempting to gather determines when the sensor collects it. Some RFCs may require a specific collection time frame based on operational needs, while others simply require collection within a 24-hour period with no time requirements.

Periodicity

4-13. In addition to time requirements, an RFC must specify collection periodicity (how often or how many times the sensor must collect within a given timeframe). RFCs express periodicity in different ways:

- Two numbers separated by a colon convey reconnaissance-focused collection periodicity (1:1, 1:3, 2:1, 1:7). The first number represents the number of collections, while the second number represents collection frequency. For example, 1:1 indicates a request for one collection every day, 1:3 indicates one collection every 3 days, and 2:1 represents two collections every day.
- A block of time in Zulu format (such as 0200Z–0500Z) conveys surveillance-focused collection periodicity. The time block should reflect the critical hours requested per day. For example, a request can specify a 4-hour time block in the morning and a 6-hour time block in the evening.

Sensor Actions

4-14. The RFC should also define what the sensor is going to do when it arrives on the target or point of collection. This information tells the sensor operator what the requester expects from the collection and provides parameters for building a collection strategy. For example, a surveillance focused RFC identifies whether the sensor is required to loiter over the same target for the duration of the collection period or switch between multiple objectives.

Justify the Collection Requirement

4-15. The collection requirement justification explains why the unit needs the collection and how the intelligence gained will aid in filling identified intelligence gaps in the supported commander's PIRs. For example, an RFC to support operations must explain what the operation is, why the collection is needed, and how it will support the operation. The collection management team uses the justification to support the request prioritization. Justification for collection support should consider—

- **Specific operation.** What operation will the sensor support?
- **Target development.** What is the specific target?
- **Intelligence requirement.** What PIR is the request attempting to answer?
- **Force protection.** What is the force protection condition associated with the request?

Submit the Request for Collection

4-16. GEOINT sections at the BCT submit the RFC through the brigade collection management team for validation and processing. The brigade collection management team processes the RFC if available GEOINT collection assets can satisfy the collection requirement. If the team determines available GEOINT collection assets cannot satisfy the requirement, it routes the request through the collection management team at corps and division levels for further processing. The collection management team at theater army and corps levels conducts the final validation and approves the RFC from subordinate units.

Request for Collection Submission Timeline

4-17. RFCs should be submitted as early as possible to allow time for request processing, prioritization, and GEOINT collection asset mission planning. Units should follow specific theater guidance when submitting new requests to ensure collection meets the timeline of the requirement. Urgent day of execution or next day requests should be submitted as noninterference-based or ad hoc RFCs.

Noninterference-Based Requests for Collection

4-18. Noninterference-based RFCs maximize collection opportunities by leveraging GEOINT collection assets while in transit to assigned tasks. This type of request is typically reserved for collection requirements not listed on the day of execution target deck. It is only executed once all other available GEOINT collection assets have been tasked. A noninterference-based RFC attempts to either use or only slightly deviate from a current GEOINT collection asset's flight plan to collect points of interest along the transit route without changing the asset's ability to support the current tasking.

4-19. Units should not depend exclusively on noninterference-based RFCs. They should use these requests to enhance the information collection plan for the execution day and coordinate them no less than 24 hours in advance to provide the supporting unit enough time to factor the deviation into mission planning. Although these requests are considered dynamic in nature, they do not change the GEOINT collection asset's day of execution tasking.

Ad Hoc Collection Requirements

4-20. Ad hoc requirements change a GEOINT collection asset's assigned tasks during the day of execution. Ad hoc requirements are necessary when changes within the OE dictate the need to redirect a GEOINT collection asset to support emerging, new, or existing requirements.

4-21. Ad hoc requirements require additional levels of coordination. Because the GEOINT collection asset may transit to a new location, higher echelons ensure factors such as transit times, airspace deconfliction, and current payloads are considered before the asset is reassigned.

Dynamic Requirements

4-22. Dynamic requirements, like ad hoc requirements, change a GEOINT collection asset's assigned task. Dynamic requirements are different because the GEOINT collection asset is already actively collecting against its scheduled mission and requires approval before moving from that location to support the new requirement. This change is called dynamic retasking.

4-23. Dynamically retasking an available GEOINT collection asset requires coordination through the G-3/S-3 or operations staff as the owner of the information collection plan. Dynamic retasking of a GEOINT collection asset assigned to support by a higher headquarters must be coordinated through the appropriate collection management channels.

VALIDATE THE COLLECTION REQUIREMENT

4-24. Unless an exception has been granted, requesters must follow theater- or national-level standard operating procedures when submitting an RFC. The GEOINT cell leader should consult the unit's collection management team for proper request format prior to development. If an RFC is not in compliance, the collection management team may return it to the requester for revision or reject the RFC entirely.

4-25. The requester's next higher headquarters reviews the RFC before submitting to the corps or theater army collection management team. When the theater army team generates a requirement from a submitted RFC, it becomes an active requirement for future collection tasking. The team provides the requesting unit with a unique collection nomination name and identification number after request approval. If the RFC was submitted via email, the collection nomination name and identification number are emailed to the requester.

PRIORITIZE THE COLLECTION REQUIREMENT

4-26. The theater army collection management team may be responsible for hundreds of active collection requirements every day. To ensure a limited number of GEOINT collection assets are leveraged against the theater command's top PIR, the collection management team must prioritize active requirements.

4-27. Prioritizing active requirements may result in some collection requirements not having a high enough priority to obtain GEOINT collection asset coverage, causing a lack of information to answer the supported commander's PIR. A collection requirement may be valid but not compete well against another organization's requirements due to the nature of its mission, line of effort, or operational phase. GEOINT imagery analysts must understand how a collection requirement relates to a PIR as well as current and future operations. This allows the requester to articulate how the unit's mission, PIR, and RFCs are integrated into the operation's larger scheme and objectives.

4-28. Any request for new GEOINT collection must also be prioritized in accordance with the National Intelligence Priorities Framework (also known as NIPF), which establishes national intelligence priorities. GEOINT requests requiring national collection are ranked by priority against the National Intelligence Priorities Framework and receive collection based on that prioritization.

TASK THE COLLECTION REQUIREMENT

4-29. The collection management team collaborates with the GEOINT cell and other requesting units to determine the appropriate asset for each collection requirement. GEOINT collection tasking is divided into three main categories: national-level, commercial, and theater. (For domestic imagery collection requests, see appendix B.)

National-Level Tasking

4-30. GEOINT collection requirements for national-level tasking are submitted through GIMS—the only requirements management system used for tasking national satellite systems. A departmental requirements officer (also called DRO) processes validated collection requirements to deconflict and prioritize requirements from other national agencies, Services, and CCMDs. The officer coordinates with NGA to determine the requirements' prioritization and collection method. CCMDs use local collection or requirements managers to consolidate, prioritize, and coordinate GEOINT collection requirements before going through a departmental requirements officer.

Commercial Tasking

4-31. Commercial tasking requirements are submitted via GIMS and approved through a path similar to national-level tasking. NGA validates and authorizes new commercial imagery collection. (Appendix B discusses NGA approval and other policy considerations.)

Theater Tasking

4-32. Airborne GEOINT collection assets are tasked through theater tasking systems, such as the Planning Tool for Resource, Integration, Synchronization, and Management (also known as PRISM), for collection management mission applications. These web-based applications provide users at every echelon the ability to integrate GEOINT collection assets with joint collection requirements. Although GEOINT personnel must have access to these theater tasking systems, tasking is a function of the J-2 collection manager.

4-33. While theater tasking systems are the primary requirements management tools, they are not the only means of airborne tasking. Other means include voice communications, direct messaging, email, and message traffic. Although the collection management authority normally resides at the CCMD, it can be delegated to a subordinate task force, as required.

4-34. The collection management authority approves airborne collection requirements as well as—

- Establishes, prioritizes, and validates theater collection requirements.
- Establishes sensor tasking guidance.
- Develops collection plans.

ASSESS THE COLLECTION AND PROVIDE FEEDBACK

4-35. The GEOINT cell provides feedback to the collection management team to aid in the assessment of the information collection effort. Assessment of and feedback on GEOINT collection enable the commander and staff to determine if intelligence requirements have been satisfied or are still relevant.

4-36. The collection management team and intelligence staff continuously evaluate the information collection plan based on assessment results consisting of measures of performance and effectiveness. Performance measures determine whether a given collection task was executed and determine the degree to which the information satisfies the collection requirement. For example, a request for national imagery may successfully collect over the requested target, fulfilling the measure of performance. However, if weather effects obscure the target, the measure of effectiveness is unsatisfied.

4-37. The GEOINT cell provides both negative and positive feedback to the collection management team as collection requirements are tasked and collected. The cell should communicate with the collection management team concerning discrepancies between requested collection and actual results. The cell should discuss specific characteristics that prevent collected imagery from producing intelligence products that satisfy the RFC. For example, collection that yields an image with lower-quality resolution than requested

may prevent a GEOINT imagery analyst from conducting tactical identification of the threat's weapon systems or vehicles. This feedback allows the collection management team to identify solutions on a case-by-case basis for unsatisfactory collection, and to adjust collection strategy going forward. By providing feedback to the collection management team, the GEOINT cell increases its chances of obtaining high-quality imagery for processing and exploitation.

MANAGING GEOINT COLLECTION REQUIREMENTS

4-38. The collection management team interacts with other collection management sections within the organization and interfaces with outside organizations throughout each collection day. The collection management team will have the most current information regarding the status of RFCs. When a request has been validated and selected for action, the collection management team can provide additional information, such as what GEOINT collection assets have been tasked and projected collection dates.

INTERACTIONS AND UPDATES

4-39. Threat reporting, shifting operational dynamics, and changes within the OE can alter operations or adjust intelligence priorities, causing changes to current collection strategies. All these factors play a role in a collection requirement's life. Competing requirements for other units may cause an assigned GEOINT collection asset's priorities to shift, resulting in a lower priority or cancelled valid RFC. Operations and intelligence update briefs provide critical information that allows the GEOINT cell leader and responsible collection management team to determine if existing RFCs remain valid, anticipate changes to requests, and adjust their unit's collection requirements accordingly.

EXPLOITATION REQUIREMENTS

4-40. An exploitation requirement articulates the collection requirement's production needs. Routinely, an exploitation requirement is developed if an outside organization (such as a theater ground intelligence center or national agency) is required to process and exploit the new collection for the requester.

4-41. A GEOINT exploitation requirement defines who will exploit the GEOINT data yielded from the collection requirement to which it is linked. An exploitation requirement can identify a unit or individual analyst as the responsible exploiting entity. When exploitation is conducted internally, the exploitation requirement serves as an aid in tracking requirements. When exploitation cannot be conducted internally, the exploitation requirement serves as the method through which exploitation can be requested from and assigned to an outside agency or unit. Collection managers with access to the systems of record assist the requester in creating exploitation requirements for both national-level and theater-level requirements.

4-42. Exploitation requirements are managed as part of the requirements management process. Upon determining the need for an exploitation requirement, the requesting organization, GEOINT imagery analyst, requirements manager, collection manager, and any requested exploitation support collaborate to develop the exploitation requirement. The requesting organization should coordinate with the organization it wants to conduct the exploitation and production before entering the exploitation requirement into the system of record. When preparing an exploitation requirement, GEOINT imagery analysts should determine—

- If the requirement will be immediate, ad hoc, or standing.
- Its duration (for example, 9 days) and frequency (for example, one look in 3 days).
- The appropriate organization to perform the exploitation.
- The organization that will perform the exploitation.
- The essential elements of information (also called EEIs) to be exploited, based on collaboration with the analyst. (In other words, what the analyst wants.)

4-43. When the exploitation requirement has been coordinated between the requester and exploiter, it is submitted for review via the same process as the collection requirement. The collection manager reviews, validates, and approves the requirement before its submittal to the next approval authority. The resulting exploitation is integrated with current data and used to create products based on multidiscipline intelligence to satisfy RFIs. The requesting organization provides feedback to the exploiting agency, as needed, to satisfy the exploitation requirement.

GEOINT COLLECTION ACTIVITIES

4-44. The commander's information needs and intelligence gaps drive GEOINT collection. The commander focuses collection by establishing priorities and approving intelligence requirements the staff recommends during the MDMP. The intelligence staff develops requirements and the collection management plan, supports tasking and directing, assesses collection, and updates the collection management plan, as needed. GEOINT collection operations focus on those collection requirements that can be satisfied using GEOINT collection assets.

4-45. The GEOINT cell assists the requester with submitting the requirement to the GEOINT requirements manager. The cell does this by either converting the RFI into a collection requirement or by advising the requester on how to convert it. The GEOINT requirements manager refines the collection requirement for the collection manager to review. The GEOINT requirements manager should consider these factors when evaluating incoming collection requirements:

- Intelligence requirements.
- Emerging GEOINT capabilities.
- RFI prioritization.
- GEOINT collection asset availability.
- Requirement process validation.
- Requirement research and tracking.
- Weather.

4-46. The GEOINT cell leader is responsible for adjudicating incoming collection requirements from the requirements manager and other requesting units, depending on unit structure. The collection manager submits adjudicated collection requirements to the next approving authority. RFC considerations include but are not limited to—

- Required GEOINT types.
- Indicators and specific information requirements.
- Collection priorities.
- Collection strategies.
- The latest time information is of value.
- Required types of GEOINT collection assets or systems.
- Required levels of fidelity.
- Required frequencies.
- Required amounts of data.

4-47. The GEOINT cell should assist the collection manager in understanding how to submit RFCs for national, commercial, and airborne GEOINT systems. Additionally, the collection manager should be able to identify and resolve technical errors in collection requirement submissions.

GEOINT COLLECTION METHODS

4-48. A GEOINT cell leader selects the appropriate GEOINT collection asset once all RFI values have been validated and a collection requirement is determined to exist. Collection feasibility research identifies capabilities and limitations of available GEOINT collection assets. The collection manager matches GEOINT collection asset capabilities and limitations to collection and exploitation requirements and determines which GEOINT collection asset best satisfies the requirements. The collection manager should also obtain weather effects on GEOINT collection assets from analysis the USAF SWO provides.

4-49. Satellite and aircraft systems produce the majority of GEOINT data, but ground systems, such as engineer reconnaissance equipment, may also be used. GEOINT unites a broad range of evolving and emerging capabilities that include geospatial sources associated with gravimetric, geomagnetic, and hydrographic surveying data and advanced GEOINT collection assets that provide hyperspectral, polarimetric, enhanced gravimetric, and high-definition data.

NATIONAL AND COMMERCIAL SATELLITE SYSTEMS

4-50. National satellite systems are a primary source of GEOINT imagery. The NRO manages all data collection from these systems. (See paragraphs A-10 and C-4 through C-6.) Commercial satellite systems can provide a range of information and products for NSG requirements. High-resolution commercial imagery provides an advantage in multinational operations because it can be shared with multinational partners without compromising the capabilities and operating characteristics of U.S. satellite systems.

Note. U.S. Government intelligence agencies can coordinate access to foreign government data.

AIRCRAFT SYSTEMS

4-51. U.S. Government systems are normally operated, controlled, and managed in theater and at tactical levels. Aircraft GEOINT collection assets provide a critical and responsive GEOINT source for the tactical commander. GEOINT collection assets include manned aircraft and UASs.

MULTISENSOR COLLECTION AND MULTIDISCIPLINE INTELLIGENCE COLLECTION

4-52. Electro-optical (EO) sensors are the primary means of collecting geospatial data. Integrating multiple GEOINT collection assets (such as LIDAR and interferometric SAR) gives GEOINT imagery analysts the ability and flexibility to depict targets with greater resolution and identify specific target conditions. GEOINT collaborates with other intelligence disciplines (such as signals intelligence [SIGINT], human intelligence, and measurement and signature intelligence) to generate multidiscipline knowledge that ensures a more comprehensive GEOINT product.

4-53. Cueing involves using one or more GEOINT collection asset systems to provide data that directs collection by other systems. A cueing opportunity may be recognized while developing and synchronizing the information collection plan or after receiving intelligence from another discipline, initiating the employment of a complementary GEOINT collection asset. Cueing maximizes the efficient use of finite GEOINT collection assets to support multiple, often competing, intelligence collection priorities.

Border Crossing

A GEOINT cell receives a request to conduct GEOINT analysis to support border security operations. The analysis assists in determining if current border assets are placed in the most advantageous positions:

- Geospatial engineers identify passable and impassable areas along the border to identify border crossing routes by analyzing elevation, slope, vegetation, and soil composition data.
- The collection management team submits a request for MTI collection along the border and examines MTI data to identify smuggling routes and the flow of traffic across the border.
- GEOINT imagery analysts analyze current imagery to identify breaks in the border through which traffic may pass.

The collection management team uses multidiscipline intelligence data to layer the final product. This information allows the commander to decide on border asset positions. The team uses measurement and signature intelligence to cue UASs to potential crossings as they occur, SIGINT to track movement to and from the border, and human intelligence to validate and confirm crossing activity.

The GEOINT cell coordinates with the collection management team to request collection from aerial MTI GEOINT collection assets using the following strategies: SAR and enhanced SAR, sector search, attack planning and attack control, fixed target indicator, and small area target classification.

After considering GEOINT collection asset capabilities and limitations, the collection management team determines parameters and service requests using RFI details. Parameters include the desired coverage area, center point, geographic length and width, and coverage orientation. Service requests may require a single GEOINT collection asset, multiple GEOINT collection assets, a single sensor, or multiple sensors. Once the request is submitted, the GEOINT cell monitors its status and conducts target analysis in response to reporting.

SECTION II – PROCESSING

4-54. *Processing* is a system of operations designed to convert raw data into useful information (JP 2-0). Processing distributes collected information for further analysis and provides combat information to commanders and staffs to answer information requirements to support military operations.

4-55. GEOINT processing consists of converting and refining collected GEOINT data into usable information to develop GEOINT reporting and products to support analysis. Once data has been collected and processed, users can exploit the data and produce either general intelligence or mission-specific products.

DATA MANAGEMENT

4-56. Every GEOINT cell is responsible for maintaining georeferenced information and vector files created at the cell level and below. Data managers within each GEOINT cell have two major responsibilities:

- Establish a flow of information from all subordinate units.
- Update imagery of the AO (acquire, format, and upload high-resolution imagery to the cell server).

UPDATING IMAGERY

4-57. Dividing responsibility of imagery updates creates a more manageable area to maintain, allowing managers to understand specific problem sets and distinct characteristics in their respective regions. Managers use this knowledge to determine where and how often imagery is updated.

4-58. Managing data before processing is vitally important. GEOINT imagery analysts manage data by—

- Managing the GEOINT background to create a COP for all information system users.
- Entering imagery, geospatial data, and IMINT in standardized databases accessible to consumers.
- Populating metadata so information is discoverable.
- Using editing software to import geospatial data and prepares, displays, and enhances digital images for mapping use in geospatial information systems software.

4-59. GEOINT imagery analysts and/or geospatial engineers use GEOINT and/or GI&S software to—

- Create, manage, analyze, map, and visualize all types of data (to include intelligence) that can be tied to the surface of the Earth.
- Connect data to a map, integrating location data (where things are) with all types of descriptive information (what things are like there).
- Assist commanders, staffs, and decision makers in understanding patterns, relationships, and geographic context; improving communications and efficiency; and influencing management and decision making.

4-60. Updating and synchronizing databases at all levels is crucial. The GEOINT imagery analyst and/or geospatial engineer review current holdings, update holdings, and archive older data. They check for updates using data websites, including but not limited to AGC, NGA, and the U.S. Geological Survey, which can be accessed via NIPRNET. If the analyst finds data that is more current, they import it and archive outdated data in the corresponding geodatabase scheme.

DATA ACCURACY

4-61. The GEOINT cell uses NGA data accuracy standards for all products and data to maintain data integrity. Data accuracy descriptions may also be in a specific product's metadata.

GEOINT Standards

4-62. GEOINT standards focus on GEOINT data and their structures. They support specific elements of the NSG architecture and represent approximately 25 percent of all standards relevant to the architecture. NGA is responsible for ensuring the selection and adoption of GEOINT standards for the NSG.

4-63. GEOINT standards characterize GEOINT data, data constructs, data services, products, and interfaces. They enable the collection, processing, analysis, and exploitation of GEOINT. GEOINT standards also govern GEOINT access, dissemination, and storage. Applying common GEOINT standards reduces the use of multiple and incompatible datasets and makes it possible to create and share suitable, accurate, comprehensive, and timely GEOINT.

4-64. GEOINT imagery analysts extract current dispositions and significant activities of threat and friendly units from intelligence reports, sources, and the current situation map. The intelligence required may be partially or completely posted on a current situation map. They translate this information into graphics on an overlay that only depicts significant items that answer the exploitation requirement or RFI.

Metadata

4-65. Metadata is a critical component of GEOINT. It provides valuable information, such as when the data was created, its precision level, source materials used, intelligence purposes, and security level. Metadata is generally described as data about data. It describes data's structure, such as whether it comprises text or numbers, and characteristics of the data content, such as the data's purpose, lineage, and quality. Metadata enables data producers and users to properly manage, find, assess, and use data.

4-66. Relative data accuracy often corresponds to the product's purpose. For example, targeting products have a higher degree of accuracy than topographic line maps. These metadata categories describe a product's suitability for the intended use:

- **Data currency** refers to either the time of data collection or the time of product completion.
- **Data consistency** refers to the dataset's fidelity, or how well the parts fit together. This is especially relevant in digital map and image mosaic products, which may contain sections derived from different sources.

DATA MANIPULATION

4-67. GEOINT cells manipulate imagery data values and positions to see features that would not normally be visible and to determine geolocation. Manipulation includes but is not limited to—

- Using software to perform linear feature extraction of map data from imagery and import and export the data in a variety of data formations.
- Altering the level of brightness or reflectance of light from the surfaces in the image to assist in analyzing vegetation and soil composition.

POSITIONAL REFERENCE SYSTEMS

4-68. GEOINT cells use source grid reference systems, such as MGRS, the universal transverse Mercator (UTM) system, and the world geographic reference system, and determine position reference procedures. At a minimum, they report the accuracy or reliability of the coordinates' source, along with the series number, sheet number, edition, and mapmaker.

4-69. MGRS is an extension of the UTM system. Although commonly referred to as UTM coordinates, Soldiers using coordinate conversion tools and the Global Positioning System note MGRS coordinates are different. MGRS uses the UTM grid zone designator, followed by two alphabetic characters for the 100,000-meter square identification. The UTM system does not use a 100,000-meter square identifier.

4-70. Geographic coordinates appear on all standard military maps. On some maps, these coordinates may be the only method of locating and referencing a specific point. Figure 4-1 on page 42 depicts a grid reference box for a 1:1,000,000 scale map.

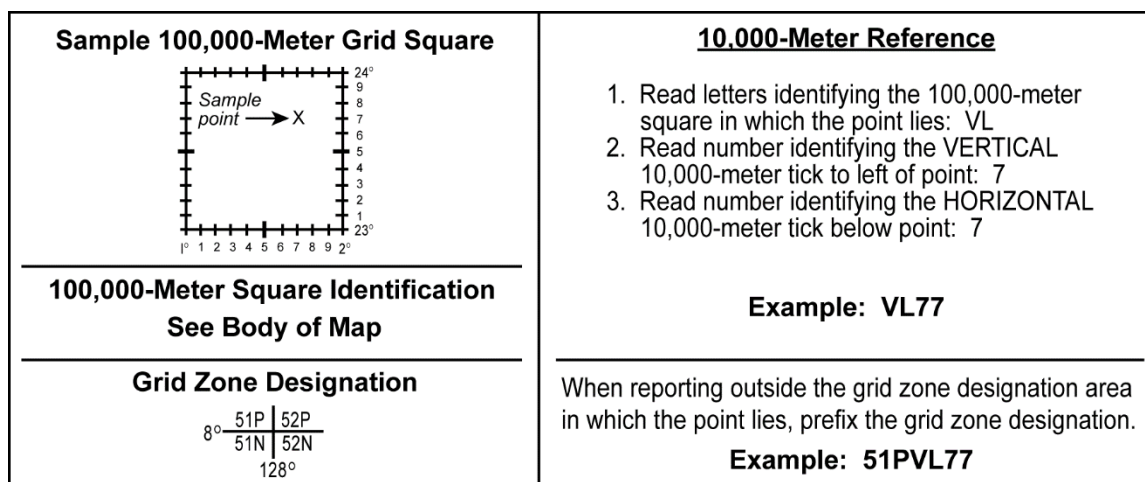


Figure 4-1. Grid reference box (example)

4-71. WGS 84 is the official DOD dataset. Datasets may come from multiple sources, including multinational forces, archived datasets, or the local populace. Coordinates should be converted whenever possible as requested products determine the coordinate system. These requirements may not be WGS 84.

GEOREFERENCING

4-72. *Georeferencing* refers to the process of aligning geographic data to a known coordinate system so it can be viewed, queried, and analyzed with other geographic data. Georeferencing may involve shifting, rotating, scaling, skewing, and, in some situations, warping, rubber sheeting, or orthorectifying the data. Georeferencing is the recording of the absolute location of data points. Georeferencing is a requisite task for all spatial data, since spatial data cannot be positioned in space or evaluated regarding other data without being assigned a spatial coordinate within a defined coordinate system.

Note. *Rubber sheeting* refers to a procedure for adjusting the coordinates of all the points in a dataset to allow a more accurate match between known locations and a few points within the set. Rubber sheeting preserves the interconnectivity between points and objects through stretching, shrinking, or reorienting their interconnecting lines.

RECTIFICATION

4-73. Orthorectification and georectification are rectification methods that vary due to the different techniques, algorithms, and software used.

Orthorectification

4-74. *Orthorectification* refers to a process of correcting imagery for geometric distortions caused by errors related to topography, atmosphere, maps, measuring instruments, GEOINT collection assets, or sensors. It is an accurate method of correcting an image to align with real-world coordinates, creating a true map.

4-75. Orthorectification involves precisely measuring the photo center's exact location and the exact camera angle when the image is acquired. The process requires camera calibration parameters to remove camera- and lens-included distortions, and a digital elevation model of the earth to remove terrain-included distortions. Orthorectification results in very high positional accuracy or uniform scale throughout the image. For example, after orthorectification, it becomes appropriate to measure or precisely locate features in the image, to collect information for a geographic information system (also called GIS), or to combine the image with other accurately rectified images for analyses.

Georectification

4-76. *Georectification* (not to be confused with georeferencing) refers to a method of stretching and warping an image to align with a map or other spatial data in a geographic information system, thus removing geometric distortions between sets of data points. The process matches image data to a set of geographical coordinates so that each pixel of the image is assigned a geographical coordinate.

4-77. Georectification does not require the location of points with space; only adjustment of the relative position of those points to one another. Several corresponding control points, such as street intersections, are marked on both the image and the map. These locations become reference points in the subsequent processing of the image. On the image, x, y coordinates are determined for each corresponding ground control point. The relationship of the x, y image coordinates to real-world ground control points is used to determine the algorithm to warp the image to the map. The resulting accuracy can never exceed the accuracy of the reference data (a 1:50,000 topographic line map is only accurate to within 66 feet or 23 meters). This method may be acceptable for projects with low accuracy requirements.

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Chapter 5

Production and Dissemination and Integration

SECTION I – PRODUCTION

5-1. *Intelligence production* is the integration, evaluation, analysis, and interpretation of information from single or multiple sources into finished intelligence for known or anticipated military and related national security consumer requirements (JP 2-0). Production provides comprehensive, highly detailed, and precise intelligence products to tactical commanders.

5-2. GEOINT production includes traditional geospatial information and imagery-based products, as well as multidimensional products created by combining geospatial information, imagery, and other geospatial products. GEOINT products and related services and support can be standard or tailored.

5-3. The main intelligence producers include Service exploitation and production centers, NGA, the Defense Intelligence Agency, Service component commands, AGC, and CCMDs. MI brigade GEOINT cells enable the analysis of integrated databases for specific applications. They add valuable information or update features and attributes within databases and strengthen database content to meet the commander's tailored mission requirements.

5-4. Three factors influence GEOINT product development:

- GEOINT analysis.
- Product type.
- Consumer product feedback.

GEOINT ANALYSIS

5-5. GEOINT leaders manage and supervise the production of GEOINT analytical products—whether single-source or layered multiple intelligence discipline products. Products must be accurate, concise, and current. All products undergo quality assurance checks before they are considered final. Quality checks include confirming the product meets the requirements for which the product was initiated. GEOINT imagery analysts validate source materials to ensure databases and libraries are as current as possible, and they check the quality of graphic products, ensuring the products meet analytical standards and production quality standards and answer the requester's requirements.

5-6. If the analysis is misleading, incoherent, or not in accordance with established standards, the senior GEOINT imagery analyst returns the product for additional processing and analysis. The returned product includes comments, critiques, and suggestions for refining the analysis product development. The analysis results in a new product that must meet the senior GEOINT imagery analyst's quality standards.

PRODUCT TYPE

5-7. Data can be combined in many ways to develop products for specific mission requirements. Intelligence users or requesters should coordinate with producers to ensure products meet mission needs. There are two types of GEOINT products:

- Standard.
- Tailored.

STANDARD GEOINT PRODUCTS

5-8. GEOINT imagery analysts develop standard products using vector data or raster data, including panchromatic, SAR, infrared, or multispectral sensor data. These may be simple products, such as imagery readouts, reports, maps, and charts, or more complex products containing many layers of data, ranging from geographic to intelligence information. Many of these products are two-dimensional but products can be three-dimensional in certain circumstances.

TAILORED GEOINT PRODUCTS

5-9. Tailored GEOINT products are concisely constructed visual packages of intelligence information. GEOINT imagery analysts create them to assist in answering a requirement or to provide the requester additional information. While the base elements of GEOINT products may be similar, the details are unique to the specific requirements or RFI that drive product development.

5-10. GEOINT imagery analysts can tailor GEOINT products to support the different activities of the operations process. Analysts digitally generate products that consist of base imagery, map, or elevation backgrounds overlaid with various data layers. These digital layers have database tables associated with each component that allow them to be queried, analyzed, and displayed to create the desired product. The digital layers (as digital overlay files) can be displayed in numerous information systems.

5-11. Analysts can combine or layer GEOINT products into a fused product to enhance battlefield visualization. They can also combine multiple products in layers to provide more comprehensive final products designed to answer specific requirements. Tailored GEOINT products, as listed in table 5-1, provide key geospatial, mobility corridor, and AA information.

Table 5-1. Tailored GEOINT products

Imagery intelligence reports (See table 5-2 on page 48 for examples.)
Imagery derived product: A literal (imagelike) or graphic representation of an original image or information extracted from the image. Analysts can create these products using digital or analog techniques and store them in hardcopy or softcopy format.
Print: A representation of a portion of a frame of imagery produced from a negative or by a softcopy workstation. Most prints feature a target of interest.
Mosaic: An assembly of overlapping aerial photographs that have been matched to form a continuous photographic representation of a portion of the Earth's surface. (Also referred to as aerial mosaic.)
Video clip: A short portion of video, usually part of a longer recording. The term also more loosely refers to any short video taken from the stream captured during the length of an unmanned aircraft system orbit.
Target folder: Contains target intelligence and related materials prepared for planning and executing action against a specific target.
Combined obstacle product: Includes data about obstacles, surface configuration, surface material, vegetation, hydrology, and some line of communications (LOC) information. Combined obstacle products may also include cross-country mobility surface conditions, soil conditions, and water levels.
Cover and concealment product: Shows a visualization of the vegetation and terrain in an area that may allow for cover and concealment of vehicles, personnel, and facilities. It may stand alone or fuse into other products.
River-crossing product: Allows for the selection of multiple crossing methods (fording sites, amphibious vehicles, and assault launch bridges) and provide data and analysis on the slope and soil type of riverbanks, width and velocity of rivers, surface and bottom conditions, and weather effects on the terrain. It may stand alone or fuse into other products.
LOC product: Provides data on highways, roads, railways, bridges, tunnels, and runways in the area of operations. It may stand alone or fuse into other products.
Mobility product: Annotates the mobility that meets the mission objective. It may stand alone or fuse into other products.
Image map: Simple hardcopy map employed universally.
Three-dimensional visualization product: A fly-through of an area of operations or approach.
Visibility product: Shows visibility of a given position from another position or visibility of an area from a given position.
Slope analysis: Determines areas of acceptable slope for drop zones, landing zones, and artillery assets.

CONSUMER PRODUCT FEEDBACK

5-12. At the initial phase of dissemination, GEOINT imagery analysts return the final product to the consumer with a request for feedback. They close the RFI if the final product meets the consumer's needs and satisfies the requirement. At the last phase of dissemination, analysts post the final product to GEOINT databases and make it available for broad consumer access and use. The product is available to higher, adjacent, and subordinate elements.

Note. The GEOINT imagery analyst must solicit timely and relevant feedback from the consumer on intelligence products. Feedback is critical to ensuring GEOINT meets consumer requirements.

5-13. If the consumer determines the product does not meet the original requirement, GEOINT imagery analysts must determine the point of failure. A feedback process ensures the requirement is clear, concise, and complete and ensures GEOINT leaders understand the consumer's intent. GEOINT leaders may need to conduct an additional assessment of the original RFI or completely restart the GEOINT RFI process.

SECTION II – DISSEMINATION AND INTEGRATION

5-14. *Dissemination* is the delivery of intelligence to users in a suitable form (JP 2-0). Commanders must receive combat information and intelligence products in time and in an appropriate format to facilitate situational understanding and support decision making. The timely distribution of intelligence products in the appropriate format is accomplished by providing hardcopy or digital products to users. Dissemination occurs through push and pull principles:

- The push principle allows analysts to transmit intelligence to requesters along with other relevant information. Typically, the intelligence cell at each echelon manages intelligence dissemination.
- The pull principle provides intelligence organizations at all levels with direct reach capability via electronic access to central databases, intelligence files, or other repositories containing intelligence data and products.

GEOINT DISSEMINATION AND INTEGRATION

5-15. GEOINT dissemination refers to the timely conveyance of GEOINT products in an appropriate form and by suitable means, whether in hardcopy form, softcopy form, or both. The distribution of standard hardcopy geospatial information products (such as topographic maps) is a supply function. Intelligence channels commonly distribute nonstandard or tailored products. (See paragraphs 2-24 through 2-26 for Army GEOINT integration.)

5-16. The GEOINT cell performs critical and noncritical tasks that are synchronized to disseminate GEOINT and geospatial products. Critical tasks within the GEOINT cell include briefing and developing GEOINT and associated geospatial products (a GEOINT imagery analyst critical task) as well as geospatial visualization, data, and products (a geospatial engineer critical task).

DISSEMINATION METHODS AND TECHNIQUES

5-17. When the GEOINT cell develops a product they ensure dissemination, either electronically or via physical delivery. Direct dissemination encompasses posting information to the appropriate websites, intelligence reach, and updating the COP. Producing organizations disseminate GEOINT and geospatial products directly by granting access to databases, information, or intelligence, and by sharing; they may disseminate information directly in verbal, written, or graphic format. The Defense Logistics Agency and the Army supply system distribute standard geospatial products.

5-18. When disseminating GEOINT products, GEOINT imagery analysts determine what products and format best suit the requester. In doing so, they consider several factors:

- Type of products needed.
- Size of the products.
- Number of products.
- Product classification.
- Best dissemination method.
- How to track disseminated products.

5-19. Upon determining the best format, GEOINT imagery analysts—

- Create the product in that format (for example, burn to a compact disc, post to a website, send via email, or create a hardcopy product).
- Archive all products and data to reference or re-disseminate later.
- Create an archive scheme to allow easy and understandable pathways to the desired data.
- Create a reference log.
- Import data and products to the appropriate location within the archive scheme.

5-20. GEOINT imagery analysts record and disseminate analysis results to the U.S. IC via different types of IMINT reports. (See table 5-2 on page 48.) Consumers may request reports directly or use them as a data input means for maintaining intelligence databases.

Table 5-2. Types of imagery intelligence reports

Imagery interpretation report	
• A DOD account of imagery-derived information pertaining to one or more specified targets, resulting from the exploitation of imagery from one or more collection systems and missions. • May include a mission report component, item type component, and target report component, and may be produced with a tactical report. • May stand alone or become the base for other products. • Initial phase interpretation report: A national target report or a DOD imagery interpretation report that contains imagery-derived information from first phase (response to immediate requirements) exploitation of satellite or airborne imagery. • Supplemental phase interpretation report: A national target report or a DOD imagery interpretation report that contains imagery-derived information from second phase (7-day turnaround on the requirement) exploitation of overhead imagery.	
Reconnaissance exploitation report	
• The standard message format used to report the results of a tactical air reconnaissance mission. Whenever possible, the report should include the interpretation of sensor imagery. • A DOD reporting product that serves as an abbreviated imagery interpretation report, created mainly from tactical imagery collection tasked at the local command level. • The format is installation oriented or event oriented and may contain theater-unique data elements. It may be issued in response to five imagery collection mission types: ▪ Tactical air reconnaissance (land). ▪ Tactical air reconnaissance (maritime). ▪ Armed reconnaissance (maritime). ▪ DOD theater. ▪ National airborne reconnaissance missions.	
Inflight report	
Given by the aircrew while still in flight.	
Preliminary exploitation phase interpretation report	
• Contains imagery-derived information from the first exploitation of satellite imagery. • Accomplished within two hours of time-over-target.	
Internet relay chat	
• Typical first phase reporting is handled through chat (live interactive text communications on the internet). • Multiuser internet relay chat and Jabber are the most used chat applications (both provide rapid response functionality, a significant benefit during a crisis). • A major chat deficiency is its inability to provide a database-accessible breakdown of the message content. • Chat logs can be stored, but communitywide access to unique values is limited to manual transfer.	
GETS	
GETS is a GEOINT reporting database and the Army's GEOINT requirement and production tool. It provides web-based reporting with linkage to the Global Broadcast Service. It is mandated by INSCOM for the Army National to Theater Program.	
DOD	Department of Defense
GEOINT	geospatial intelligence
GETS	Geospatial Intelligence Enterprise Tasking, Processing, Exploitation, and Dissemination Service
INSCOM	United States Army Intelligence and Security Command

ELECTRONIC DELIVERY

5-21. Electronic delivery is the most common means of disseminating GEOINT products. Each GEOINT cell component has its own GEOINT database accessible by supported units (for example, imagery product library or map server). (See appendix E for the GEOINT integrated architecture.)

5-22. The GEOINT imagery analyst ensures the requester receives direct dissemination of the requested product in a format that suits the requirement. This may be an email notifying the requester of the product's availability or providing a digital copy of the product.

5-23. GEOINT imagery analysts can disseminate GEOINT products through GETS. GETS is a web-based platform used to manage GEOINT tasking, processing, exploitation, and dissemination (TPED) processes for GEOINT production and report dissemination.

5-24. GETS disseminates intelligence products to the Library of National Intelligence as well as its own web, email, and subscription services. The GETS development team collaborates with NGA's online GEOINT services to expose GETS content through NSG exploitation services and a unified web presence, allowing GETS to publish its intelligence products to the national GEOINT database. (See appendix E for GETS.)

PHYSICAL DELIVERY

5-25. GEOINT imagery analysts may distribute hard drives, compact discs, digital video discs, or other specialized products via personal delivery couriers, other mail systems, or briefings. Upon request, geospatial engineers produce and distribute hardcopy maps and overlays to requesters.

Briefings

5-26. Briefings are a means of presenting GEOINT to commanders, staffs, or other audiences. The briefing's purpose, the briefer's role, and the desired response combine to determine the techniques employed. Briefings are a method of direct physical dissemination.

5-27. GEOINT leaders prepare for four types of briefings:

- **Information brief** informs the listener and increases understanding.
- **Decision brief** obtains an answer or a decision.
- **Mission brief** provides information, gives specific instructions, or instills appreciation of a mission.
- **Staff brief** secures a coordinated or unified effort.

5-28. GEOINT leaders determine an audience's composition to decide how much and what type of information to brief. Composition includes the size and makeup of the audience, such as whether members are single- or multi-Service military, civilians, or foreign nationals. It also considers audience members' official positions, level of professional knowledge, and any special interests or personal preferences.

5-29. GEOINT leaders establish how much time to allocate and the state of the facility in which the briefing will occur. In doing so, they—

- Determine location deficiencies and make recommendations to overcome them.
- Determine if the briefing will contain or lead to a discussion of classified material.
- Ensure the facility has been approved or cleared for discussion of classified material.

5-30. GEOINT leaders construct the briefing by—

- Collecting required materials and becoming familiar with the subject.
- Stating the problem (if necessary).
- Isolating key points and arranging them in a logical order.
- Providing supporting data to substantiate key points.
- Selecting GEOINT products and visual aids that support the briefing.
- Determining conclusions and recommendations.
- Reviewing and revising the initial draft brief after preparing it.
- Ensuring the briefing includes all necessary facts and answers to anticipated questions.

5-31. When presenting the brief, GEOINT leaders—

- State classification of the briefing and at which discussions may occur within the facility.
- Explain the purpose and scope of the briefing, alternatively providing the mission statement of the unit or mission.
- Emphasize the essentials.
- Ensure the analysis and conclusions are understood, and restate any decisions or directives made during the briefing to illustrate understanding.

Hardcopy Dissemination

5-32. Hardcopy dissemination may require mass production. The GEOINT cell determines the size and required quantity of the product. It also determines the supplies needed and calculates the amount required to satisfy the requirement. For mass production, the GEOINT cell initially produces a single product to ensure the quality is satisfactory.

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PART THREE

GEOINT OPERATIONAL CONSIDERATIONS

Chapter 6

GEOINT Across the Army Strategic Contexts

COMPETITION BELOW ARMED CONFLICT

6-1. Competition below armed conflict exists when two or more state or nonstate adversaries have incompatible interests, but neither seeks armed conflict. Nation-states compete using the instruments of national power to gain and maintain advantages to achieve their goals. Low levels of lethal force can be part of competition. Adversaries often employ cyberspace capabilities and information warfare to destroy or disrupt infrastructure, interfere with democratic processes, and conduct other malign activities in a way that does not cause the United States and its allies to respond with force. Competition also includes cooperation with allies and partners to counter pacing threats.

6-2. During competition, GEOINT focuses on training, target development, and support to IPOE. All activities should support the transition to armed conflict. GEOINT imagery analysts must know how to ingest data types, manipulate that data using the GEOINT architecture, and understand how raw data and finished intelligence assist the commander and staff in visualizing the enemy and potential OEs. (See FM 2-0.)

6-3. GEOINT analysis establishes a baseline understanding of what environments look like and better prepares analysts to determine change during armed conflict. Table 6-1 describes the collection focus for GEOINT elements across the levels of warfare (at echelon) during competition.

Table 6-1. GEOINT activities during competition below armed conflict

Level of warfare (echelon of employment)	Collection focus during competition
Theater strategic (MIB-T)	• Provide recurring GEOINT support to IPOE steps 3 and 4 within the regional AO to prepare for transition to armed conflict, if necessary. • Create and update standing GEOINT collection decks for national and theater GEOINT collectors in the appropriate requirements management system. • Support target development to ensure relevant targets are validated on a joint target list and properly prioritized. • Facilitate joint, multinational, and interagency GEOINT support to the regional AO. • Certify GEOINT imagery analysts in NGA's GEOINT Professional Certification Fundamentals and Imagery Analysis–II. (For more information, access the NGA website on NIPRNET.)
Operational (ASCC and corps)	• Develop Appendix 5 (GEOINT) to Annex B (Intelligence) for inclusion in the operation or concept plan for which the unit has an assigned role should either plan be executed. • Leverage joint and combined regional exercises to optimize synchronization laterally, vertically, and across domains (for example, Defender Pacific and Defender Europe). • Exercise GEOINT architecture, knowledge management procedures, and PACE plans. • Support target development to ensure relevant targets are validated on a joint target list and properly prioritized. • Certify GEOINT imagery analysts in NGA's GEOINT Professional Certification Fundamentals and Imagery Analysis–II.

Table 6-1. GEOINT activities during competition below armed conflict (*continued*)

Level of warfare (echelon of employment)		Collection focus during competition	
Tactical (E-MIB, division, and BCT)		• Maintain mission readiness and task proficiency via the unit mission-essential task list, collective, and GEOINT task training. • Leverage IEWTPT to aid realistic training simulation of GEOINT tasks and architecture. • Use and maximize Foundry to support military intelligence training and certification. • Collaborate with outside agencies, units, and complementary military occupational specialties to maximize complexity and realism in unit and element training scenarios. • Exercise GEOINT architecture, knowledge management procedures, and PACE plans. • Use LET opportunities to diversify GEOINT imagery analyst skills and ability to support all mission types.	
AO	area of operations	IPOE	intelligence preparation of the operational environment
ASCC	Army Service component command	LET	live environment training
BCT	brigade combat team	MIB-T	military intelligence brigade-theater
E-MIB	expeditionary-military intelligence battalion	NGA	National Geospatial-Intelligence Agency
GEOINT	geospatial intelligence	NIPRNET	Nonsecure Internet Protocol Router Network
IEWTPT	Intelligence Electronic Warfare Tactical Proficiency Trainer	PACE	primary, alternate, contingency, and emergency

CRISIS

6-4. A *crisis* is an emerging incident or situation involving a possible threat to the United States, its citizens, military forces, or vital interests that develops rapidly and creates a condition of such diplomatic, economic, or military importance that commitment of military forces and resources is contemplated to achieve national and/or strategic objectives (JP 3-0). Commanders must consider that overt military action may escalate a crisis toward armed conflict, which requires rapid response by forces prepared to fight if deterrence fails. Using space and cyberspace capabilities provides other options less likely to cause escalation. Crisis is also the period between armed conflict and the return to competition.

6-5. From competition to potential armed conflict, GEOINT elements prepare for combat and contingency operations. GEOINT support focuses on visualizing the potential OE, including the physical environment, threat composition, disposition, and strength, and assessed enemy COAs. During the MDMP, GEOINT must ensure Appendix 5 (GEOINT) to Annex B (Intelligence) to the OPORD is updated and refined to align with the commander's mission and intent.

6-6. During crisis, one action from either friendly or enemy forces may lead to armed conflict. The intelligence staff must identify enemy indicators of that transition; once identified, GEOINT visualization of the AO monitors established NAIs to determine if any enemy criteria for transition to armed conflict have been met. Table 6-2 describes the collection focus for GEOINT elements across the levels of warfare (at echelon) during crisis.

Table 6-2. GEOINT activities during crisis

Level of warfare (echelon of employment)		Collection focus during crisis	
Theater strategic (MIB-T)		• Coordinate with national agencies and FORSCOM to align GEOINT priorities and collection strategies. • The integrated GEOINT division solidifies knowledge management procedures for echelons. Procedures must identify dissemination systems and reporting formats that units can ingest. • Create GEOINT data repository for access by supported FORSCOM elements. • Source operations shift to focus on developing events and crises to support the combatant commander's information requirements.	
Operational (ASCC and corps)		• Finalize Appendix 5 (GEOINT) to Annex B (Intelligence) that aligns to the commander's mission. • Intelligence staffs leverage GEOINT information derived from national and theater standing GEOINT requirements to answer the supported commander's priority intelligence requirement. • Update validated targets with data relevant to ongoing activity within the AO; refine priorities based on assessed enemy courses of action. • Analyze current and historical AO imagery to create a baseline to build from during current operations. • Ensure the GEOINT architecture is ready for deployment, has connectivity, and meets communications security requirements.	

Table 6-2. GEOINT activities during crisis (*continued*)

Level of warfare (echelon of employment)		Collection focus during crisis	
Tactical (E-MIB, division, and BCT)		- Require predeployment unit mission-essential task list, collective, and GEOINT task training and certification for deployment. - Ensure GEOINT architecture is ready for deployment, has connectivity, and meets communications security requirements. - Analyze current and historical imagery of the AO to a baseline to build from during current operations. - Provide GEOINT input to the initial information collection plan that covers all levels of the enemy depth to ease the transition to armed conflict. - Conduct predeployment inspections to ensure equipment serviceability and personnel readiness. - Finalize Appendix 5 (GEOINT) to Annex B (Intelligence) to the OPOD that aligns to the commander's mission and higher headquarters OPOD.	
AO	area of operations	FORSCOM	United States Army Forces Command
ASCC	Army Service component command	GEOINT	geospatial intelligence
BCT	brigade combat team	MIB-T	military intelligence brigade-theater
E-MIB	expeditionary-military intelligence battalion	OPOD	operation order

ARMED CONFLICT

6-7. Armed conflict occurs when a state or nonstate actor uses lethal force as the primary means to satisfy its interests. Entering and terminating armed conflict is a political decision. Army forces may enter conflict with some advanced warning during a prolonged crisis or with little warning during competition. Army forces' preparedness to enter a conflict ultimately depends on decisions and preparations made during competition and crisis.

6-8. During armed conflict, GEOINT elements are deployed to an OE where U.S., and possibly multinational forces, are engaged in large-scale combat operations against a peer or near peer threat force. Due to the high tempo and rapid movement of large-scale combat operations, GEOINT imagery analysts must rapidly process, exploit, and disseminate GEOINT, provide timely, relevant, accurate, predictive, and tailored data for visualization of the AO, and update enemy activities. GEOINT elements must be highly proficient with their intelligence architecture and properly understand relevant U.S. message text format report types. Large visual products do not provide the necessary timeliness, and reliance on these products can lead to disinformation and dated intelligence. Table 6-3 describes the collection focus for GEOINT elements across the levels of warfare (at echelon) during armed conflict.

Table 6-3. GEOINT activities during armed conflict

Level of warfare (echelon of employment)		Collection focus during armed conflict	
Theater strategic (MIB-T)		- Integrated GEOINT divisions serve as liaisons between tactical elements and the national GEOINT community. - Create a commercial imagery collection strategy to leverage unclassified data to support Soldiers and enable multinational sharing solutions. - Supplement supported organization PED capabilities and conduct second and third phase GEOINT analysis to provide big picture context to enemy activity. - Forward deploy MIB-Ts or other theater collectors to support the combatant commander and joint task force commander's information requirements.	
Operational (ASCC and corps)		- Identify and report location of enemy long-range precision fires capabilities and associated radar systems and conduct BDA after target engagement. - GEOINT informs the information collection plan to keep new collection current with activity in the AO. - Prioritize requests for GEOINT collection and exploitation from subordinate organizations to send to theater and national collectors.	

Table 6-3. GEOINT activities during armed conflict (*continued*)

Level of warfare (echelon of employment)	Collection focus during armed conflict
Tactical (E-MIB, division, and BCT)	• Through GEOINT's operational reach and ability to collect against all areas of the enemy depth, the primary GEOINT effort during initial entry operations is identifying enemy equipment represented on the high-payoff target list. • During large-scale combat operations, both the speed and lethality of the fight will necessitate a GEOINT focus on near real-time activity reported using USMTF to close intelligence gaps and answer PIRs. • GEOINT informs the information collection plan to keep new collection current with activity in the AO. • Process, exploit, and disseminate all organic, theater, and national GEOINT capabilities to give maneuver elements the most up-to-date picture of the battlefield. • GEOINT leader is an active member of the targeting working group to ensure PED continuously supports the targeting process, both for target identification and BDA. • Organic GEOINT serves as the liaison with higher and adjacent GEOINT cells to maximize PED efforts and avoid redundant exploitation.
<i>Transition back to competition</i>	• Upon completion of armed conflict or redeployment from an armed conflict mission, GEOINT elements will refocus on their units enduring mission. • Tactical units will refit and return to a focus of mission readiness training. • Operational elements will monitor potential points of escalation around the world. • Strategic elements will continue to conduct GEOINT operations to support theater strategic and national strategic level information requirements.
AO area of operations ASCC Army Service component command BCT brigade combat team BDA battle damage assessment E-MIB expeditionary-military intelligence battalion	GEOINT geospatial intelligence MIB-T military intelligence brigade-theater PED processing, exploitation, and dissemination PIR priority intelligence requirement USMTF United States message text format

Chapter 7

Joint and Multinational Operations and Unique Environments

JOINT AND MULTINATIONAL OPERATIONS

7-1. Modern OEs require that Army forces be able to operate under a variety of circumstances:

- As part of a joint force.
- As part of a combined or multinational force.
- In close coordination with interagency and intergovernmental organizations.

JOINT OPERATIONS

7-2. Most operations occur as part of a coalition effort. Joint and multinational capabilities enable Army forces, and Army capabilities enable joint and multinational forces to achieve objectives. Army forces employ joint capabilities to the degree they are available. However, because peer threats can contest the force in all domains, Army forces must be prepared to conduct operations when some or all joint capabilities are unavailable to support mission accomplishment. Army forces must also account for threat capabilities in all relevant domains.

7-3. The joint intelligence architecture interconnects collectors, producers, and customers in an information network. It provides a dynamic and flexible structure for global access to information from all intelligence sources at all echelons. All intelligence made available to the network from any source is stored and communicated as data (a text file, graphic, imagery, or other formatted information). Standard-compliant file servers that interface with the communications network store this data and associated metadata.

7-4. The joint intelligence operations objective is to integrate Service and national intelligence capabilities into a unified effort. This endeavor should surpass any single organizational effort and provide commanders with the most accurate and timely intelligence. GEOINT support to joint operations includes rapidly responding to worldwide threats by providing georeferenced visual and data products that serve as a foundation and common frame of reference for any joint operation.

Joint Intelligence Architecture Support

7-5. The joint intelligence architecture facilitates support to the joint force commander and subordinate joint force components through the U.S. IC and integrates any required assistance from nondefense agencies and nongovernmental organizations. It provides the baseline data needed to support joint operations and establishes a common way to provide theater and tactical commanders the intelligence required for operations.

7-6. GEOINT support to the joint intelligence architecture includes but is not limited to—

- Creating traditional and specialized hardcopy geospatial products and electronic data.
- Providing safety of navigation products and services.
- Populating and maintaining national databases that provide the visualization and analytical framework to support decision making.
- Managing the acquisition of commercial and foreign government geospatial and remote sensing data for DOD users.
- Maintaining crisis-specific GEOINT products and data on NGA websites posted on SIPRNET to complement direct support activities.

Joint Force GEOINT Support

7-7. GEOINT provides a common framework for supporting joint operations to better enable mission accomplishment across operational themes—from peacetime military engagement to major combat operations. GEOINT support to joint operations is the multidirectional flow and integration of geospatially referenced data from all sources. GEOINT provides shared situational awareness of the OE, near real-time tracking, and collaboration between forces. The GEOINT cell contains geospatial and imagery capabilities

required to oversee GEOINT support to the joint force. The cell at the CCMD coordinates with the joint force GEOINT cell to ensure operational continuity across functions, organization levels, and levels of warfare.

7-8. The joint force GEOINT cell directly communicates with mission partners to define user requirements. It then interfaces with the NSG to obtain the best quality GEOINT and provide it directly to Soldiers. This process satisfies the broad range of GEOINT requirements associated with joint operations. The GEOINT cell supports joint operations by—

- Defining GEOINT requirements.
- Obtaining mission-essential GEOINT.
- Evaluating GEOINT content and form.
- Using and sharing GEOINT.
- Maintaining GEOINT data.

Defining GEOINT Requirements

7-9. The GEOINT cell collects and prioritizes GEOINT mission requirements using a standardized process that defines—

- Mission partners and other trusted data sources.
- Baseline data layers that form the starting point for analysis, visualization, and sharing scenario dependent layers that typically change frequently or are somewhat specialized.
- Search parameters.
- Forms.
- Geographic area coverage requirements.

Obtaining Mission-Essential GEOINT

7-10. The GEOINT cell identifies, validates, and accesses authoritative sources to obtain GEOINT data and resources. They identify geospatial information sources, imagery, and sensors and access unified operations directories, catalogs, or web-mapping services and libraries.

Evaluating GEOINT Content and Form

7-11. Evaluating GEOINT content and form includes conflating, augmenting, and adding value to data; putting data into usable forms and formats; and managing production. Evaluating GEOINT content and form also identifies gaps in coverage, existing sources, and planned-source acquisition and determines if data is available, meets requirements, and is in the required form. If the required intelligence does not exist, or if existing intelligence does not fully meet requirements, the GEOINT cell may need to generate new intelligence. The cell may then monitor and track provider capabilities and distribution channel capacity.

7-12. The GEOINT cell uses standardized processes that allow the joint Soldier to decide the most effective approach to meeting requirements within timelines. To meet requisite timelines, the GEOINT cell may leverage NSG GEOINT collection assets by—

- Submitting RFCs to NGA via CCMDs and the Defense Intelligence Agency in accordance with NSG processes, ensuring requirements are documented in the appropriate GEOINT system.
- Collecting geospatial data using available GEOINT collection assets and exploiting the data to extract required information.
- Requesting CCMD support in tasking other NSG GEOINT collection assets.

Using and Sharing GEOINT

7-13. GEOINT's ultimate objective in joint operations is enabling the Soldier to complete the mission more efficiently. GEOINT elements share information with the NSG, CCMDs, and across the JTF (down to the lowest tactical level) to achieve shared awareness of the OE. GEOINT developed below the JTF level is made available to the JTF and CCMD to augment the theater geospatial database and NSG library holdings. Theater-level storage, retrieval, and production capability is required for all GEOINT components.

Maintaining GEOINT Data

7-14. As changes and updates are received, current, relevant imagery and geospatial information must be maintained and continuously validated for accuracy and consistency. Data element changes must be conflated with current data and shared across forces to maintain consistency and a shared awareness of the OE. The GEOINT cell, in conjunction with NGA, may—

- Receive updated imagery and geospatial information.
- Purge databases and archive deleted holdings as necessary.
- Establish version control and naming conventions.
- Check reliability, availability, metadata, and authentication.
- Monitor and track holdings throughout the command to ensure use of the most current and relevant imagery and geospatial information in accordance with mission requirements.

MULTINATIONAL OPERATIONS

7-15. *Multinational operations* is a collective term to describe military actions conducted by forces of two or more nations, usually undertaken within the structure of a coalition or alliance (JP 3-16). The multinational OE is unique due to the mission's nature. The purpose of the operation and participants involved determine what GEOINT products are needed. Multinational operations require interoperable GEOINT data, applications, and data exchange capabilities. Participants should agree to a standard datum and ensure all products use it. A multinational GEOINT plan must coordinate all products for member force use and blend assets into a cohesive production program with consideration for disclosure and release issues as well as accessibility approval and procedures.

7-16. GEOINT products that support multinational operations include but are not limited to—

- MTIs that provide warning intelligence.
- Planning maps, nautical and aeronautical charts, imagery, and digital data.
- Preassembled packages of select maps, charts, and other geographic materials of various scales to support planning and conducting noncombatant evacuation operations in specific countries or areas.
- Aeronautical products that include aim point graphics, automated air facilities information files, aeronautical charts and graphics, and airfield products.
- Nautical or hydrographic products that include digital nautical charts, digital bathymetric databases, hydrographic charts, and port graphics.
- Geospatial products that include CIB, DTED, shuttle radar topography mission elevation data, mission specific datasets, tactical terrain data, and image city maps.
- Detailed characteristics of potential detention facilities, recovery sites, and other operationally significant features.
- Precision positioning, collateral damage assessment, targeting, and battle damage assessment (BDA) products.
- Proper product release authorities.

UNIQUE ENVIRONMENTS

7-17. GEOINT supports operations in unique physical environments:

- Urban.
- Cold weather and mountain.
- Jungle.
- Desert.
- Littoral.

URBAN

7-18. Units conduct reconnaissance of urban areas to gain information on objectives and provide timely, accurate, and relevant information to their higher headquarters. Depending on time available, units develop the urban situation progressively from the surrounding areas toward the city. The commander refines objectives and routes upon analyzing information. Using data obtained from GEOINT collection assets, units can request airborne GEOINT collection asset support to conduct aerial reconnaissance of routes within their objective area. Airborne GEOINT collection assets can focus on entry points to assist units in developing the

environment—such as obstacles, rooftop ambush positions, or movement of threat personnel and vehicles—as friendly units approach.

Finding Snipers

A GEOINT cell receives a request from the S-3 to analyze an urban area for potential enemy sniper positions. The cell officer in charge tasks geospatial engineers to conduct LOS analysis and identify AAs and mobility corridors. The engineers generate LIDAR products and reverse LOS products to determine the most likely locations of past activities as well as potential future sniper positions. GEOINT imagery analysts collect high-resolution EO imagery and exploit MTI data to assess activity trends and determine chokepoints along the routes. The GEOINT cell produces three-dimensional models showing potential target areas and positions from which an enemy sniper could strike. The models provide the staff, patrol leaders, and convoy commanders an effective tool that reduces risk on the ground.

7-19. Units orient on the objective by occupying or observing key terrain and threat AAs. Reconnaissance platoons use sensors and optics to conduct long-range surveillance and locate threat positions and vehicles before forces enter an urban area. The commander compiles and analyzes information collected by the platoons. The commander then assesses the AO according to the mission and commander intent. Information requirements tasked to GEOINT cells to support urban operations often include but are not limited to the following:

- Threat composition and activity.
- Areas of vulnerability to friendly forces.
- Key terrain.
- Approach routes for mounted and dismounted forces.
- Entry points or points of penetration.
- Support positions for direct and indirect fire assets.
- Civilian disposition.
- Urban area density and composition.
- Hazard areas (such as fuel storage, natural gas lines, and chemical, biological, radiological, nuclear, or explosives research, development, testing, production, or storage sites).
- Key infrastructure (communications facilities, water treatment facilities, religious, or culturally sensitive sites).

COLD WEATHER AND MOUNTAIN

7-20. Many tasks required to operate successfully in cold weather also apply to mountain regions. Operating in mountainous terrain, however, requires specialized training. Factors such as slope, soil composition, and surface configuration differentiate mountain operations from cold weather operations. Information requirements tasked to GEOINT cells to support cold weather or mountain operations often include but are not limited to the following:

- OAKOC factors (military aspects of terrain).
- Slope.
- Soil composition.
- Surface configuration.
- Foliage density.

Observation Posts

A GEOINT cell receives a request from the S-3 to analyze several proposed landing plans and operations in a mountain region. The proposed sites overlook a valley and are in sparsely populated areas suspected to be frequented by enemy forces.

The cell officer in charge directs geospatial engineers to conduct LOS analysis and identify AAs, mobility corridors, and potential threats at each proposed landing plan and operation site. The engineers use topographic maps, LIDAR, and other tools to provide LOS data from each landing plan and operation, as well as data on how visible each landing plan and operation is from the enemy perspective. The imagery section reviews FMV and national and commercial imagery of each proposed site to identify vertical obstacles and potential hazards. After analyzing MTI, GEOINT imagery analysts identify high-traffic areas.

UASs are tasked to collect the most up-to-date information and assist GEOINT imagery analysts in providing flythroughs to the S-3. The final GEOINT products assist the staff in emplacing landing plans and operations to provide the best information while avoiding detection by enemy forces.

JUNGLE

7-21. Units operating in a jungle environment may need highly accurate latitude, longitude, and elevation positions for targets, navigation, and other requirements that are frequent in jungle areas and featureless on maps. GEOINT can assist units in jungle areas by providing products that support fires. Information requirements tasked to GEOINT cells to support jungle operations often include but are not limited to route analysis, image maps, survey (ground truth) maps, and foliage penetration.

DESERT

7-22. Key terrain in the desert largely depends on restrictions to movement. If the desert floor does not support wheeled vehicle traffic, roads and desert tracks become key terrain. Crossroads are vital as they control military operations in a large area. Desert warfare is often a battle for control of LOCs. Water sources are vital, especially if a force is incapable of long-distance resupply of its water requirements. Information requirements tasked to GEOINT cells to support desert operations often include but are not limited to—

- Infrared heat maps to identify man-made activity.
- Terrain-based vehicle trafficability assessments.
- Change detection analysis.
- Weather penetrating SAR imagery.

LITTORAL

7-23. GEOINT collaborates with the U.S. IC to ensure information and analytical capabilities are leveraged to improve maritime domain awareness. GEOINT (combined with all-source capabilities and traditional sources and methods) supports the U.S. Coast Guard by providing unprecedented maritime domain awareness of vessels, people, cargo, weather, environment, and infrastructure.

7-24. GEOINT products that operationally leverage the U.S. Coast Guard provide situational awareness to Coast Guard commanders, planners, and operators before and during coastal operations. Information requirements tasked to GEOINT cells to support coastal operations often include but are not limited to—

- Products that define the AO.
- Visual intelligence.
- Three-dimensional models of ports.
- Locations of vessels in search operations.
- Port closures and waterway conditions.
- Locations of near-surface obstructions.

River Crossing

In preparation for possible river crossing operations, a GEOINT cell receives a request for a product containing river data. The GEOINT or geospatial engineering technician tasks geospatial engineers to collect and process hydrology and vector data to show soil composition at the riverbanks, vegetation growth, and slopes leading to and from the river, as well as to identify depth and underwater obstacles.

GEOINT imagery analysts collect and analyze MTI data to determine crossing points used by locals. They analyze current imagery and combine it with geospatial data to provide a better visualization of potential crossing points. UASs obtain the most up-to-date imagery of the area before the operation. By combining data derived from GEOINT, the staff ascertains the most accessible river crossing point and the location at which enemy forces will most likely conduct waterborne operations.

Appendix A

Imagery Types and Characteristics

OVERVIEW

A-1. GEOINT collection assets gather data across the EMS, using a variety of sensors to produce different types of imagery. The characteristics of each imagery type determine how it is used.

Note. This appendix uses terms from NSG’s *Geospatial Intelligence (GEOINT) Basic Doctrine, Publication 1.0* (hereinafter referred to as Pub 1.0) to describe imagery sensors and types. The terms do not appear in the Glossary.

IMAGERY ALONG THE ELECTROMAGNETIC SPECTRUM

A-2. The EMS is the entire range of electromagnetic radiation frequencies. It is a maneuver space essential for facilitating control within the OE, affecting all portions of the OE and military operations. Military operations and training are executed in an environment complicated by increasingly challenging demands and constraints on the EMS. Just as in the physical domains and in cyberspace, military forces maneuver and conduct operations within the EMS to achieve tactical, operational, and strategic advantages. Freedom of maneuver and action within the EMS are essential to U.S. and multinational operations. (For doctrine on the EMS, see JP 3-85.)

A-3. The EMS comprises many subranges (referred to as portions) that are named based on the behavior of waves’ emission, transmission, and absorption and on their applications. Boundaries between contiguous portions of the EMS are undefined, so the ranges overlap. The EMS—from the lowest frequency to the highest (longest to shortest wavelength)—includes radio waves, infrared radiation, visible light, ultraviolet radiation, x rays, and gamma rays.

A-4. GEOINT systems generally collect data in the following EMS ranges (exact values depend on the sensor) (see figure A-1):

- **Visible light** (400–700 nanometers) is the visible portion of the EMS.
- **Near infrared (also known as NIR)** (700–1,100 nanometers) is used primarily for imaging vegetation and detecting camouflage.
- **Short-wave infrared (also known as SWIR)** (1,100–3,000 nanometers) is used for imaging vegetation, soil moisture content, forest fires, geological features, silicates, and clays.
- **Mid-wave infrared (also known as MWIR)** (3,000–5,000 nanometers) is used for night studies, imaging geological structures, and detecting thermal differences in water currents, fires, and effluents (uses emitted radiation and some reflected radiation during daytime imaging).
- **Long-wave infrared (also known as LWIR)** (7,000–14,000 nanometers) is used for environmental studies and for imaging subtle differences in temperature between objects, gas plumes, and effluents using emitted energy.
- **Microwave** (1 millimeter–1 meter) is used for radar, SAR, and related technologies; useful for mapping terrain, day and night all-weather imaging, detecting objects under canopy, and communication paths.

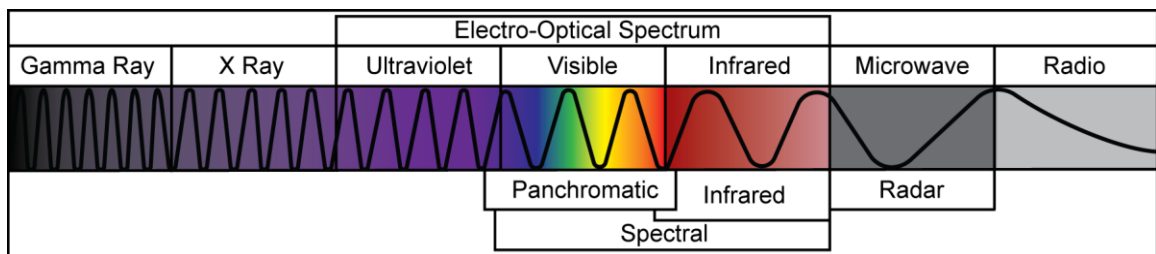


Figure A-1. Portions of the electromagnetic spectrum used for GEOINT collection

IMAGERY SENSORS

A-5. All forms of imaging occur through interaction between a sensor and the object being observed. Some GEOINT collection assets can be configured to carry a variety of sensor packages. Each sensor and GEOINT collection asset have a unique capability with distinct advantages and disadvantages. The requirements manager must understand each capability to select sensors and GEOINT collection assets that best meet requirements, allowing analysts to answer intelligence requirements and provide quality GEOINT products. There are two types of imagery sensors: active and passive.

ACTIVE SENSORS

A-6. *Active sensors* are sensors that generate the electromagnetic energy needed to illuminate the object that is being imaged. They broadcast energy and process returning portions of it in order to image a targeted area. Active imagery sensing capabilities include real-aperture radar, synthetic aperture radar, interferometric synthetic aperture radar, moving target indicator, and light detection and ranging systems (Pub 1.0).

PASSIVE SENSORS

A-7. *Passive sensors* are sensors that use natural electromagnetic energy sources such as the Sun, naturally occurring radiation, or objects that generate heat. They generate data by capturing the reflected or emitted electromagnetic energy from an object. Passive sensors, which operate in *receive* mode and do not emit energy, diminish the adversary's ability to detect the fact that they are being imaged. Electro-optical sensors (except light detection and ranging) are passive (Pub 1.0).

IMAGERY CHARACTERISTICS

A-8. Imagery characteristics include imagery sources, imagery resolution, and imagery types.

IMAGERY SOURCES

A-9. Imagery sources include—

- Satellite systems:
 - National satellite systems.
 - Commercial or civil satellite systems.
- Airborne systems.
- Terrestrial systems:
 - Ground reconnaissance.
 - Unattended ground sensors.
 - PAI.

Satellite Systems

A-10. Satellite systems have been a critical component of imagery collection since their inception in the 1960s. The NRO is a U.S. IC organization responsible for developing, launching, and operating the Nation's signals, imagery, and communications intelligence satellites to meet U.S. security needs.

A-11. Domestic imagery collected by national reconnaissance assets and commercial or civil imaging satellites operates under various legal and policy limitations. NGA is responsible for imagery dissemination, and the appropriate departmental requirements office is responsible for managing and clarifying domestic imagery requirement guidance and policy. Limitations on the use of domestic imagery derive from several sources:

- The U.S. Constitution.
- Executive Order 12333 as amended.
- Goals, direction, duties, and responsibilities with respect to the national intelligence effort.
- The National Security Act of 1947 as amended.
- Other applicable laws and policy directives.

Note. Users must be aware of legal and policy concerns associated with domestic imagery, particularly if they pertain to private property. Individuals may be held responsible for any violation of law or inappropriate use of domestic imagery.

National Satellite Systems

A-12. National satellite systems provide timely, relevant, and accurate imagery and GEOINT to support national policymakers, military forces, and civil users. They are a primary source of GEOINT imagery. Imagery collected from national satellite systems is classified and requires release authorization, thereby increasing dissemination timelines. NGA has authorized the release of some national satellite system data at the Controlled Unclassified Information (also called CUI) level. National satellite imagery and products are disseminated via classified and unclassified DOD information networks.

Commercial or Civil Satellite Systems

A-13. Commercial satellite systems collect GEOINT data, including but not limited to panchromatic, hyperspectral, multispectral, video, infrared, and SAR data. Commercial satellite systems operated by U.S. and foreign vendors increasingly contribute GEOINT data and products to support NSG requirements. These systems enhance GEOINT support to multinational operations because the raw data they produce is unclassified and can be disseminated to multinational partners, reducing operations security concerns.

Note. A bilateral exchange and cooperation agreement is required before exchanging any commercial data acquired from NGA, including unclassified data. Commercial or civil imagery may be unclassified but depicting or adding classified data classifies the product. Additionally, commercial satellite imagery may be protected by copyright laws that could hinder dissemination.

A-14. Commercial satellite systems have been augmenting national satellite systems more and more due to their—

- Growing quantity.
- Rapid development of new systems.
- Constantly improving spatial resolution. For example, they assist in planning civil and military support to humanitarian efforts, natural disasters, and special security events.

A-15. Army National Guard units must use ADRO to obtain commercial imagery. Accounts and routing schemes are available to units that have access to GIMS.

Note. Some states have given consent to collect overhead imagery. Neither ADRO nor NGA can grant this consent.

CAUTION

While commercial or civil imagery is unclassified, exercise caution when requesting and obtaining imagery from commercial or civil organizations for operations security reasons. Acquiring commercial or civil imagery is generally not a secure process, and the imagery collected can be placed in a publicly available archive for resale by the vendor.

Airborne Systems

A-16. Airborne systems, such as aerial GEOINT collection assets, include all manned aircraft and UASs that collect still and motion imagery, including but not limited to panchromatic, thermal, multispectral, hyperspectral, and LIDAR. Airborne systems can locate and recognize major threat forces, moving vehicles, weapon systems, and other targets that contrast with their surroundings. They can also locate and confirm the position of friendly forces. Current airborne system missions include intelligence, reconnaissance, surveillance, security, cooperative engagements, and communications relay.

A-17. Airborne GEOINT systems' numerous capabilities provide near real-time reconnaissance, surveillance, and target acquisition, as well as reconnaissance and attack resolution. These systems—

- May be equipped with laser designators to mark targets (some systems may also be armed).
- Support target acquisition efforts and lethal attacks on threat reconnaissance and advance forces.
- Locate threat forces and assist in determining their composition, disposition, and activity.
- Maintain contact with threat forces from initial contact through BDA.
- Provide accurate target coordinates that enable immediate target handover and first-round fire-for-effect engagements.
- Provide information to other airborne systems, thus increasing survivability.
- Perform decoy, demonstration, feint, and deception operations.
- Support organic as well as higher, adjacent, and subordinate collection requirements.

Terrestrial Systems

A-18. Terrestrial systems consist of digital images (photographs or video). GEOINT imagery analysts use these images to visually enhance their products, depicting high-resolution and perspective views of equipment, buildings, or facilities. NGA maintains an extensive ground photo library to support this effort.

A-19. Terrestrial images are produced through ground reconnaissance, unattended ground sensors, and PAI. Terrestrial image examples include but are not limited to—

- Reconnaissance photographs.
- Hand-held photographs taken from ground or low-flying aircraft personnel.
- Security camera and closed-circuit television videos.
- Images captured by the Internet of Things (also known as IoT).

Note. For this publication, terrestrial systems do not include hand-held or clandestine photographs taken by or on behalf of human intelligence collection organizations. (See Pub 1.0.)

Ground Reconnaissance

A-20. Digital images taken of physical areas during ground reconnaissance can provide additional details for general planning, information collection, and targeting. Units should include these images in their imagery product library.

Unattended Ground Sensors

A-21. Unattended ground sensor systems produce imagery or activity reports using various sensor modalities, including seismic, acoustic, magnetic, and pyroelectric transducers; daylight imagers; and passive infrared imagers. These modalities automatically detect the presence of people or vehicles and transmit data via radio frequency or satellite communications links to a remote PED station. The systems are packaged for concealed emplacement in the field and for long-duration unattended operations.

Publicly Available Information

A-22. Digital images taken by news organizations, civil governments, and other groups that post or store images on the internet can provide additional details supporting the development of GEOINT products. These images fall under the category of PAI, which is another source for terrestrial images used by GEOINT imagery analysts.

Note. Users must not violate copyright laws when using PAI.

IMAGERY RESOLUTION

A-23. Imagery resolution is a set of parameters that determines the overall quality of an image. Resolution may also be referred to more specifically as radiometric resolution, temporal resolution, spectral resolution, and spatial resolution.

Note. While the U.S. IC tracks resolution in feet and inches, the Army converts it to meters for reporting consistency.

Radiometric Resolution

A-24. Radiometric resolution describes how precisely a system can represent or distinguish differences of intensity. It is usually expressed as bits or levels. For example, 8 bits or 256 levels is typical for computer image files. In theory, higher radiometric resolution results in more discernable subtle differences in intensity or reflectivity. In practice, however, effective radiometric resolution is typically limited by noise levels rather than by the number of bits represented.

Temporal Resolution

A-25. Temporal resolution refers to the frequency at which images are recorded. Temporal resolution for motion imagery is expressed in frames per second.

Spectral Resolution

A-26. Spectral resolution describes a sensor's ability to define finite wavelength differences. The finer the spectral resolution, the narrower the wavelength range for a channel or band. Spectral resolution is determined by the number and width of bands and their location along the EMS.

Spatial Resolution

A-27. Spatial resolution is expressed in terms of ground sample distance for EO systems and impulse response for radar collections:

- **Ground sample distance** refers to the specific measurements between two objects required for a sensor to distinguish them as independent objects. These measurements are limited to passive EO collection systems.
- **Impulse response** is the term SAR uses to express measurements similar to ground sample distance. While positioning is critical to determining ground sample distances, SAR image scale and resolution are largely determined by bandwidth (wavelength) and aperture size (dwell time).

IMAGERY TYPES

A-28. There are two types of imagery:

- **Still imagery** refers to an individual image of an object or location.
- **Motion imagery** refers to a series of images collected sequentially at varying frames per second.

Still Imagery

A-29. Still imagery provides information about specific areas or objects at varying levels of detail. Sensors may collect still imagery in various formats, including but not limited to the following:

- Panchromatic.
- Polarimetric.
- Infrared.
- Spectral (multispectral and hyperspectral).
- LIDAR.
- SAR.
- OPIR.

Panchromatic

A-30. *Panchromatic* uses the visible part of the electromagnetic spectrum, along with a portion of infrared spectrum, and produces an image similar to a black and white aerial photograph. The images can provide a very detailed and accurate likeness of an area or object, and can show shadows and other effects that help identify characteristics of an object or environment (Pub 1.0).

A-31. A sensor that is sensitive to all or most of the visible spectrum acquires a panchromatic or visible image. Panchromatic imagery is black and white, displayed as a grayscale image, and generally has a higher spatial resolution than infrared or spectral imagery. Panchromatic imagery can be merged with other imagery data sources. CIB is an example of a standard output format that uses panchromatic imagery. Numerous C2 systems throughout the Army use CIB (see paragraph 3-5).

Polarimetric

A-32. Polarimetric imagery measures the polarization states of energy coming from all points in a scene. This information can assist in classifying materials and identifying objects of interest for remote sensing and military applications. Polarization also provides useful information about the surface properties of an object. Fabricated objects have unnaturally smooth surfaces that emit and reflect radiation that is more polarized. Natural objects such as grass, trees, dirt, and sand emit and reflect radiation that is less polarized.

Infrared

A-33. *Infrared* is an electro-optical phenomenology that uses the infrared part of the spectrum to detect heat and radiance/reflection, and displays that information in a gray-scale image (Pub 1.0). The different shades of gray represent differences in temperatures and emissivity of objects in the image.

A-34. Infrared imagery comprises one of three types:

- **Mid-wave infrared** is used for GEOINT thermal infrared imagery. It combines infrared reflective and emissive data with the highest spatial resolution for thermal imagery. Mid-wave infrared imagery can be collected day or night, but reflectance properties may skew thermal data collected during the day.
- **Long-wave infrared** can be used to obtain the greatest amount of pure emittance data. Long-wave infrared imagery can be collected day or night and is not subject to reflectance properties. The spatial quality of long-wave infrared suffers substantially, however, due to the associated longer wavelengths.
- **Near infrared and short-wave infrared** are used to primarily identify soil and vegetation reflective properties. They are only collected during the day because the Sun or another strong illumination source needs to be apparent upon the target.

A-35. Applications for infrared imagery include—

- Characterizing foliage.
- Detecting camouflage.
- Assessing crop health.
- Determining a land-water boundary.
- Analyzing coastal hydrography.
- Analyzing cloud cover and cloud heights.
- Determining target background contrast and thermal crossover.

Note. Military applications for infrared imagery include target acquisition, surveillance, reconnaissance, night vision, homing, and tracking.

Spectral

A-36. A series of sensors sensitive to specific wavelengths (usually grouped in bands) collects spectral imagery. Adding or subtracting different bands generates color imagery. Spectral imagery is categorized into two groups mainly distinguished by the number and width of spectral bands present: multispectral and hyperspectral.

A-37. *Multispectral imaging* is an electro-optical phenomenology that provides a color image by using tens of additional bands of the visible and infrared portions of the spectrum. It also provides more detail to images, showing characteristics that are not otherwise visible (Pub 1.0). The bands can be combined in various ways to highlight different man-made or natural features.

A-38. *Hyperspectral imaging* is an electro-optical phenomenology similar to multispectral imaging but uses hundreds, rather than tens, of bands of the visible and infrared portions of the electromagnetic spectrum. These extra bands can provide greater fidelity and characteristics than multispectral imaging. For example, hyperspectral imagery allows users to detect materials by their unique spectral signature (Pub 1.0).

A-39. Spectral imaging collects multiple bands across the EMS to sense reflective and emissive values of matter within an area. The higher the band's resolution, the more capable the sensor is of identifying specific details like rock or soil types, camouflage net manufacturers, and vegetation. Spectral imaging and reflectance value libraries provide signatures that resolve a range of matters, such as distinguishing between real aircraft and decoys, characterizing substances by various types of emissions, and detecting vehicles based on their unique signatures.

Light Detection and Ranging

A-40. *Light detection and ranging* is an electro-optical phenomenology that uses laser pulses in the visible and infrared portions of the electromagnetic spectrum. The sensor emits a pulse of laser light toward an object, which reflects back some of the energy to the sensor. The time it takes for the energy to return provides elevation values—shorter roundtrips indicate higher elevations and longer roundtrips indicate lower elevations. This process helps measure distances to, heights of, and other characteristics of an object. The process can be used to develop three-dimensional models of an object, such as a building (Pub 1.0). LIDAR is an active sensor, similar to radar, that includes scanning and nonscanning systems capable of three-dimensional imaging.

Synthetic Aperture Radar

A-41. *Synthetic aperture radar* is a radar phenomenology that illuminates an object with pulses of microwave energy. It applies signal processing to a series of many pulses to produce imagery (Pub 1.0). SAR imagery detects and identifies activity based on reflected radar energy. SAR systems emit and collect radar energy, recording returned energy to create a history of frequency shifts (known as a phase history) as they move in relation to the desired target. These shifts are reprocessed to create an image with consistent resolution, whether objects are closer to or farther from the sensor. Due to the sensor's active nature, SAR imagery can illuminate objects day or night and in most weather conditions, providing tactical commanders with imagery that complements visible and infrared sensor capabilities.

A-42. Processed radar data is depicted in a map-like format, enabling the use of unprocessed or complex radar data to collect and automatically detect motion within a single radar image and automatically detect change between two radar images. SAR imagery is nonliteral, which means it does not appear as a photo that shows an exact likeness of an object or environment. It identifies characteristics of an area or object that a literal image would not show, similar to an x ray.

A-43. *Interferometric synthetic aperture radar* is a phenomenology that uses a synthetic aperture radar sensor to observe a location from two separate positions. This method generates elevation data of the Earth's surface (Pub 1.0).

Overhead Persistent Infrared

A-44. *Overhead persistent infrared* is, within geospatial intelligence, a capability that provides on-demand, persistent, global, and/or localized coverage of high- to low-intensity infrared events to detect energy radiation from various tactical to strategic objects (JP 2-0). It is a defense program of remotely sensed data provided in large spatial samples. According to Pub 1.0, *overhead persistent infrared* sensors operate in the visible and near infrared, short-wave infrared, and mid-wave infrared portions of the spectrum to characterize this energy as an event or process it as a scene. Overhead persistent infrared sensors provide persistent (that is, continuous) coverage of mission areas and energy radiation detection from both tactical and strategic targets including missiles launches and events, wildfires, enemy hostilities, weather, volcanos, and weapons characterization (Pub 1.0).

Motion Imagery

A-45. Motion imagery is a valuable tool for ongoing operations and surveillance. Collecting motion imagery from sensors mounted on airborne GEOINT collection assets with long-mission endurance allows GEOINT imagery analysts to monitor high-interest activities in the OE. It also allows analysts to track moving, fleeting, and emerging targets. Near real-time FMV enables observation of rapidly developing events.

A-46. According to Pub 1.0, in *motion geospatial intelligence*, both electro-optical and radar sensors can create or capture motion by collecting two or more frames per second. The most common types of motion geospatial intelligence are full-motion video, moving target indicator, and wide area motion imagery (Pub 1.0).

A-47. A motion imagery system, as described by the Motion Imagery Standards Board, is any imaging system that collects at a rate of one frame per second (1 hertz) or faster over a common field of regard. This includes but is not limited to panchromatic, infrared, multispectral, and hyperspectral systems. Motion imagery, as with still imagery, is collected in various portions of the EMS.

Note. The Motion Imagery Standards Board is the official standards body responsible for reviewing and recommending standards for motion imagery, associated metadata, audio, and other related systems for use within the NSG and DOD. The Motion Imagery Standards Board makes no formal distinction between the terms motion imagery and FMV. Some entities call video with no metadata FMV, but motion imagery must contain metadata. Historically, FMV has been the subset of motion imagery at television-like frame rates (24 to 60 hertz). (Obtain information on the Motion Imagery Standards Board through the Geospatial-Intelligence Standards Working Group website on NIPRNET.)

Full-Motion Video

A-48. *Full-motion video*, typically an electro-optical capability, uses many still frames per second, creating a near-real-time video capability, which may be displayed as visible or infrared and may be recorded by digital or analog means. This near-real time capability allows observation of rapidly developing events and is a valuable tool for ongoing operations. It is also useful for continuous observation of an object to analyze its patterns over a long period of time (Pub 1.0). FMV enables commanders to make decisions and execute missions from a safe distance without endangering their troops. Aerial GEOINT collection assets can transmit a direct video feed to a nearby ground control station or broadcast video via satellite to command centers around the globe. They provide an on-demand, near real-time, close-up view of the combat zone that would not otherwise be possible.

Moving Target Indicator

A-49. *Moving target indicator* is an electro-optical and radar capability that shows only those objects that are in motion, and provides information on the speed and direction that objects are moving (Pub 1.0). This capability provides commanders with better situational awareness of the AO.

A-50. MTI imagery quality considerations include target range resolution and temporal resolution:

- **Target range resolution** is sometimes referred to as high-range resolution in the case of MTI imagery. It determines whether two or more targets moving in proximity will be detected as individual targets.
- **Temporal resolution** refers to the frequency with which a radar beam passes over an area. Frequent revisits allow radar to achieve track continuity and increase the likelihood of target detection by reducing the probability of obscuration from screening by trees, buildings, or other objects.

Wide Area Motion Imagery

A-51. *Wide area motion imagery* is a set of motion imagery sensors with a wide field of view diameter of up to 8 kilometers, which provides persistent surveillance of a broad area of coverage. Wide area motion imagery data enables forensic backtracking of kinetic events, network analysis, pattern of life studies, and security monitoring (Pub 1.0). Wide area motion imagery (WAMI) also refers to a surveillance, reconnaissance, and

intelligence-gathering capability that uses specialized software and a powerful system that is usually airborne and operates for extended periods of time. It detects and tracks movement out in the open throughout a city-sized area kilometers in diameter. For this reason, WAMI is sometimes referred to as wide-area airborne surveillance or wide-area persistent surveillance.

A-52. A WAMI sensor captures an entire area in real time, recording and archiving imagery for real-time and forensic analysis. WAMI can perform the work of a larger workforce; it identifies activity otherwise missed by standard sensors with narrower fields of view, analyzes these activities in context, and distinguishes threats from normal patterns of behavior. WAMI is typically used for force protection, security, route reconnaissance, counterterrorism, disaster response, traffic pattern analysis, wildlife protection, and law enforcement.

Imagery Type Characteristics

A-53. Table A-1 lists imagery types and their characteristics.

Table A-1. Characteristic matrix of imagery types

<i>Still imagery</i>		
<i>Imagery type</i>	<i>Advantages</i>	<i>Disadvantages</i>
Panchromatic (passive) <i>Best tool for—</i> - Daytime. - Clear weather. - Detailed analysis. - Single-frame acquisition.	- Affords a familiar view of a scene. - Offers system spatial resolution unachievable in other optical systems or in thermal images. - Preferred for detailed analysis and mensuration. - Offers stereoscopic viewing. - Less susceptible to denial attacks.	- Is restricted by terrain and vegetation. - Is limited to daytime use. - Produces degraded imagery in other than clear weather. - Is subject to denial and deception (camouflage).
Infrared (passive) <i>Best tool for—</i> - Nighttime. - Clear weather. - Detection of human activity. - Forensic analysis.	- Less susceptible to denial attacks. - Includes camouflage penetration. - Has nighttime imaging capability.	- Is ineffective during thermal crossover periods. - Is not easy to interpret, requires skilled analysis. - Cannot penetrate clouds.
Multispectral (active) <i>Best tool for—</i> - Acquiring spectral signatures used in geospatial product generations. - Frequent use when standard mapping, charting, and geodesy products are not available. Note. Can be used for bathymetry.	- Is easily interpretable. - Provides naturally colored images of a desired site. - Provides context to otherwise black and white panchromatic imagery. - Provides wide-area coverage. - Provides detailed analysis, including spectral signature and material category information. - Provides spectral capability from both aircraft and space-based geospatial intelligence (GEOINT) collection assets.	- Has lower spatial resolution than panchromatic imagery. - Is limited to daytime use. - Produces degraded imagery in other than clear weather.
Hyperspectral (passive) <i>Best tool for—</i> - Differentiating material types. - Detecting and identifying plumes. - Search and rescue. - Producing cues for high spatial resolution spot collection sensors. Note. Used for ecology, mining, and denial and deception (camouflage).	- Is easily discernible. - Is preferred for GEOINT application as compared to multispectral imagery. - Provides detailed analysis, including spectral signature and material category information. - Provides spectral capability from both aircraft and space-based GEOINT collection assets.	- Usually has poorer spatial resolution than panchromatic or multispectral imagery. - Produces degraded imagery in other than clear weather. - Has large cubes of data that inhibit rapid or wide-area coverage. - Has bands that are more sensitive to noise than multispectral or panchromatic imagery.

Table A-1. Characteristic matrix of imagery types (*continued*)

<i>Still imagery (continued)</i>		
<i>Imagery type</i>	<i>Advantages</i>	<i>Disadvantages</i>
Light detection and ranging (LIDAR) (active) <i>Best tool for—</i> • Providing high-spatial resolution, three-dimensional imaging for either three-dimensional visualization or digital elevation mapping. • Using three-dimensional overlay on panchromatic; red, green, blue; and visible imagery. Note. Can use in shallow water for bathymetry.	• Has good spatial resolution in both ground sampling and range accuracy. • Provides three-dimensional target imaging regardless of material properties. • Collects in the visible spectrum, near infrared, and short-wave infrared. • Is for daytime and nighttime use. • Can locate man-made and natural targets (dismounts).	• Has limited area acquisition. • Has standoff limitations. • Is not all-weather imaging. • Is heavily absorbed by water.
Radar (active) <i>Best tool for—</i> • Detecting objects at night and in bad weather. • Including synthetic aperture radar (SAR) still frame imagery.	• Is all-weather; penetrates fog, haze, clouds, and smoke. • Is for daytime and nighttime use. • Does not rely on visible light or thermal emissions. • Is the best sensor for change detection (good standoff capability). • Provides large area coverage.	• Is not easily interpretable; does not produce a literal representation of the imaged area. • Requires skilled analysis. • Does not easily allow positive identification or classification of equipment. • Has terrain masking.
Interferometric SAR (active) <i>Best tool for digital elevation extraction for wide areas to meet nighttime collection requirements.</i>	• Acquires accurate terrain elevation in real time. • Includes active, all-weather acquisition. • Maps large areas.	• Terrain posting and range accuracy. • Not as fine as LIDAR.
<i>Motion imagery</i>		
<i>Imagery type</i>	<i>Advantages</i>	<i>Disadvantages</i>
Full-motion video (FMV) (passive) <i>Best tool for—</i> • Daytime (red, green, blue) or nighttime (infrared; depends on if imagery is red, green, blue, or infrared FMV). • Clear weather. • Detailed temporal analysis. • Persistent surveillance. • Forensic analysis.	• Red, green, blue FMV offers color over black and white panchromatic still imagery. • Infrared FMV offers nighttime persistent imaging. • Includes temporal information. • Allows near real-time tracking. • Future systems offer high-definition imaging. • Less susceptible to denial attacks.	• Has a limited field of view. • Is restricted by terrain and vegetation. • Red, green, blue FMV is limited to daytime only. • Produces degraded imagery in other than clear weather. • Airborne GEOINT collection assets are affected by weather.
Moving target indicator (MTI) (active) <i>Best tool for—</i> • Locating, tracking, classifying, and identifying moving targets. • Target profiling based on speed. • All weather. • Daytime or nighttime.	• Is based on radar technology, providing all-weather, nighttime, and wide-area surveillance capabilities. • See radar advantages.	• Requires automatic processing for tracking and analysis. • See radar disadvantages.
Wide area motion imagery (active) <i>Best tool for—</i> • Daytime. • Clear weather. • Wide-area event detection. • Forensic analysis.	• Offers near real-time, persistent surveillance. • Allows large field-of-view event detection, providing big picture for intelligence preparation of the operational environment. • Enables event tracking from start to finish.	• Enables limited wide-area persistent surveillance analysis and requires no formal training. • Has limited tools for automated exploitation. • Produces large datasets, making dissemination and storage difficult.

Appendix B

Policy Considerations

OVERVIEW

B-1. Performing GEOINT operations may require various types of approvals depending on location, mission, organization, or other aspects of the OE. As the functional manager for GEOINT, NGA will typically manage GEOINT mission approval based on current policies. Any GEOINT operation requiring approval from NGA includes a formal request process that must be vetted and validated.

DOMESTIC IMAGERY REQUIREMENTS

B-2. When requesting domestic imagery from NGA for purposes other than intelligence, NSG members must justify their request based on identified mission requirements. The subsequent use of such imagery for intelligence purposes requires an NDL account (see paragraph 3-66), obtained through the ADRO, which manages Army domestic GEOINT policies and access (see paragraph 2-19).

DOMESTIC IMAGERY REQUEST

B-3. Domestic imagery collection or access requests should include a pertinent justification that—

- Provides a description of the requirement and the geolocation of the requested areas (including a concept of operations, if possible).
- Identifies exempted mission area being supported.
- Identifies mission duration with a start and end date.
- Specifies whether the mission requires temporary or ongoing access.
- Provides NDL account user information.
- Identifies the classification level of network access the user requires.

DOMESTIC IMAGERY RELEASE DOCUMENT

B-4. A domestic imagery release document is required when Army intelligence functions obtain imagery of private property. The property owner may be an individual or a city. The owner or chief executive of a municipality must sign the release.

B-5. A private property release statement should include a statement of consent granting the U.S. Government permission to take overhead photographs and collect remote sensing data of specific targets and surrounding area. The statement should also include the following: the landowner's name, a list of the specific targets, the full address of the location, a statement of understanding that photography and data collection will occur on specific dates, and the landowner's signature.

DOMESTIC IMAGERY LIMITATIONS

B-6. Organizations performing intelligence activities may not violate the rights of any *United States person*—a United States citizen; an alien known by the concerned intelligence agency to be a permanent resident alien; an unincorporated association substantially composed of United States citizens or permanent resident aliens; or a corporation incorporated in the United States, except for those directed and controlled by a foreign government or governments (JP 2-0). Collecting information on specific targets within the United States raises policy and legal concerns that require careful considerations, analysis, and coordination with legal counsel. Information collection within the United States must be in accordance with Executive Order 12333 as amended, the National Security Act of 1947 as amended, DODD 5240.01, and AR 381-10.

B-7. NSG members will collect and access only the amount of domestic imagery necessary to meet mission requirements. When accessing or requesting domestic imagery, NSG members will identify the specific location necessary to meet mission requirements and limit the scope of their activities accordingly. Domestic imagery collected incidentally to an approved collection activity cannot be exploited unless additional approval is granted.

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Appendix C

GEOINT Collection Assets

OVERVIEW

C-1. GEOINT collection systems include space-based imaging satellites, manned aircraft systems, and UASs. GEOINT imagery analysts use available collection assets (organic, assigned, and attached as well as supporting) (see paragraph 2-6) to support requirements across the domains. Nonorganic GEOINT capabilities further support this by providing additional aerial and space support to satisfy both organic and subordinate unit requirements.

In this publication—

- *Collection asset* is a collection system, platform, or capability that is supporting, assigned to, or attached to a particular commander (JP 2-0). In many contexts, *collection assets* are referred to as intelligence assets, intelligence collection assets, or information collection assets to describe units, systems, and sensors that perform information collection.
- A *collection resource* refers to a collection system, platform, or capability that is not assigned or attached to a specific commander, unit, or echelon and is requested and coordinated through the chain of command of the unit that directs and controls them. A collection asset is subordinate to the requesting unit or echelon, while a collection resource is not.

IMAGING SATELLITES

C-2. According to Pub 1.0, imagery data is collected from space by two types of satellites: national systems (also known as government-owned satellites) and commercial systems (also known as private industry-owned satellites). National systems are used for both unclassified and classified purposes. Satellites orbit at a far higher altitude than airborne assets, allowing them to collect above any airspace, including the sovereign airspace of any nation, hostile territory, and areas to which airborne assets are denied access.

NATIONAL SYSTEMS

C-3. Sensors attached to national systems are a significant source of GEOINT data used for intelligence and national security purposes. National systems are used when the U.S. Government needs discretion in when or where it collects data. They are used for more sensitive data collection and circumstances.

Note. Executive Order 12333 as amended restricts U.S. Government systems from imaging U.S. persons.

COMMERCIAL SYSTEMS

C-4. Commercial imagery is collected by remote sensing satellites and aircraft owned and operated by commercial imagery providers. Commercial imagery is unclassified. The U.S. Government uses unclassified commercial data whenever possible, and increasingly uses these systems as a primary data source. NGA produces the Commercial GEOINT Strategy for the NSG and, along with the NRO, conducts data assessments to identify vendors that provide desirable capabilities.

C-5. The NRO sees private sector data acquisition as potentially less costly but equally valuable to what national assets can provide. According to Pub 1.0, products derived from commercial system data can also be more easily shared with government and military organizations outside the U.S. IC, nongovernmental organizations (such as nonprofit relief organizations), private industry partners, foreign allies, and other foreign nations. This allows the NRO to meet a larger number of customer requirements more quickly and dedicate national systems to the most challenging and sensitive missions.

C-6. Commercial imagery includes the EO commercial layer (also known as EOCL) and commercial SAR (also known as COMSAR). The EO commercial layer involves a substantial increase in requirements for foundation data, intelligence points, and untaskable data collection; short-wave infrared, nighttime, and non-Earth imaging; and direct downlink to theater-based remote ground terminals. The NRO manages the EO commercial layer and commercial SAR programs by selecting and awarding contracts to commercial data providers. These contract awards are the result of close collaboration with stakeholders across the defense and intelligence communities. Contracts allow collection of data supporting NSG efforts, licensing agreements to share data, and dedicated and secure communications with NGA for sensitive tasking and rapid dissemination of imagery to the NDL for community storage and retrieval via NIPRNET. (Access the NRO website via NIPRNET.)

AIRCRAFT SYSTEMS

C-7. GEOINT collection assets consist of manned aircraft and UASs equipped with a variety of sensors. Individual asset capabilities depend on the sensor and the aircraft's ability to equip it. Some assets include a weapons payload to engage confirmed enemy targets.

MANNED AIRCRAFT SYSTEMS

C-8. Manned aircraft systems from sources such as the USAF and Navy can be leveraged to support Army intelligence requirements. Examples of manned aircraft systems include but are not limited to the U-2 and the MC-12 Enhanced-Medium Altitude Reconnaissance and Surveillance System (also known as EMARSS). Currently, the Army is modernizing the manned aerial layer, divesting of systems designed for counterinsurgency support, for platforms and capabilities that can fly faster and higher while improving sensor standoff ranges.

U-2

C-9. The U-2 provides high-altitude, all-weather surveillance and reconnaissance day or night in direct support of U.S. and allied forces. (See figure C-1.) It delivers critical imagery and SIGINT to decision makers throughout all the Army strategic contexts. The U-2 can gather a variety of imagery, including multispectral, panchromatic, infrared, and SAR products that can be stored or sent to ground exploitation centers.

Enhanced Medium Altitude Reconnaissance and Surveillance System

C-10. The MC-12 Enhanced-Medium Altitude Reconnaissance and Surveillance System is a worldwide, self-deployable, airborne information collection system designed for timely, accurate, and assured support to tactical forces. (See figure C-1.) It provides direct support to tactical forces by providing persistent area assessments, situation development, and mission overwatch. The system's performance is optimized to support irregular warfare operations. It has a simultaneous multisensory capability (EO, infrared, and SIGINT), and is equipped with point-to-point (sensor-to-shooter) and network communications and an intelligence analysis system workstation. The system can cue onboard and off-board sensors.

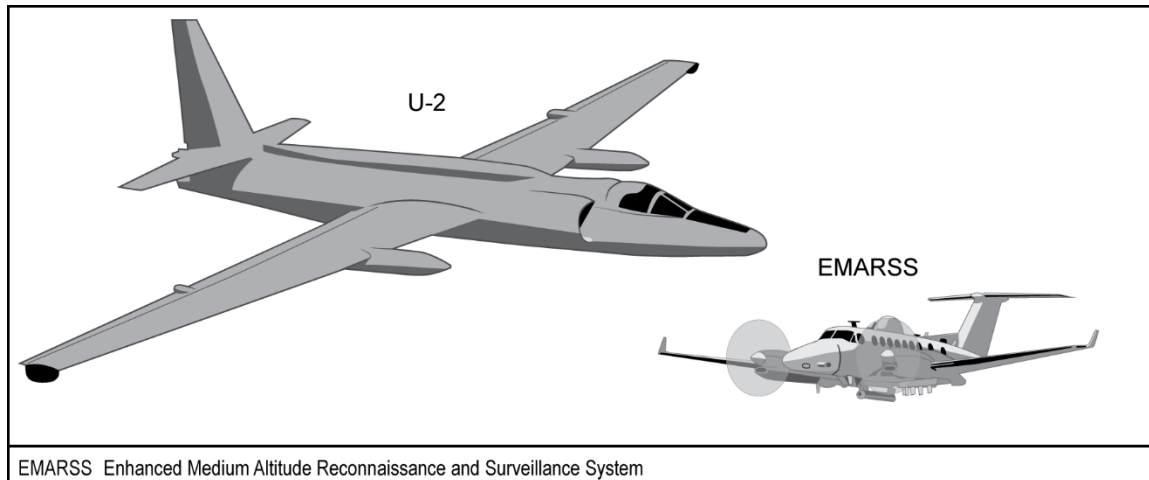


Figure C-1. Manned aircraft systems

UNMANNED AIRCRAFT SYSTEMS

C-11. UASs perform some or all the following:

- Information collection.
- Enhanced targeting through acquisition, detection, designation, suppression, and destruction of threat targets.
- Support to BDA and effects assessment.

C-12. UASs support commanders and staffs as they conduct operations by increasing situational awareness through information collection. Additionally, armed UASs provide direct-fire capabilities that allow commanders to prosecute the close fight and influence shaping of the AO.

C-13. UASs support maneuver commanders by contributing to the effective tactical operation of smaller units. Ground control stations, including the following, enhance situational awareness and the COP:

- Common data links.
- Remote video terminals and remote-operations, video-enhanced receivers.
- One-system remote video transceivers.
- Portable ground control stations.

C-14. While their characteristics and capabilities vary greatly, UASs fit into three main classes: rucksack-portable, tactical, and theater. Additional system-unique characteristics include—

- Expendability.
- Day and night imagery and operations.
- Low noise signature.
- Interchangeable payloads and components.
- Mobile launch capability.
- Sensors:
 - EO.
 - SAR.
 - SAR MTI.

C-15. DOD assigns alphanumeric designations to UASs:

- **C:** Cargo designation.
- **R:** Reconnaissance designation.
- **M:** Multirole designation.
- **Q:** UAS designation.

C-16. The MQ-1C Gray Eagle is a medium-altitude, long-endurance UAS. (See figure C-2.) It is satellite communications-capable and air data relay-capable to support operations beyond LOS. It also has external hard points that can support payload pods or weapons. The Gray Eagle UAS provides—

- Combatant commanders a near real-time, responsive capability to conduct enduring, persistent stare, wide-area reconnaissance, surveillance, target acquisition, communications relay, and attack missions.
- Divisions and MI brigades with improved situational awareness, allowing forces to maneuver from a position of advantage with greater speed and precision.

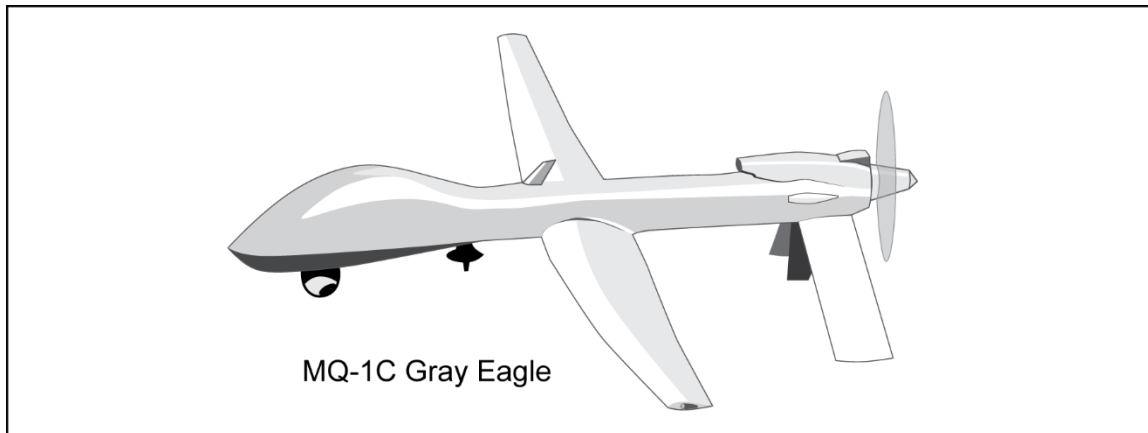


Figure C-2. MQ-1C Gray Eagle unmanned aircraft system

C-17. Gray Eagle has a 20-hour endurance (30-hour endurance if it is an extended range variant) and a 25,000-foot altitude ceiling. It can operate at ranges of up to 1,200 kilometers from the controlling ground control station. The Gray Eagle can carry many payload combinations, including sensors, munitions, and fuel; the limiting factor is weight. It can be configured with an EO, SAR, MTI, laser designator, laser pointer, and laser spot tracker payload that provides an FMV capability.

C-18. In addition to GEOINT and SIGINT payloads that provide multidiscipline intelligence collection and support to divisions, Gray Eagle UASs can also carry AGM-114 Hellfire missiles, which are compatible with the AH-64. The Hellfire missile is redesigned specifically for high-altitude launch profiles. Hellfire missile flight profiles support an increased engagement envelope of up to 360 degrees, more efficient impact angles for a higher kill probability, and an improved stability that prevents errors inherent with gyro-based missiles.

C-19. A Gray Eagle UAS can carry up to four Hellfire laser-designated missiles on four wing-store pylons. However, the added weight and parasitic drag of the missiles greatly reduce aircraft speed and endurance time. The normal armament configuration is armed reconnaissance with two Hellfire missiles (110 pounds each). This configuration allows for longer aircraft endurance and on-station time. Required on-station coverage time may significantly restrict weapons loads and station time associated with carrying more than two Hellfire missiles.

Appendix D

GEOINT Support to Targeting

OVERVIEW

D-1. *Targeting* is the process of selecting and prioritizing targets and matching the appropriate response to them, considering operational requirements and capabilities (JP 3-0). The Army targeting process applies the decide, detect, deliver, and assess (D3A) methodology to facilitate the engagement of the right target with the right asset at the right time. (See FM 3-60 for doctrine on the D3A methodology.)

D-2. GEOINT supports targeting throughout the D3A methodology through analysis and production during the decide, detect, and assess functions by identifying and nominating potential targets and assessing post-operational results. GEOINT monitors deliberate strikes during the deliver function. Accurately identifying and analyzing activities and assessing battle damage in the AO can significantly affect the COA selected by the commander. GEOINT shapes the commander's decision-making process by providing a valuable tool for visualizing and analyzing threat COAs.

DECIDE

D-3. The decide function begins the targeting process. During this function, the commander and staff use their knowledge of the enemy along with the results of IPOE to determine targeting priorities. The GEOINT cell supports this effort by developing IPOE products and refining and updating products throughout each function of the operation.

D-4. GEOINT products focus on the effects of terrain on friendly and enemy COAs. GEOINT analysis identifies enemy TTP and how the enemy may use the terrain to their advantage. GEOINT also aids in identifying and characterizing HPTs and HVTs through imagery exploitation.

D-5. GEOINT input to the information collection plan is vital to the decide function. While GEOINT collection assets engage with targets during the detect function, successful targeting depends on ensuring the appropriate GEOINT capability is selected to collect in the correct location at the proper time. In addition to supporting the initial information collection plan, GEOINT production and planning to support HVT and HPT lists influences decision making related to the collection and engagement of targets.

D-6. During competition below armed conflict, targeting continuously refines the decide function. Priority targets must remain valid throughout the target development process to ensure they are authorized for engagement if a transition to armed conflict occurs. Target development occurs in three stages: basic, intermediate, and advanced. The intelligence staff, with significant GEOINT support, provides input at each stage. (CJCSI 3370.01D provides more information on target development.)

BASIC TARGET DEVELOPMENT

D-7. Basic target development involves identifying a target, providing its location, and determining its primary function:

- **Target identification** provides the name and unique identifier for the target.
- **Target location** provides the coordinates for where the target is located and distance and direction to other targets within the area.
- **Target function** establishes the primary function of the target (what it was created to do).

D-8. GEOINT provides products that enable visualization of target locations. GEOINT analysis and pattern-of-life support assessments determine a target's function.

INTERMEDIATE TARGET DEVELOPMENT

D-9. Intermediate target development provides in-depth information about a target:

- **Target significance** addresses why a particular target is important to the overall target system.
- **Target description** provides a physical description of the recognizable attributes of the target.

- **Functional characterization** elaborates on the overall function of the target.
- **Critical target elements** determine the specific components of the target that allow it to perform its function.
- **Expectation statement** describes how engaging a target will impact the threat and how the threat is expected to react.
- **Source description** documents all sources associated with the research and reporting conducted to support each of the previous steps.

D-10. GEOINT data such as activity levels, visual target modifications, and equipment quantity support all-source assessments of a specific target's significance. GEOINT provided by aerial and space-based systems is the primary source for target description information. Analysis of multidisciplined intelligence, to include current and historic GEOINT data, provides the most accurate assessment of a target's function. Imagery analysis provides specific locations for each critical target element.

ADVANCED TARGET DEVELOPMENT

D-11. Advanced target development completes the target development process and establishes the plan for target engagement. Both the fires and intelligence warfighting functions have responsibilities within advanced target development. Advanced target development consists of three processes:

- **Weaponneering** designates the munitions type and quantity required to achieve a specific effect against a target.
- **Aim point selection and development** involves identifying, developing, and documenting the geophysical or virtual locations and the specific time to engage on the target.
- **Collateral damage estimation** assists the commander in staying within the law of armed conflict and rules of engagement by ensuring only legitimate military objects are targeted.

D-12. The GEOINT analysis and visualization of targets are critical to developing aim points; they allow for precision measurements and analysis of collateral objects in the area. GEOINT imagery analysts monitor targets to ensure they meet legal criteria throughout the targeting process.

D-13. All information required for target development is stored in a target folder. For joint targeting, information is stored in an electronic target folder within the Modernized Integrated Database. Table D-1 lists items that can be found in an electronic target folder.

Table D-1. Items in an electronic target folder (example)

Heading		
• Classification	• Category code	• Country code
• Target identification	• Suffix	• Last update
• Target name	• Geocoordinates	
Target summary		
• Critical elements	• Collateral damage assessment	• Modernized Integrated Database data
• Expectation		
Supporting materials		
• GEOINT data	• Other	
Weaponneering support		
• JDPI graphics (mensurated points)	• Critical element description	• Weaponneering solutions
Combat assessment		
• Weapon system video	• Munitions effects assessment	• Reattack recommendation
• Battle damage assessment		
Associated or collocated targets		
• Units	• Facility	• Other
• Equipment		
Objectives and guidance		
Folder notes and related information		
GEOINT	geospatial intelligence	JDPI joint desired point of contact impact

DETECT

D-14. The detect function focuses on detecting and tracking HPTs identified during the decide function of D3A through the implementation of the information collection plan. The detect function for GEOINT involves exploiting data from GEOINT collection assets tasked to support the targeting process and recommending new collection based on that exploitation. As GEOINT imagery analysts exploit data from GEOINT collection assets within tasked NAIs, they may identify unexpected enemy activity that meets targeting criteria.

D-15. Target tracking is inherent to detection. Once a target is identified, GEOINT data is used to track the target until effects are levied. GEOINT supports tracking mobile targets via FMV, MTI, two-color multiview, and change detection analysis. FMV and MTI analysis can track a target's movement in real time, while change detection and two-color multiview provide information concerning changes in vehicle or equipment location. These capabilities assist the G-2/S-2 in developing estimates of the enemy target's location as they prepare to engage the target during the deliver function.

DELIVER

D-16. The deliver function executes the target engagement guidance developed during the decide function. This function requires decisions concerning the time of engagement, desired effect, which asset to use, and potential for collateral damage. GEOINT supports this effort by providing precise target mensuration to reduce target location error.

D-17. During target engagement, analysts monitor live FMV feeds to provide analysis of the target and its surroundings. The analysis conducted during this stage requires continuous tracking of the target to maintain positive identification, addresses potential targets of opportunity, and should include considerations of the environment as targets move. This assists the commander in engaging targets within the rules of engagement and mitigating the effects of collateral damage.

ASSESS

D-18. *Combat assessment* is the determination of the effectiveness of friendly attack (JP 3-60). It comprises BDA, munitions effectiveness assessment, and reengagement recommendation. The GEOINT cell analyzes imagery and develops BDA products to—

- Aid in assessing the effectiveness of weapon systems.
- Determine physical and functional damage.
- Inform the commander's decision regarding potential reengagement of a target.

D-19. BDA consists of three types of assessments: physical damage, functional damage, and target system. GEOINT is instrumental in assessing physical damage, as this assessment requires interpreting damage based on visual analysis of the target post engagement. Physical damage assessment lays the groundwork for functional damage assessment, which informs target system assessment.

PHYSICAL DAMAGE ASSESSMENT

D-20. *Physical damage assessment* is the estimate of the quantitative extent of physical damage to a target (JP 3-60). The GEOINT imagery analyst assists the commander and staff in determining physical damage to a target via exploitation of imagery collected before, during, and after engagement. The analyst reviews previous BDA reports and considers the relationship between the target and the munition used to engage it as they make their physical damage assessment. The analyst identifies additional or collateral damage resulting from the operation, and any denial and deception activities observed.

D-21. The GEOINT cell prepares a physical damage assessment report and imagery interpretation report for each target engaged. As they do so, they must also determine at what confidence level they can assess the damage.

D-22. Table D-2 describes the three levels of confidence used in physical damage assessments.

Table D-2. Physical damage assessment confidence levels

Confidence level	Likelihood of accuracy	Inference required
Confirmed	95 percent; assured via IMINT, SIGINT, or HUMINT	None
Probable	Greater than 50 percent	Minimal
Possible	Less than 50 percent	Considerable
HUMINT	human intelligence	SIGINT
IMINT	imagery intelligence	signals intelligence

FUNCTIONAL DAMAGE ASSESSMENT

D-23. Once physical damage has been assessed, the GEOINT imagery analyst conducts a functional damage assessment using the same imagery. *Functional damage assessment* is the estimate of the effect of military force to degrade or destroy the functional or operational capability of the target to perform its intended mission and on the level of success in achieving operational objectives established against the target (JP 3-60). This assessment is based on multisource information and includes an estimation of the time required to recuperate or replace the target function.

D-24. When conducting the functional damage assessment, the GEOINT imagery analyst assists in—

- Reviewing physical battle damage on imagery as well as any previous BDA reports for the target.
- Validating initial physical damage assessment reporting using additional multisource information.
- Completing a detailed physical damage assessment and functional damage assessment using the appropriate reference for each target element or aim point attacked.
- Determining how the—
 - Physical damage relates to the functional damage.
 - Function of the target element or aim point has been affected.
 - Recuperation time of the target element or aim point is affected.
- Completing a justification for the target element's functional damage assessment.
- Considering how important each target element is to the whole target and whether the target element should be struck again (if it was not destroyed).

Note. Confidence levels are not associated with the functional damage assessment due to its subjective nature.

TARGET SYSTEM ASSESSMENT

D-25. *Target system assessment* is the broad assessment of the overall impact and effectiveness of military force applied against the operation of an enemy target system, significant subdivisions of the system, or total combat effectiveness relative to the operational objectives established (JP 3-60). GEOINT provides physical and functional BDA reporting to support target system assessment. GEOINT reporting and analysis may assist in identifying functional damage, recuperation times, and reattack recommendations for the BDA target system assessment report.

Appendix E

GEOINT Integrated Architecture

ARMY GEOINT ARCHITECTURE

E-1. The GEOINT integrated architecture is a vast net-centric environment, a framework for full human and technical connectivity and interoperability, allowing DOD users and mission partners to share the information they need, when they need it, and in forms they can understand and act on with confidence. Additionally, the net-centric environment protects access to information, via individual network adjudication processes, against those who should not have it.

Note. The *Department of Defense information network* is the information capabilities and associated processes owned or leased by the Department of Defense for collecting, processing, storing, disseminating, and managing information, whether interconnected or stand-alone. (JP 6-0).

E-2. The Army's reachback concept of support is critical to the Nation's ability to build a campaign-quality force with joint and expeditionary capabilities. Within a modular force design, GEOINT imagery analysts examine imagery received from a multitude of GEOINT collection assets—from national satellite systems to tactical UASs—to answer commanders' PIRs and information requirements.

E-3. GEOINT imagery analysts within corps main and tactical command posts receive, process, exploit, and disseminate GEOINT products to support current corps operations and to provide direct support to division, combat aviation brigade, and BCT operations.

E-4. Intelligence analysis systems provide division main and tactical command posts with communications downlinks and secure internet interfaces, along with raw and processed imagery. GEOINT imagery analysts exploit, analyze, and disseminate selected theater or organic data in real or near real time to support intelligence operations at division and supported BCTs.

E-5. GEOINT imagery analysts, through TGS, provide current threat information and input for the combat aviation brigade, E-MIB, and BCT AO or area of interest COP. To support target development, GEOINT imagery analysts receive, process, database, and graphically present information on threat forces, facilities, and capabilities.

PROCESSING SYSTEMS

E-6. Imagery must be transformed into IMINT through an analytical process. GEOINT imagery analysts conduct analysis using software applications within processing systems.

TACTICAL AND OPERATIONAL INTELLIGENCE GROUND STATIONS

E-7. TGS is based on an up-armored high mobility multipurpose wheeled vehicle (also called HMMWV) that can support operations on the move. It operates at the Secret level and is located at corps, divisions, E-MIBs, and BCTs (one per BCT). TGS has antennas for directly receiving data from sensors, including integrated broadcast service, MTIs from multiple aerial and space-based sensors, and EO, infrared, and MTIs from UAS sensors (Secret only).

E-8. The Operational Intelligence Ground Station (OGS) is configured in multiple portable container shelters. Ground stations have multiple antennas and servers for directly receiving Secret and Top-Secret data from SIGINT and GEOINT collection assets. Ground stations are located at corps and with INSCOM aerial exploitation battalions. OGS does not include workstations, but GEOINT Workstation can be used with it.

E-9. TGS and OGS components—

- Provide imagery exploitation and targeting tools to GEOINT imagery analysts for FMV, imagery, MTIs, and SAR.
- Distribute imagery products via secure local area networks.
- Develop products based on the commander's requirements.
- Generate precision coordinates or geopositions from video and fixed frames.
- Receive MTI track archives or live feeds to integrate with video and fixed frames.
- Display current UAS positions with friendly and threat locations.

Note. TGS and OGS use different antennas for different sensors, but they use the same software for antennas, imagery, MTI, and FMV.

GEOINT WORKSTATION

E-10. GEOINT Workstation is for geospatial analysis and production as well as IMINT PED. It provides geospatial GEOINT imagery analysts within tactical and operational Army units with the ability to process, view, exploit, manage, transmit, and store geospatial and imagery information via Army area communications from echelons above corps to brigade.

JOINT DEPLOYABLE INTELLIGENCE SUPPORT SYSTEM

E-11. A *joint deployable intelligence support system* is a transportable workstation and communications suite that electronically extends a joint intelligence center to a joint task force or other tactical user (JP 2-0). A joint deployable intelligence support system (JDISS) ensures the Services and national agencies have an intelligence system that is both compatible and interoperable using a common data standard. It is the principal means by which intelligence flows throughout the JTF.

E-12. Local area networks or Service-specific systems support GEOINT collection assets. Trojan Special Purpose Integrated Remote Intelligence Terminal (also known as SPIRIT) communications systems provide divisions with organic, JDISS connectivity, but the additional bandwidth required—which should be identified as early as possible—to support joint operations may require augmentation from theater and national sources.

E-13. The joint intelligence architecture uses JDISS and Joint Worldwide Intelligence Communications System the (JWICS) as the joint standard and foundation for commonality among support systems:

- JDISS provides automated support by—
 - Transmitting and receiving specific intelligence requests.
 - Accessing national, Service, and theater intelligence databases.
 - Supporting digitized imagery exchange.
 - Accessing automated record message processing systems, warning intelligence systems, and collection management systems.
 - Inputting intelligence data into a variety of operations and intelligence systems.
 - Performing multimedia functions such as voice electronic publishing and video teleconferencing.
 - Transmitting and disseminating large imagery files.
- JWICS—
 - Satisfies the requirement for secure, high-speed, multimedia transmission services for intelligence information.
 - Incorporates advanced networking technologies that permit greater throughput and capacity, making possible the use of applications that take advantage of multimedia technologies, including video teleconferencing.
 - Has nodes that can create, receive, transmit, and store video images as well as voice, text, graphics, and data.
 - Allows information to be either broadcasted or shared interactively among JWICS subscribers on a point-to-point or multipoint basis.

- Has circuits that can be managed via the allocation of bandwidth, allowing simultaneous use of the link for multiple applications.
- Is an integral part of the Defense Information Systems Network's sensitive compartmented information portion.

SOFTWARE AND APPLICATION TOOLS

E-14. GEOINT imagery analysts use many different software and application tools, including but not limited to the following, which vary by unit:

- GETS.
- GEOINT Search and Retrieval.
- Information Store.

GEOSPATIAL INTELLIGENCE ENTERPRISE TASKING PROCESSING, EXPLOITATION, AND DISSEMINATION SERVICE

E-15. GETS, accessible via SIPRNET, is a DA G-2 system within the NSG and Army National to Theater Program. It is a U.S. Government-owned, integrated TPED platform that provides intelligence producers, analysts, and customers the ability to create, curate, and consume TPED artifacts. GETS provides tools for all TPED users, including the tactical user submitting an RFI and the mission planner using available metadata to gauge a GEOINT collection asset's impact.

E-16. GETS streamlines the handling of GEOINT requirements, report generation, GIS mapping, production, dissemination, and customer feedback. The system integrates with existing and emerging reporting requirements, to include direct publication to Office of the DNI-mandated intelligence databases. GETS provides an enterprise solution that fulfills GEOINT mission requirements for Army, Marine Corps, Navy, USAF, and other U.S. Government and intelligence agencies.

GEOINT SEARCH AND RETRIEVAL

E-17. GEOINT Search and Retrieval, accessible via SIPRNET, allows users to search and retrieve content from NSG, NGA, ASG, and emergent NGA sources. GEOINT Search and Retrieval provides regular access to new data sources and—

- Consolidates multiple search tools.
- Provides access to more content.
- Provides focused search functionality to drive users to more mission-relevant content.
- Replaces and enhances current NGA search services, including Enterprise Search.

INFORMATION STORE

E-18. NGA's Information Store (also known as iSToRE) is a software application that incorporates advanced geospatial capabilities for archival, discovery, retrieval, and dissemination of GEOINT and imagery products. Use of this software spans multidomain operations.

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Glossary

The glossary lists acronyms and terms with Army or joint definitions. Where Army and joint definitions differ, (Army) precedes the definition.

SECTION I – ACRONYMS AND ABBREVIATIONS

AA	avenue of approach
ACE	analysis and control element
ADP	Army doctrine publication
ADRO	Army Departmental Requirements Office
AGB	Army Geospatial Intelligence Battalion
AGC	Army Geospatial Center
AO	area of operations
AOR	area of responsibility
AR	Army regulation
ASCC	Army Service component command
ASG	Allied System for Geospatial Intelligence
ATP	Army techniques publication
BCT	brigade combat team
BDA	battle damage assessment
C2	command and control
CCMD	combatant command
CIB	controlled image base
CJCSI	Chairman of the Joint Chiefs of Staff instruction
COA	course of action
COP	common operational picture
D3A	decide, detect, deliver, and assess
DA	Department of the Army
DA G-2	Army Deputy Chief of Staff for Intelligence
DA G-3/5/7	Army Deputy Chief of Staff for Operations, Plans, and Training
DNI	Director of National Intelligence
DOD	Department of Defense
DODD	Department of Defense directive
DODM	Department of Defense manual
DTED	digital terrain elevation data
E-MIB	expeditionary-military intelligence brigade
EMS	electromagnetic spectrum
EO	electro-optical
FM	field manual

FMV	full-motion video
FORSCOM	United States Army Forces Command
G-2	assistant chief of staff, intelligence
G-3	assistant chief of staff, operations
G-6	assistant chief of staff, signal
GEOINT	geospatial intelligence
GETS	Geospatial Intelligence Enterprise Tasking, Processing, Exploitation, and Dissemination Service
GI&S	geospatial information and services
GIMS	Geospatial Intelligence Information Management Services
GPC	geospatial planning cell
HUMINT	human intelligence
HPT	high-payoff target
HVT	high-value target
IC	intelligence community
IDP	imagery derived product
IEW	intelligence and electronic warfare
IMINT	imagery intelligence
INSCOM	United States Army Intelligence and Security Command
IPOE	intelligence preparation of the operational environment
J-2	intelligence directorate of a joint staff
JDISS	joint deployable intelligence support system
JP	joint publication
JTF	joint task force
JWICS	Joint Worldwide Intelligence Communications System
LIDAR	light detection and ranging
LOC	line of communications
LOS	line of sight
MASINT	measurement and signature intelligence
MDMP	military decision-making process
MGRS	military grid reference system
MI	military intelligence
MIB-T	military intelligence brigade-theater
MTI	moving target indicator
NAI	named area of interest
NDL	National Data Library
NGA	National Geospatial-Intelligence Agency
NGIC	National Ground Intelligence Center
NIPRNET	Nonsecure Internet Protocol Router Network
NRO	National Reconnaissance Office
NSG	National System for Geospatial Intelligence
NST	National Geospatial-Intelligence Agency support team

OAKOC	observation and fields of fire, avenues of approach, key terrain, obstacles, and cover and concealment [military aspects of terrain]
OB	order of battle
OE	operational environment
OGS	Operational Intelligence Ground Station
OPIR	overhead persistent infrared
OPORD	operation order
PAI	publicly available information
PED	processing, exploitation, and dissemination
PIR	priority intelligence requirement
Pub 1.0	Geospatial Intelligence (GEOINT) Basic Doctrine, Publication 1.0
RFC	request for collection
RFI	request for information
S-2	battalion or brigade intelligence staff officer
S-2X	battalion or brigade counterintelligence and human intelligence staff officer
S-3	battalion or brigade operations staff officer
S-6	battalion or brigade signal staff officer
SAR	synthetic aperture radar
SIGINT	signals intelligence
SIPRNET	SECRET Internet Protocol Router Network
SOP	standard operating procedure
SWO	staff weather officer
TENCAP	Tactical Exploitation of National Capabilities Program
TGS	Tactical Intelligence Ground Station
TPED	tasking, processing, exploitation, and dissemination
TTP	tactics, techniques, and procedures
UAS	unmanned aircraft system
U.S.	United States
USAICOE	United States Army Intelligence Center of Excellence
USASMD	United States Army Space and Missile Defense Command
USASOC	United States Army Special Operations Command
UTM	universal transverse Mercator
WAMI	wide area motion imagery
WGS 84	World Geodetic System 1984

SECTION II – TERMS

avenue of approach

A path used by an attacking force leading to its objective or to key terrain. Avenues of approach exist in all domains. (ADP 3-90)

collection

In intelligence usage, the acquisition of information and the provision of this information to processing elements. (JP 2-0)

collection asset

A collection system, platform, or capability that is supporting, assigned to, or attached to a particular commander. (JP 2-0)

collection operations management

The authoritative direction, scheduling, and control of specific collection operations and associated processing, exploitation, and reporting resources. (JP 2-0)

collection requirements management

The authoritative development and control of collection, processing, exploitation, and/or reporting requirements that normally result in the direct tasking of requirements to units over which the commander has authority, or the generation of tasking requests to collection management authorities at a higher, lower, or lateral echelon to accomplish the collection mission. (JP 2-0)

combat assessment

The determination of the effectiveness of friendly attack. (JP 3-60)

common operational picture

(DOD) A single, identical display of relevant information shared by more than one command that facilitates collaborative planning and assists all echelons to achieve situational awareness. (JP 3-0)

concealment

Protection from observation or surveillance. (FM 3-96)

cover

(Army) Protection from the effects of fires. (FM 3-96)

crisis

An emerging incident or situation involving a possible threat to the United States, its citizens, military forces, or vital interests that develops rapidly and creates a condition of such diplomatic, economic, or military importance that commitment of military forces and resources is contemplated to achieve national and/or strategic objectives. (JP 3-0)

decisive terrain

Key terrain whose seizure and retention is mandatory for successful mission accomplishment. (ADP 3-90)

Department of Defense information network

The information capabilities and associated processes owned or leased by the Department of Defense for collecting, processing, storing, disseminating, and managing information, whether interconnected or stand-alone. (JP 6-0)

dissemination

In intelligence usage, the delivery of intelligence to users in a suitable form. (JP 2-0)

field of fire

The area that a weapon or group of weapons may cover effectively from a given position. (FM 3-90)

functional damage assessment

The estimate of the effect of military force to degrade or destroy the functional or operational capability of the target to perform its intended mission and on the level of success in achieving operational objectives established against the target. (JP 3-60)

geospatial information

(DOD) Information that identifies the geographic location and characteristics of natural or constructed features and boundaries on or about the Earth, including: data and information derived from, among other things, remote sensing, mapping, and surveying technologies; and mapping, charting, geomatics data, and related products and services. (JP 2-0)

geospatial information and services

The collection, information extraction, storage, dissemination, and exploitation of geodetic, geomagnetic, imagery (both commercial and national source), gravimetric, aeronautical, topographic, hydrographic, littoral, cultural, and toponymic data accurately referenced to a precise location on the surface of the Earth. (JP 2-0)

geospatial intelligence

The exploitation and analysis of imagery and geospatial information to describe, assess, and visually depict physical features and geographically referenced activities on or about the Earth. Geospatial intelligence consists of imagery, imagery intelligence, and geospatial information. (JP 2-0)

imagery

A likeness or presentation of any natural or man-made feature or related object or activity, and the positional data acquired at the same time the likeness or representation was acquired, including: products produced by space-based national intelligence reconnaissance systems; and likeness and presentations produced by satellites, airborne platforms, unmanned aerial vehicles, or other similar means (except that such term does not include handheld or clandestine photography taken by or on behalf of human intelligence collection organizations). (JP 2-0)

imagery intelligence

The technical, geographic, and intelligence information derived through the interpretation or analysis of imagery and collateral materials. (JP 2-0)

intelligence production

The integration, evaluation, analysis, and interpretation of information from single or multiple sources into finished intelligence for known or anticipated military and related national security consumer requirements. (JP 2-0)

joint deployable intelligence support system

A transportable workstation and communications suite that electronically extends a joint intelligence center to a joint task force or other tactical user. (JP 2-0)

key terrain

(Army) An identifiable characteristic whose seizure or retention affords a marked advantage to either combatant. (ADP 3-90)

multinational operations

A collective term to describe military actions conducted by forces of two or more nations, usually undertaken within the structure of a coalition or alliance. (JP 3-16)

National System for Geospatial Intelligence

The combination of technology, policies, capabilities, doctrine, activities, people, data, and organizations necessary to produce geospatial intelligence in an integrated, multi-intelligence environment. (JP 2-0)

observation

The condition of weather and terrain that permits a force to see the friendly, enemy, and neutral personnel and systems, and key aspects of the environment. (FM 1-02.1)

obstacle

Any barrier designed or employed to disrupt, fix, turn, or block the movement and maneuver, and to impose additional losses in personnel, time, and equipment. (JP 3-15)

overhead persistent infrared

Within geospatial intelligence, a capability that provides on-demand, persistent, global, and/or localized coverage of high- to low-intensity infrared events to detect energy radiation from various tactical to strategic objects. (JP 2-0)

physical damage assessment

The estimate of the quantitative extent of physical damage to a target. (JP 3-60)

processing

A system of operations designed to convert raw data into useful information. (JP 2-0)

request for information

Any specific, time-sensitive, ad hoc requirement for intelligence information or products to support an ongoing crisis or operation not necessarily related to standing requirements or scheduled intelligence production. (JP 2-0)

target system assessment

The broad assessment of the overall impact and effectiveness of military force applied against the operation of an enemy target system, significant subdivisions of the system, or total combat effectiveness relative to the operational objectives established. (JP 3-60)

targeting

The process of selecting and prioritizing targets and matching the appropriate response to them, considering operational requirements and capabilities. (JP 3-0)

United States person

A United States citizen; an alien known by the concerned intelligence agency to be a permanent resident alien; an unincorporated association substantially composed of United States citizens or permanent resident aliens; or a corporation incorporated in the United States, except for those directed and controlled by a foreign government or governments. (JP 2-0)

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Index

Entries are by paragraph number.

A

ADRO, 2-19, A-15
AGC, 2-19 (special segment). *See also* NSG, Army organizations support to.
as intelligence producer, 5-3
as tool, 4-7, 4-60
aircraft systems, C-7. *See also* GEOINT collection, methods.
manned, C-8–C-10
unmanned, C-11–C-19
Allied System for Geospatial Intelligence. *See* ASG.
architecture, Army GEOINT, E-1–E-5
Army Departmental Requirements Office. *See* ADRO.
Army GEOINT Battalion (AGB). *See* NSG, Army organizations support to.
Army GEOINT Office, 2-23. *See also* NSG, Army organizations support to.
Army Geospatial Center. *See* AGC.
Army Geospatial Enterprise, 2-22, 2-23
Army GEOINT integration, 2-24–2-26
corps and below GEOINT elements, 2-37–2-51
theater army GEOINT elements, 2-27–2-36
Army National to Theater Program, 2-8–2-10
NGIC support, 2-11
U.S. Army Space and Missile Defense Command, 2-12, 2-13
Army Service component command (ASCC). *See also* GEOINT elements, theater army.
during armed conflict, 6-8
during competition below armed conflict, 6-3
during crisis, 6-6

Army strategic contexts, 1-2, 2-22
armed conflict. *See* GEOINT.
competition below armed conflict. *See* GEOINT.
crisis. *See* GEOINT.
ASG, 2-43, E-17. *See also* NSG, multinational partners, collaboration with.

B

brigade combat team (BCT). *See* corps and below levels; GEOINT elements, corps and below.

C

collection, 4-1. *See also* GEOINT collection.
multisensory/multidiscipline intelligence, 4-52, 4-53
collection management, 3-78–3-82. *See also* GEOINT collection.
combat assessment. *See* targeting, GEOINT support to, assess function of D3A.
common operational picture (COP). *See* GEOINT.
corps and below levels
during armed conflict, 6-8
during competition below armed conflict, 6-3
during crisis, 6-6

D

D3A methodology. *See* targeting, GEOINT support to.
data, 3-5. *See also* GEOINT processing; joint operations, joint force GEOINT support.
dissemination. *See* GEOINT dissemination.

division. *See* corps and below levels; GEOINT elements, corps and below.

E

Enhanced-Medium Altitude Reconnaissance and Surveillance System (EMARSS). *See* aircraft systems, manned.
expeditionary-military intelligence brigade (E-MIB). *See* corps and below levels; GEOINT elements, corps and below.

F

FORSCOM, 2-7, 2-34. *See also* NSG, Army organizations support to.
full-motion video (FMV). *See* imagery types, motion imagery.
functional assessment. *See* targeting, GEOINT support to, assess function of D3A.

G

GEOINT
collection asset. *See* GEOINT collection asset.
defined, 1-1–1-3
focus during armed conflict, 6-7, 6-8
focus during competition below armed conflict, 6-1–6-3
focus during crisis, 6-4–6-6
primary GEOINT providers, 2-24
requirement. *See* collection management.
resources
(organic/nonorganic), 1-13
software and application tools, E-14–E-18
standards. *See* GEOINT processing, data management.
support to the common operational picture (COP), 3-5
GEOINT analysis, 5-5, 5-6
GEOINT cell, 2-26, 2-37, 3-3. *See also* information collection plan. and the intelligence staff, 2-45–2-50

Entries are by page number.

- GEOINT cell (*continued*)
 capabilities, 2-40–2-44
 data, 3-5
 functions, 2-39
 support to IPOE, 3-18–3-65
 support to the MDMP, 3-2–3-17
 systems, 3-6
 personnel, 3-4
- GEOINT collection, 4-2–4-4
 activities, 4-44–4-47
 assessing the collection and provide feedback, 4-35, 4-37
 developing the request for collection (RFC), 4-5–4-23
 managing collection requirements, 4-38–4-43
 methods, 4-48–4-51
 prioritizing the collection requirement, 4-26–4-28
 tasking the collection requirement, 4-29–4-34
 validating the collection requirement, 4-24, 4-25
- GEOINT collection asset, A-1, C-1. *See also* aircraft systems; collection management; imaging satellites.
 integrating, 4-52, 4-53
- GEOINT dissemination, 5-14
 and integration, 5-15, 5-16
 methods and techniques, 5-17–5-32
- GEOINT elements
 corps and below, 2-37–2-51
 theater army, 2-27–2-36
- GEOINT imagery analyst, 2-37, 2-40, 2-42. *See also* GEOINT, personnel.
 conducting research, 3-66
 in the U.S. Army Special Operations Command, 2-5
 organizing data, 3-5
- GEOINT operations, 3-73
 architecture evaluation, 3-76
 bandwidth support, 3-74
 systems improvements, 3-75
- GEOINT policy, B-1
 domestic imagery limitations, B-6, B-7
 domestic imagery release document, B-4, B-5
 domestic imagery request, B-3
 domestic imagery requirements, B-2
- GEOINT processing, 4-54, 4-55
 data management, 4-56–4-66
 data manipulation, 4-67–4-77
 systems, E-6–E-13
- GEOINT production, 5-1–5-4, 5-7.
See also multinational operations.
 consumer product feedback, 5-12, 5-13
 standard GEOINT products, 5-8
 tailored GEOINT products, 5-9–5-11
- GEOINT Search and Retrieval.
See GEOINT, software and application tools.
- GEOINT support, 7-17. *See also* joint operations; multinational operations.
 in cold weather and mountain environments, 7-20
 in desert environments, 7-22
 in jungle environments, 7-21
 in littoral environment, 7-23, 7-24
 in urban environment, 7-18, 7-19
- GEOINT Workstation. *See* GEOINT processing, systems.
- georeferencing. *See* GEOINT processing, data manipulation.
- geospatial engineer, 2-42, 2-44
 during IPOE, 3-19
 in the U.S. Army Special Operations Command, 2-5
- geospatial information, 1-8
- geospatial information and services (GI&S), 1-7
- geospatial intelligence. *See* GEOINT.
- Geospatial Intelligence Enterprise Tasking, Processing, Exploitation, and Dissemination Service (GETS). *See* GEOINT, software and application tools.
- geospatial planning cell (GPC), 2-27, 2-29, 2-30, 2-43
- Gray Eagle. *See* aircraft systems, unmanned.
- ground reconnaissance. *See* imagery sources, terrestrial systems.
- I**
- imagery
 along the electromagnetic spectrum, A-2–A-4
 defined, 1-4
 updating, 4-57–4-60
- imagery intelligence, 1-5, 1-6
- imagery intelligence reports. *See* GEOINT dissemination, methods and techniques.
- imagery resolution, A-23
 radiometric, A-24
 spatial, A-27
 spectral, A-26
 temporal, A-25
- imagery sensors, A-5
 active sensors, A-6
 passive sensors, A-7
- imagery sources, A-9
 airborne systems, A-16, A-17
 satellite systems, A-10–A-15
 terrestrial systems, A-18–A-22
- imagery types, A-28
 characteristics, A-53
 motion imagery, A-45–A-52
 still imagery, A-29–A-44
- imaging satellites, C-2
 commercial systems, C-4–C-6
 national systems, C-3
- information collection plan, 3-67
 GEOINT input to, 3-68
 validating GEOINT portions of, 3-69–3-72
- Information Store (ISToRE). *See* GEOINT, software and application tools.
- infrared. *See* imagery types, still imagery.
- INSCOM, 2-7, 2-16. *See also* NSG, Army organizations support to.
 aerial exploitation battalion, E-8
 functional command, 2-16–2-18, 2-32
 MIB-Ts, 2-10, 2-27
 support to regional treaty partners and allies, 2-33
- intelligence preparation of the operational environment. *See* IPOE.
- intelligence process, GEOINT nested within, 1-2
 step 1, plan and direct, 3-1, 3-77
 step 2, collect and process, 4-1, 4-54, 4-55
 step 3, produce, 5-1–5-4
 step 4, disseminate and integrate, 5-14
- intelligence production. *See* GEOINT production.

Entries are by page number.

intelligence reach. See GEOINT imagery analyst, conducting research.

intelligence staff. See GEOINT cell.

IPOE. See GEOINT cell; geospatial engineer.

J

joint deployable intelligence support system. See GEOINT processing, systems.

joint operations, 7-1–7-4
joint force GEOINT support, 7-7–7-14
joint intelligence architecture support, 7-5, 7-6

L

light detection and ranging (LIDAR). See imagery types, still imagery.

M

manned aircraft system. See aircraft systems, manned.

MIB-T. See *also* GEOINT elements, theater army; INSCOM.
during armed conflict, 6-8
during competition below armed conflict, 6-3
during crisis, 6-6

military decision-making process (MDMP). See GEOINT cell.

military intelligence brigade-theater. See MIB-T.

moving target indicator (MTI). See imagery types, motion imagery.

multinational operations, 7-15, 7-16

N

National Data Library (NDL), 3-66, B-2, B-3, C-6

National Geospatial-Intelligence Agency (NGA), 1-15
director, 1-16
NGA-Army 1-20
support team, 1-17–1-19, 2-16

National Ground Intelligence Center. See NGIC.

National Reconnaissance Office (NRO), 4-50, A-10, C-4–C-6

National System for Geospatial Intelligence. See NSG.

NGIC, 3-61. See *also* Army National to Theater Program; Army organizations support to.

NSG

defined, 1-9, 1-10
members, 1-11, 1-14. See *also* GEOINT policy, domestic imagery requirements and limitations.
multinational partners, collaboration with, 1-12, 1-13

NSG, Army organizations support to, 2-1
AGC, 2-14
Army GEOINT Battalion, 2-19
Army GEOINT Office, 2-7
Army National to Theater Program, 2-8–2-10
Army TENCAP Office, 2-15
FORSCOM, 2-2
INSCOM, 2-3, 2-4
NGIC, 2-16–2-19
U.S. Army Engineer School, 2-21
U.S. Army MI School, 2-20
U.S. Army Special Forces Command, 2-5, 2-6

O

Operational Intelligence Ground Station (OGS). See GEOINT processing, systems.

overhead persistent infrared (OPIR). See imagery types, still imagery.

P

panchromatic. See imagery types, still imagery.

physical damage assessment. See targeting, GEOINT support to, assess function of D3A.

polarimetric. See imagery types, still imagery.

positional reference systems. See GEOINT processing, data manipulation.

processing. See GEOINT processing.

publicly available information (PAI), 3-39. See *also* imagery sources, terrestrial systems.

R

rectification. See GEOINT processing, data manipulation.

regionally aligned forces assistance from MIB-Ts, 2-34–2-36

request for collection (RFC). See GEOINT collection.

request for information (RFI) process, 3-83, 3-84
documenting and managing the request, 3-86
fulfilling the request, 3-85

requirements. See *also* collection management; GEOINT collection; joint operations, joint force GEOINT support.
ad hoc, 4-20, 4-21
dynamic, 4-22, 4-23

S

satellite system. See GEOINT collection, methods.

spectral. See imagery types, still imagery.

synthetic aperture radar (SAR). See imagery types, still imagery.

T

Tactical Exploitation of National Capabilities Program (TENCAP). See NSG, Army organizations support to.

Tactical Intelligence Ground Station (TGS). See GEOINT processing, systems.

target development. See targeting, GEOINT support to, decide function of D3A.

target system assessment. See targeting, GEOINT support to, assess function of D3A.

targeting, GEOINT support to, D-1, D-2
assess function of D3A, D-18–D-25
decide function of D3A, D-3–D-13
deliver function of D3A, D-16, D-17
detect function of D3A, D-14, D-15

U

U.S. Army Forces Command. See FORSCOM.

U.S. Army Intelligence and Security Command. See INSCOM.

Entries are by page number.

U.S. Army Space and Missile
Defense Command. *See* Army
National to Theater Program.

U.S. Army Special Operations
Command. *See* NSG, Army
organizations support to.

U-2. *See* aircraft systems,
manned.

unattended ground sensor. *See*
imagery sources, terrestrial
systems.

United States person. *See*
GEOINT policy, domestic
imagery limitations.

unmanned aircraft system (UAS).
See aircraft systems,
unmanned.

W

wide area motion imagery
(WAMI). *See* imagery types,
motion imagery.

ATP 2-22.7

09 June 2025

By Order of the Secretary of the Army:

RANDY A. GEORGE

*General, United States Army
Chief of Staff*

Official:

A handwritten signature in black ink, appearing to read 'Mark F. Averill', written in a cursive style.

MARK F. AVERILL

*Administrative Assistant
to the Secretary of the Army
2514805*

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